EUROPEAN MEDICINES AGENCY
SCIENCE MEDICINES HEALTH

23 April 2015
EMA/CHMP/76688/2015
Committee for Medicinal Products for Human Use (CHMP)

Assessment report

OPDIVO

International non-proprietary name: nivolumab

Procedure No. EMEA/H/C/003985/0000

Note

Assessment report as adopted by the CHMP with all information of a commercially confidential nature deleted.

Administrative information

Name of the medicinal product:	OPDIVO
Applicant:	Bristol-Myers Squibb Pharma EEIG Uxbridge Business Park Sanderson Road Uxbridge UB8 1DH UNITED KINGDOM
Active substance:	nivolumab
International Nonproprietary Name/Common Name:	nivolumab
Pharmaco-therapeutic group (ATC Code):	L01XC17
Therapeutic indication(s):	OPDIVO as monotherapy is indicated for the treatment of advanced (unresectable or metastatic) melanoma in adults
Pharmaceutical form(s):	Concentrate for solution for infusion
Strength(s):	10 mg/ml
Route(s) of administration:	Intravenous use
Packaging:	Vial (glass)
Package size(s):	- Pack with 1 vial of 4 ml - Pack with 1 vial of 10 ml

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List of abbreviations

Abs	antibodies
ADA	anti-drug antibody
ADCC	antibody dependent cell-mediated cytotoxicity
AE	adverse event
AJCC	American Joint Committee on Cancer
ALC	absolute lymphocyte count
ALP	alkaline phosphatase
ALT	alanine aminotransferase
ANC	absolute neutrophil count
AST	aspartate aminotransferase
BOR	best overall response
CDC	complement dependent cytotoxicity
CHMP	Committee for Medicinal Products for Human Use
CI	confidence interval
CMV	cytomegalovirus
CR	complete response
CT	computerized tomography
CTC	common toxicity criteria
CTCAE	common terminology criteria for adverse events
CTLs	cytotoxic T lymphocytes
CTLA-4	cytotoxic T-lymphocyte antigen 4
CYP	Cytochrome P450
DMC	Data Monitoring Committee
DOR	duration of response
DTIC	dacarbazine
EC	European Commission
EC50	half maximal effective concentration
ECOG	Eastern Cooperative Oncology Group
ELISA	Enzyme linked immunosorbent assay

EORTC	European Organisation for Research and Treatment of Care
ePPND	enhanced pre- and post-natal development
EU	European Union
FPFV	first patient first visit
GCP	Good Clinical Practice
GD	gestational day
GI	gastrointestinal
GLP	Good laboratory practice
Hep B	hepatitis B
HLA	human leukocyte antigen
HIV	human immunodeficiency virus
HR	hazard ratio, heart rate
HRQoL	health-related quality of life
ICH	International Conference on Harmonization
IHC	immunohistochemistry
ICS	intracytoplasmic staining
IFN	interferon
Ig	immunoglobulin
IL-2	interleukin-2
IV	intravenous/intravenously
IVRS	interactive voice response system
K-M	Kaplan-Meier
LDH	Lactate dehydrogenase
LLN	lower limit normal
LPLV	last patient last visit
MAb	monoclonal antibodies
MLR	Mixed Lymphocyte Reaction
MRI	magnetic resonance imaging
NCA	non-compartmentalised analysis
NSCLC	non-small cell lung cancer
ORR	objective response rate

OS	overall survival
PASS	post-authorisation safety study
PAES	post-authorisation efficacy study
PBMCs	peripheral blood mononuclear cells
PD	progressive disease
PD-1	programmed cell death 1
PD-L1	programmed cell death ligand 1
PD-L2	programmed cell death ligand 2
PD	progressive disease, pharmacodynamics
PDCO	Paediatric Committee
PFS	progression-free survival
PK	pharmacokinetic(s)
popPK	population PK analysis
PR	partial response
PRAC	Pharmacovigilance Risk Assessment Committee
PT	preferred term
QW	weekly
Q2W	every 2 weeks
Q3W	every 3 weeks
QLQ-C30	Quality of Life Questionnaire - Core 30
RCC	renal cell carcinoma
RECIST v1.1	Response Evaluation Criteria in Solid Tumors version 1.1
RMP	risk management plan
RO	Receptor Occupancy
RSDV	reduced source data verification
SAE	serious adverse event
SAP	statistical analysis plan
SD	stable disease, standard deviation
SmPC	summary of product characteristics
SPR	surface plasmon resonance
SNP	single nucleotide polymorphism

TSH	thyroid stimulating hormone
TTR	time to response
ULN	upper limit of normal
USA	United States of America
2QW	twice a week

1. Background information on the procedure

1.1. Submission of the dossier

The applicant Bristol-Myers Squibb Pharma EEIG submitted on 2 September 2014 an application for Marketing Authorisation to the European Medicines Agency (EMA) for Opdivo, through the centralised procedure falling within the Article 3(1) and point 1 of Annex of Regulation (EC) No 726/2004. The eligibility to the centralised procedure was agreed upon by the EMA/CHMP on 20 March 2014.

The applicant applied for the following indication: "treatment of advanced (unresectable or metastatic) melanoma in adults".

The legal basis for this application refers to:

Article 8.3 of Directive 2001/83/EC - complete and independent application. The applicant indicated that nivolumab was considered to be a new active substance.

The application submitted is composed of administrative information, complete quality data, non-clinical and clinical data based on applicant's own tests and studies and/or bibliographic literature substituting/supporting certain test(s) or study(ies).

Information on Paediatric requirements

Pursuant to Article 7 of Regulation (EC) No 1901/2006, the application included an EMA Decision(s) P/0064/2014 on the agreement of a paediatric investigation plan (PIP).

At the time of submission of the application, the PIP P/0064/2014 was not yet completed as some measures were deferred.

Information relating to orphan market exclusivity

Not applicable.

Similarity

Pursuant to Article 8 of Regulation (EC) No. 141/2000 and Article 3 of Commission Regulation (EC) No 847/2000, the applicant did not submit a critical report addressing the possible similarity with authorised orphan medicinal products because there is no authorised orphan medicinal product for a condition related to the proposed indication.

New active Substance status

The applicant requested the active substance nivolumab contained in the above medicinal product to be considered as a new active substance in itself, as the applicant claims that it is not a constituent of a product previously authorised within the Union.

Scientific Advice

The applicant received Scientific Advice from the CHMP on 18 October 2012 and 17 January 2013. The Scientific Advice pertained to quality and clinical aspects of the dossier.

Licensing status

Opdivo has been given a Marketing Authorisation in Japan on 4 July 2014 and in the USA on 22 December 2014.

1.2. Manufacturers

Manufacturer responsible for batch release

Bristol-Myers Squibb S.r.l.
Loc. Fontana del Ceraso
03012 Anagni (FR)
Italy

1.3. Steps taken for the assessment of the product

The Rapporteur and Co-Rapporteur appointed by the CHMP:

Rapporteur: Arantxa Sancho-Lopez

Co-Rapporteur: Pieter de Graeff

- The application was received by the EMA on 2 September 2014.
- Accelerated Assessment procedure was agreed-upon by CHMP on 24 July 2014.
- The procedure started on 24 September 2014.
- The Rapporteur's first Assessment Report was circulated to all CHMP members on 15 December 2014. The Co-Rapporteur's first Assessment Report was circulated to all CHMP members on 12 December 2014.
- During the meeting on 9 January 2015 the Pharmacovigilance Risk Assessment Committee (PRAC) adopted the PRAC Advice on the submitted Risk Management Plan.
- During the meeting on 22 January 2015, the CHMP agreed on the consolidated List of Questions to be sent to the applicant. The final consolidated List of Questions was sent to the applicant on 22 January 2015.
- The applicant submitted the responses to the CHMP consolidated List of Questions on 20 February 2015.
- The Rapporteurs circulated the Joint Assessment Report on the applicant’s responses to the List of Questions to all CHMP members on 12 March 2015.
- During the meeting on 12 March 2015 the Pharmacovigilance Risk Assessment Committee (PRAC) adopted the PRAC Advice on the submitted Risk Management Plan.
- During the CHMP meeting on 23 to 26 March 2015, the CHMP agreed on a List of Outstanding Issues to be addressed in writing and/or in an oral explanation by the applicant.
- The applicant submitted the responses to the CHMP List of Outstanding Issues on 30 March 2015.
- The Rapporteurs circulated the Joint Assessment Report on the applicant’s responses to the List of Outstanding issues to all CHMP members on 7 April 2015.
- During the meeting on 10 April 2015 the Pharmacovigilance Risk Assessment Committee (PRAC) adopted the PRAC Advice on the submitted Risk Management Plan.

- During the meeting on 20 to 23 April 2015, the CHMP, in the light of the overall data submitted and the scientific discussion within the Committee, issued a positive opinion for granting a Marketing Authorisation to Opdivo.

2. Scientific discussion

2.1. Introduction

Melanoma is the most aggressive form of skin cancer. Melanoma is the sixth and seventh most common malignancy in men and women, respectively.[1] The median age at diagnosis is 59 years. The incidence of melanoma varies between different European countries but the estimated incidence was about 39.6 cases /100.000 men and 42.5 cases /100.000 women in 2012. In Europe in 2012, the mortality rate was approximately 8.8 cases/100.000 in males and 6.9 cases/100.000 in females.[1] The outcome of melanoma depends on the stage at presentation[2]. Approximately 85% of patients with melanoma present with localised disease, 10% with regional disease and 5% with distant metastatic disease. The 5-year survival rates in patients who present with localised disease and primary tumours 1.0mm or less in thickness are very good, with more than 90% of patients surviving. The 5-year survival rates decrease as the tumour spreads: for tumours of more than 1.0mm in thickness, survival rates range from 50% to 90%, with regional node involvement survival rates are around 50%, for within stage III (regional metastatic melanoma) 5-year survival rates range between 20-70%, depending on primary nodal involvement. The long term survival for distant metastatic melanoma, the 5-year survival is less than 10%.

Current treatments for metastatic melanoma include systemic therapy, surgery and radiotherapy. Spontaneous regression of melanoma has been reported with an incidence of less than 1%. Complete resection of isolated metastases to one anatomic site (lung, gastrointestinal tract, bone or brain) may occasionally achieve long term survival. Systemic treatment may consist of chemotherapy, and/or immunotherapy. Palliative radiotherapy is indicated for symptomatic relief of metastases to brain, bones and viscera.

Chemotherapy with dacarbazine (DTIC) may achieve objective response rates of about 20%, of which less than 5% is complete remission. Higher response rates have been seen using combination chemotherapy, however no increase in over-all survival has been demonstrated with combination regimens when compared to dacarbazine alone. Immunotherapeutic agents used for metastatic melanoma are interferon-alfa (IFNα) and interleukine-2 (IL-2). Recurrent melanoma is resistant to most standard systemic therapy and no standardized effective second line treatment is established.

In EU Countries, dacarbazine was used for many years as standard first line treatment of patients with metastatic melanoma[3]. Clinical trials with dacarbazine have shown low response rates ranging from 11% -25%, low rate of complete responses and of short duration (3 to 6 months). The median survival time ranged from 4.5 to 6 months[4,5,6]. Ipilimumab (Yervoy), a human monoclonal antibody against CTLA-4, was approved in the EU in 2011 for melanoma patients who have received prior therapy. The approval of ipilimumab was based on the

[1] Ferlay J., Steliarova-Foucher E., Lortet-Tieulent J., et al. Cancer incidence and mortality patterns in Europe: Estimates for 40 countries in 2012. Eur J Cancer; 49, 1374– 1403, 2013
[2] Balch CM., Gershenwald JE., Soong SJ., et al. Final version of 2009 AJCC melanoma staging and classification. J Clin Oncol; 27(36):6199-206, 2009
[3] Serrone L, Zeuli M, Sega FM, et al. Dacarbazine- based chemotherapy for metastatic melanoma: Thirty year experience overview. J Exp Clin Cancer Res; 19: 21-34, 2000
[4] Luce JK, Thurman WG, Isaacs BL, et al. Clinical trials with the antitumor agent 5-(3,3–dimethy-1-triazeno)imidazole-4-carboxamide. Cancer Chemother Rep; 54:119-124, 1970
[5] Hill GJ, Moss SE, Golomb FM, et al. DTIC and combination therapy for melanoma. Cancer; 47:2556-2562, 1981
[6] Falkson G, Van der Merwe AM, Falkson HC: Clinical experience with 5-(3,3–bis(2-chloroethyl)- 1-triazeno)-imidazole-4-carboxamide (NSC 82196) in the treatment of metastatic malignant melanoma. Cancer Chemother Rep; 56:671-677, 1972

results of a phase III study performed in previously treated melanoma patients was associated with a statistically significant improvement in overall survival (OS) compared with the gp100 vaccine (10.1 versus 6.4 months; HR: 0.66; p= 0.003). In recent times, the serine-threonine kinase BRAF was discovered mutated in many cancers[7]. BRAF mutations have been found in approximately 50% of melanoma, 30-70% of thyroid carcinomas, 30% of ovarian carcinoma and 10% of colorectal carcinoma. Oncogenic mutations in BRAF result in constitutive activation of the RAF-MEK-ERK pathway which in turn stimulates cell growth, proliferation and cell survival in the absence of typical growth factors[8]. For patients with tumours harbouring the BRAFV600 mutation, kinase inhibitors vemurafenib (Zelborarf) and dabrafenib (Tafinlar) were approved in the EU in 2012 and 2013, respectively, and target directly the mutated protein BRAFV600. Vemurafenib (Zelboraf), as first line treatment was approved based on the results of the pivotal phase III study (BRIM3) and was associated with a median progression-free survival (PFS) 6.9 vs 1.6 months, respectively (HR 0.38, 95%CI: 0.32-0.46, p<0.0001) and median OS 13.6 vs 9.7 months (HR: 0.70, 95%CI: 0.57-0.87, p<0.0001) compared with DTIC. Dabrafenib (Tafinlar), as first line therapy was approved based on the results of the pivotal trial BREAK-3 that showed a median PFS of 6.9 vs 2.7 months for the dabrafenib and DTIC group respectively, (HR: 0.37; 95% CI: 0.24, 0.58; p-value<0.0001) and OS (HR 0.76, 95%CI: 0.48, 1.21); a 12 month OS rate of 70 % and 63 % for dabrafenib and DTIC treatments, respectively). Trametinib (Mekinist) was approved in 2014 to target the mitogen-activated extracellular signal regulated kinase 1 (MEK 1 and MEK 2) proteins to dampen the signalling pathway that promotes proliferation and survival in BRAFV600 mutated melanoma. Mekinist as first line of treatment was approved based on the results of a phase III study that showed a median PFS 4.8 vs 1.5 months, respectively (HR 0.45; 95%CI:0.33, 0.63; p-value <0.0001) and median OS of 15.6 vs 11.3 months (HR 0.78; 95%CI 0.57, 1.06) for trametinib and dacarbazine respectively.

The co-inhibitory receptor programmed cell death – 1 (PD-1) is a key regulator of T cell activity that belongs to the same immunoglobulin superfamily which includes the co-stimulatory receptor CD28 and the co-inhibitory receptor CTLA-4.[9,] Nivolumab is a human immunoglobulin G4 (IgG4) monoclonal antibody (HuMAb), which binds to the PD-1 receptor and blocks its interaction with PD-1 ligand (PD-L1) and PD-1 ligand 2 (PD-L2). The PD-1 receptor is a negative regulator of T cell activity that has been shown to be involved in the control of T cell immune responses. Engagement of PD-1 with the ligands PD-L1 and PD-L2, which are expressed in antigen presenting cells and may be expressed by tumours or other cells in the tumour microenvironment, results in inhibition of T cell proliferation and cytokine secretion. Nivolumab potentiates T cell responses, including anti-tumour responses, through blockade of PD-1 binding to PD-L1 and PD-L2 ligands. In syngeneic mouse models, blocking PD-1 activity resulted in decreased tumour growth.

The applicant applied for an accelerated procedure for the following indication:

"OPDIVO as monotherapy is indicated for the treatment of advanced (unresectable or metastatic) melanoma in adults."

The final approved indication was:

"OPDIVO is indicated for the treatment of advanced (unresectable or metastatic) melanoma in adults."

Treatment must be initiated and supervised by physicians experienced in the treatment of cancer.

The recommended dose of OPDIVO is 3 mg/kg administered intravenously over 60 minutes every 2 weeks. Treatment should be continued as long as clinical benefit is observed or until treatment is no longer tolerated by the patient.

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[7] Davies H, et al. Mutations of the BARF gene in human cancer. Nature; 417:949-954, 2002
[8] Garnett MJ and Marais R. Guilty as charged: B-RAF is a human oncogene. Cancer Cell; 6:313-319, 2004
[9] Nurieva RI, Liu X., Dong C., Molecular mechanism of T-cell tolerance J Immunol; 241: 133–44, 2011

Dose escalation or reduction is not recommended. Dosing delay or discontinuation may be required based on individual safety and tolerability. Guidelines for permanent discontinuation or withholding of doses are described in Table 1 of the SmPC. Detailed guidelines for the management of immune related adverse reactions are described in section 4.4 of the SmPC.

OPDIVO is for intraveous use only. It is to be administered as an intravenous infusion over a period of 60 minutes. The infusion must be administered through a sterile, non-pyrogenic, low protein binding in-line filter with a pore size of 0.2-1.2 μm.

OPDIVO must not be administered as an intravenous push or bolus injection.

The total dose of OPDIVO required can be infused directly as a 10 mg/mL solution or can be diluted to as low as 1 mg/mL with sodium chloride 9 mg/mL (0.9%) solution for injection or glucose 50 mg/mL (5%) solution for injection.

2.2. Quality aspects

2.2.1. Introduction

Nivolumab is a human monoclonal immunoglobulin G4 (IgG4) antibody directed against the programmed death-1 (PD-1) receptor. It is an IgG4 consisting of four polypeptide chains; two identical heavy chains of 440 amino acids and two identical kappa light chains of 214 amino acids, which are linked through inter-chain disulfide bonds.

Nivolumab active substance is produced from large-scale cell culture using a Chinese hamster ovary (CHO) cell line and is purified using standard chromatography and filtration steps using a manufacturing process designated as Process C. Commercial nivolumab is manufactured by Lonza Biologics, Inc., Portsmouth, New Hampshire, USA (Lonza-Portsmouth). Finished product from Process B active substance manufacturing process was used in pivotal clinical trials. Comparability of products manufactured from Process C to that manufactured from Process B was assessed by analytical comparability, product degradation profiles and in-process data comparability. Active substance is packaged in polyethylene bioprocess containers. Long-term stability studies support a 24-month shelf life for active substance when stored refrigerated (2°C to 8°C) and protected from light.

Opdivo injection is available as 100 mg/10 mL (10 mg/mL) or 40 mg/4 mL (10 mg/mL) single-use presentations. Both presentations have the same protein concentration and are packaged in the same primary packaging which consists of a 10 mL, Type I flint glass tubing vial sealed with a 20mm grey butyl stopper and an aluminium flip-off seal. The only difference between the two presentations is the fill volume. Both presentations are packaged in a paperboard folded carton. Final release will occur at the Bristol Myers Squibb facility in Anagni, Italy (BMS-Anagni). The long-term stability data support a 24-month finished product shelf life when stored refrigerated (2°C to 8°C) and protected from light.

2.2.2. Active Substance

Manufacture, characterisation and process controls

Manufacture

Active substance manufacturing takes place at Lonza Biologics in Portsmouth (USA).

Nivolumab is manufactured in bioreactors. One production bioreactor leads to one bulk active substance lot. Bulk active substance lots are not combined at the active substance stage.

Nivolumab is produced from manufacturing-scale cell culture using a Chinese hamster ovary (CHO) cell line that was transfected with an expression vector containing coding sequences for the heavy and light chains of the nivolumab IgG. A conventional two-tiered cell banking system is employed, consisting of a Master Cell Bank from which a Working Cell Bank (WCB) is derived.

The upstream manufacturing process is a conventional fermentation process. Upstream processing starts with expansion of the WCB in a pre-culture step and a subsequent seed fermentation step. The expanded cells are transferred to a bioreactor and cultured under optimised conditions by means of a fed-batch process. Cell culture conditions and in-process controls (IPCs) have been sufficiently described.

The downstream process comprises chromatography and viral inactivation/clearance steps. The concentrated clarified bulk from the primary recovery step is processed across a series of chromatography, viral inactivation, filtration, and ultrafiltration / diafiltration steps. The active substance is stored at 2ºC to 8ºC prior to shipping to the finished product manufacturing facility.

Control of materials

Raw materials used in the nivolumab commercial manufacturing process are controlled to ensure the quality and safety of the active substance and to maintain the consistency of the manufacturing process. Raw materials are purchased from qualified vendors. Raw material quality is assessed as defined in the testing specification for each raw material.

Nivolumab production cells are CHO cells transfected with an expression vector containing the coding sequences for both heavy and light chains of the nivolumab IgG, in which the variable region genes were isolated from mouse hybridoma cells. The hybridoma cells were generated by the fusion of mouse myeloma cells with spleen cells from a PD-1 receptor immunised Human Ig transgenic mouse (HuMAb-Mouse) of the HCo7+KCo5 lineage. A hybridoma clone, 5C4.B8, which produced an anti-PD-1 antibody was selected. The immunoglobulin genes coding for the variable regions from the anti-PD-1 expressing hybridoma were cloned into an expression vector, which was used to establish CHO cell line expressing the nivolumab antibody. The cell line is used in the nivolumab manufacturing process at the Lonza Biologics facility Portsmouth (USA).

The MCB is stored in the vapour phase of liquid nitrogen in a cryofreezer designated for long-term storage of released cell banks. The nivolumab MCB is re-evaluated at 5 year intervals.

The WCB was derived from a single vial of MCB. This WCB is stored in the vapour phase of liquid nitrogen in a cryofreezer designated for long term storage of released cell banks. WCBs are tested for safety and productivity in accordance with regulatory guidelines and standard operating procedures before their release for manufacturing use.

The limit of *in vitro* cell age for the nivolumab production cell line was determined through extended passages during the inoculum expansion and seed bioreactor steps for the nivolumab manufacturing process at manufacturing-scale during process performance qualification (PPQ). The results support the stability of the nivolumab production cell line have been suitably demonstrated with respect to biosafety, growth characteristics of the inoculum expansion and seed bioreactor steps, biochemical product quality assessment and genetic stability.

Process validation

The process validation strategy for nivolumab active substance is based on a lifecycle management approach that includes the process design stage, PPQ stage, and ongoing (also referred to as continued) process verification stage.

The nivolumab PPQ campaign qualified the media and buffer preparation, the upstream process, and the downstream process at Lonza Biologics, Portsmouth (USA). The PPQ protocols included prospectively defined acceptance criteria consisting of numerical limits for release testing results, (Critical) Process Parameters ((C)PPs) and (Critical) Performance Attributes ((C)PAs) as defined in the IPC strategy presented. Acceptance criteria for the upstream manufacturing process and downstream manufacturing process were met. The acceptance criteria for PPQ described in the process validation section of the dossier reflect the IPC in place at the time the PPQ studies were conducted. After PPQ was completed, the IPC ranges and classifications were reviewed and compared to manufacturing-scale data and small-scale process characterisation data. Some ranges were amended and a number of (C)PPs and (C)PAs was reclassified.

Control of critical steps

An in-process control (IPC) strategy was defined for the nivolumab active substance manufacturing process. The IPC strategy is a planned set of controls derived from current product and process understanding that ensures consistent process performance and product quality. The controls include parameters and attributes related to active substance manufacturing to ensure the active substance meets the established specifications. The IPC strategy utilises a tiered set of in-process alert ranges, action ranges and critical ranges to ensure the consistent monitoring and control of the nivolumab active substance manufacturing process. The classification of IPC levels for each input and output is based upon an assessment of the potential impact of the parameter on the manufacturing process as well as the CQAs of the nivolumab active substance. Excursions from the established ranges are investigated for product quality impact.

Manufacturing process development

The development of the nivolumab manufacturing process in CHO cells started in 2006 and progressed during clinical development from manufacture of active substance by Process A, via Process B to Process C is currently the validated nivolumab manufacturing process at the Lonza Biologics facility, located in Portsmouth, New Hampshire and is intended for commercial manufacture. All pivotal clinical studies included in the submission have been performed using nivolumab from active substance Process B.

The early development of the manufacturing process in conjunction with initial manufacturing scale experience was used to provide basic process understanding. Using the basic process understanding, upstream process characterisation was conducted to refine process understanding. Each unit operation of the manufacturing process was assessed to identify process parameters with a potential impact on critical quality attributes (CQAs). An upstream in-process control strategy was established to ensure that CQAs of the molecule were maintained within acceptable ranges.

The upstream manufacturing Process C is identical to Process B for the majority of the steps. The development of the upstream manufacturing Process B is therefore directly applicable to Process C for these steps.

The downstream manufacturing Process C was developed with a series of chromatographic steps. The downstream process also includes viral inactivation, viral filtration, and a UF/DF step which yields UFB. Polysorbate 80 is then added to UFB yielding the formulated bulk, which is subsequently filtered through a membrane ending in the final active substance. Process development activities have established several in-process control parameter ranges through a series of one-factor and multi-factor experiments. Development of process conditions was completed for each process step. Early development of the manufacturing process in conjunction with initial manufacturing scale experience provided basic process understanding. Process characterisation was then conducted to refine process understanding and establish the downstream in-process control strategy. A scale-down model was completed for design of experiment (DoE) screening and statistically-designed response surface method (RSM) studies. The parameters that were demonstrated from development or screening studies to have a potential to impact product quality or process robustness were explored in RSM characterisation studies using the appropriate scale-down models.

Comparability between Processes A, B and C

Nivolumab manufacturing Process A was modified to accommodate scale-up and transfer to a new facility. The resulting process was designated Process B. Modifications to the upstream process were made to align with the manufacturing facility and to provide improved throughput while maintaining product quality. Comparability of active substance manufactured using Process A to that using Process B was established using biochemical comparability studies that included routine release and extended characterisation tests. The purification steps of Process B were modified to further increase process robustness and provide improved process control. The resulting process was designated as Process C.

A comparison of the active substance release data from representative lots of Process A, Process B, and Process C was performed to confirm comparability of Process C active substance to Processes A and B active substance. Raw data and statistical analysis are presented. A comparison of the differences in mean responses for quantitative assays was performed between the Process B and Process C lots. The difference in means is calculated as the mean for Process C results minus the mean from the Process B results prior to rounding of the mean results. All individual Process C results fall within the Tolerance Limits (TL) of Processes A and B, and the intervals defined by the average ± 3 SD of the Process C data also fall within the TLs determined for Processes A and B.

Identification of Critical Quality Attributes and classification of process parameters

Potential CQAs were identified for product-related variants, process-related impurities and formulation related quality attributes. The available knowledge for each attribute, both specific to nivolumab and from relevant supporting class knowledge was compiled and then each attribute was assigned as critical or non-critical.

An IPC strategy was established for the nivolumab active substance manufacturing process based on an integrated control strategy, which includes process control establishment and process control elements of each unit operation for quality attributes. Information is provided on the development of the strategy; the justifications for process parameter and performance attribute (PA) designations and their numerical ranges. The establishment of the IPC strategy for the nivolumab manufacturing process was based on toxicological assessments, process development, manufacturing-scale impurity mapping, stability testing, pilot-scale experience and manufacturing-scale experience to identify process control points for each quality attribute. Process development studies were designed to assess the influence of PPs on the quality of the active substance and the performance of the nivolumab manufacturing process. The results of these studies in combination with manufacturing experience were used to classify the process variables and establish IPC ranges for the upstream and downstream process steps. The IPC strategy process control elements for quality attributes include input material testing, in-process testing, PPs and specifications. Process control elements also include procedural controls for each unit operation.

Characterisation

Physical and biochemical properties of nivolumab were characterised using a series of orthogonal techniques, using all PPQ lots of nivolumab active substance. The techniques included liquid chromatography mass spectrometry (LC-MS/MS) for the determination of primary structure; circular dichroism (CD) for determination of secondary structure; and other techniques for examination of higher order structure of nivolumab that included size exclusion chromatography (SEC), SEC-multi-angle light scattering (SEC-MALS), sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), capillary electrophoresis SDS (CE-SDS), and analytical ultracentrifugation (AUC) to characterize the size variants. Disulfide bonds were characterised by LC-MS. Protein conformation was examined by differential scanning calorimetry (DSC) and hydrogen-deuterium exchange mass spectrometry (HDX-MS).

The biological properties of nivolumab were characterised using a direct binding enzyme-linked immunosorbent assay (ELISA), binding competition ELISA (potency ELISA) and a cell-based bioassay. The binding kinetics of nivolumab to the PD-1 receptor was determined using surface plasmon resonance (SPR).

The interactions of the Fc region of the nivolumab molecule with human FcγRs (CD64, CD32 and CD16) and complement were assessed in both FcγR binding assays and in cellular cytotoxicity assays. The binding of nivolumab to the FcγRs was characterised by SPR. Nivolumab does not elicit measurable ADCC or CDC activity in vitro. Nivolumab has been shown to retain binding to neonatal Fc receptor (FcRn) in SPR binding studies.

Potential process related impurities in nivolumab have been identified and are controlled by in-process testing, release testing according to specifications and/or process performance qualification.

Potential product-related impurities have been studied through the characterisation process and have been identified as High Molecular Weight (HMW) and Low Molecular Weight (LMW) species, which are controlled at release by SE-HPLC.

Specification

Nivolumab quality control testing for batch release includes appearance, quality, pH, purity, identity, potency, process-related impurities and endotoxins/bioburden. The specification set for nivolumab active substance are in accordance with the Ph. Eur. monograph on Monoclonal Antibodies for Human Use.

The Applicant's approach for selection of the release and shelf life tests is based on a multistep process including definition of CQAs, consideration of clinical experience, patient safety limits, stability of the active substance and finished product and process and method variability.

The biological activity is tested in three assays, i.e., (1) by binding activity ELISA, (2) potency ELISA, and (3) a cell-based bioassay. Each of the three proposed assays measure a relevant biological effect of the product at different levels in its mechanism of action and are considered important for release. The three proposed biological activity release/shelf life tests are also stability indicating.

Batch Analysis

Batch (lot) information for nivolumab active substance including the process designation, site of manufacture, date of manufacture, batch size and designated use for each active substance batch is presented for all process A, B and C lots. All batch analysis data were in line with the acceptance criteria that applied at the time of testing.

Reference standards of materials

During development, a series of Research Reference Standards (RRSs) was used. The initial reference standard was established as the reference standard prior to PPQ, and will serve as the initial Working Reference Standard (WRS) for release and stability testing of commercial lots.

BMS plans to implement a two-tiered reference standard strategy, at which time the existing reference standard will become the Primary Reference Standard (PRS) and will be used to qualify future Working Reference Standards. The existing reference standard was prepared using Process B active substance; finished product manufactured from this lot was used in pivotal clinical trials for lung cancer and melanoma and thus provides a direct link to clinical experience.

Container closure system

Active substance is filled into single-use, pre-sterilised multi-layered low density polyethylene (LDPE) bioprocess containers. Bioprocess containers are made from virgin plastics specifically manufactured from non-recycled material for use by pharmaceutical and biological manufacturers.

No semi-volatile organic compounds and non-volatile organic compounds were detected. Several metal ions were detected but all were below ppm levels and therefore not reported. Leachables studies were performed by storing the bags under two real-time and accelerated conditions. Leachables identified from nivolumab active substance or formulation buffer were evaluated in a toxicological review in which the estimated maximum daily exposure was compared to the permissible daily exposure (PDE) for each leachable. Results indicated that no leachable compounds are anticipated to pose a safety risk when nivolumab is administered as directed.

Stability

Stability studies have been conducted on batches of nivolumab active substance in accordance with ICH stability guidelines and demonstrate that the active substance is stable for up to 24 months when stored at 2°C to 8°C. Some of the batches were manufactured according to Process C and three according to Process B. Active substance manufactured according to Process C was shown to be comparable to active substance manufactured according to Process B. While in the first submission only the batches manufactured according to Process B had stability data for 24 months, in the Applicant's responses new data up to 24 months was also submitted for the batches manufactured according to Process C. Stability data were evaluated according to the specification as provided. The specification evolved during the execution of the stability studies.

2.2.3. Finished Medicinal Product

Description of the product and pharmaceutical development

Two commercial presentations, Opdivo injection, 100 mg/10 mL (10 mg/mL) and Opdivo injection, 40 mg/4 mL (10 mg/mL) have been developed. Both presentations have the same protein concentration and are packaged in the same primary packaging which consists of a 10 mL, Type I flint glass tubing vial and sealed with a 20mm butyl stopper and an aluminium, flip-off seal. The only difference between the two presentations is the fill volume. Both presentations are packaged in a paperboard folded carton.

The formulation contains sodium citrate as buffering agent, sodium chloride and mannitol as tonicity modifier, pentetic acid as metal ion chelator, polysorbate 80 as surfactant, hydrochloric acid/sodium hydroxide for pH adjustment and water for injection as solvent. Choice and concentrations of the excipients and the pH of the formulation were adequately justified by formulation studies. In addition, the robustness of the formulation was studied for slight variations in pH, and concentrations of citrate buffer, sodium chloride, mannitol and pentetic acid with satisfactory results. An overfill is included in each vial to account for loss in the vial.

Opdivo injection is a clear to opalescent, colourless to pale yellow liquid which may contain light (few) particulates. The finished product is a sterile, non-pyrogenic, single-use, preservative-free, isotonic aqueous solution for intravenous (IV) administration. Opdivo injection may be administered undiluted at a concentration of 10 mg/mL or further diluted with 0.9% sodium chloride injection (sodium chloride 9 mg/mL (0.9%) solution for injection) or 5% dextrose injection (50 mg/mL (5%) glucose solution for injection) to nivolumab concentrations as low as 1 mg/mL.

Manufacture of the product and process controls

The manufacturing process for the 100/10 ml and 40 mg/ml vials is straightforward and was described in sufficient detail. Briefly, buffer solution is prepared and the active substance is diluted with filtered buffer in two steps, followed by sterile filtration, filling and finishing. Appropriate in-process tests are performed with limits that ensure that the product will meet the corresponding finished product specifications. Hold times were justified by validation studies performed during process development and were confirmed during process performance qualification.

During development no major changes were introduced in the formulation, manufacturing process or container closure system of the finished product. Only early clinical studies (phase 1) were performed with a formulation that contained lower amount of polysorbate 80 and a direct fill and finishing procedure. The change in active substance manufacturing process from Process B to C is discussed in the active substance section.

Product specification

Finished product release testing includes general quality tests, composition, identity, purity, potency, total protein, and microbial test (sterility, bacterial endotoxin, container closure integrity test). Many test procedures and specifications are the same or similar to those for the active substance.

The finished product may contain low levels of visible particles. For administration of Opdivo, an infusion set with an in-line low protein binding filter will be used. Satisfactory finished product batch analysis data of Process A batches, Process B batches and Process C batches were provided. The data show, overall, that the process is running consistently.

Stability of the product

A shelf life of 24 months at 2°C to 8°C is proposed for both presentations. Stability studies were appropriately designed. For the 100 mg/10 ml long term stability data were provided for Process B and Process C batches. For the 40 mg/4 ml presentation long term stability data were provided. The information provided in the Applicant’s responses contains data for at least 24 months in all the above 100 mg/10 ml batches and for 15 months in 40 mg/4 ml batches.

Adventitious agents

No raw material of animal origin (primary source) was used in the manufacture and cryopreservation of the Research Cell bank, MCB and WCB for nivolumab. The nivolumab active substance manufacturing process does not use any material of animal or human origin.

Viral clearance is achieved through viral inactivation, physical removal of virus by nanofiltration and three orthogonal chromatographic steps. Low-pH viral inactivation and virus removal filtration are the dedicated viral clearance steps.

2.2.4. Discussion on chemical, pharmaceutical and biological aspects

The overall standard of the Module 3 Quality dossier presented in support of this application is high, with the description of the manufacture and control of the active substance and finished product in general containing sufficient detail to permit an in-depth assessment of the marketing authorisation application.

Active substance

Manufacture

The nivolumab active substance manufacturing process was described in sufficient detail. The virus filtration (VF) and final filtration and fill steps may be reprocessed when, for example, the viral filter post-use integrity test fails or if processing is not performed as intended. The reprocessing steps were sufficiently described and were properly validated.

Information provided on control and compositions of media, buffers, solvents, reagents and auxiliary substances used in the manufacturing process of nivolumab was considered acceptable. Control of materials mainly relies on suppliers’ certificates of analysis. Specifications were set and provided for the raw materials used in the fermentation process, harvesting and purification steps, as well as for the column materials. The information on materials from biological origin was sufficiently detailed. Certificates of analysis and, where available, EDQM CEPs were provided. The data provided did not give rise to safety concerns as regards viral and/or TSE issues. The nivolumab active substance manufacturing process starting from WCB does not use any material of animal or human origin. A number of other concerns were identified regarding the raw materials and the control of the cell banks and were appropriately addressed by the Applicant.

The establishment and genealogy of the CHO cell line and the construction of the nivolumab expression vector used in the nivolumab manufacturing process at the Lonza Biologics facility were described in sufficient detail. According to the information provided in the initial submission, it was concluded that the cell banks were fully characterised, including at the limit of *in vitro* age, to evaluate their ability to consistently produce nivolumab active substance. A cell bank stability program was in place for ongoing monitoring.

Taken into consideration all information provided, it was considered that the quality of the active substance is consistently maintained at the Lonza facility. Manufacturing process changes were made during development from Process A through Process B to Process C. An extensive comparability program was performed to demonstrate comparable product quality. The lot numbers mentioned and the results from the analytical comparability testing suggest that a limited number of batches from Process A and from Process B were tested to establish comparability of these processes, which is generally not deemed sufficient to draw a sound conclusion on comparability of Process A and Process B. However, the level of comparability between Process A and Process B manufactured active substance is deemed of minor importance in demonstrating comparability of Process B (i.e. process used for pivotal clinical trial material) and Process C (i.e. intended commercial process) manufactured active substance. Therefore, requesting more information was considered of limited value.

An extensive data package was provided to demonstrate comparability of the active substance manufactured by Process B and Process C. Only minor differences were detected between active substance lots from Process B versus Process C lots. These differences were sufficiently discussed and are not expected to have a negative impact on the quality of the product. In summary, based on the comparison of the release testing data provided and further supported by a data set on extended characterisation and forced degradation, it could be concluded that active substance manufactured using Process C (commercial process) is comparable to active substance manufactured using Process B used in pivotal clinical trials.

Process validation

Process verification was based on the manufacture of multiple PPQ lots. At each step, all (Critical) Process Parameters ((C)PPs) and (Critical) Performance Attributes ((C)PAs) that were identified during upstream and downstream process parameter designation were monitored c.q. analysed. Sufficient actual numerical data were given for process controls, results of in-process testing and yields of the different steps. All PPQ lots passed release testing and met the acceptance criteria in place at the time of the PPQ.

Intentional reprocessing was performed to demonstrate that reprocessing of nivolumab through the VF and final filtration and fill steps does not impact the quality of the in-process material or the resulting active substance at release. Both the post-VF pool and the filtrated and filled nivolumab active substance met the acceptance criteria after reprocessing.

Several deviations against the PPQ protocol acceptance criteria occurred. The way these deviations were investigated and handled was sufficiently described. The impact of each deviation on the production process/product quality was discussed. All deviations appeared to be related to operational or technical errors (equipment) and did not influence CPPs or CPAs and thus did not affect any critical quality attribute (CQA). Deviations during process verification were sufficiently addressed.

Overall, the results of the validation lots were very consistent and confirmed that the final manufacturing process performs effectively and is able to produce nivolumab intermediates and active substance meeting their predetermined acceptance criteria, on an appropriate number of consecutive batches on commercial scale when operating within the Proven Acceptable Ranges.

The impact of the hold times and conditions on the product quality were appropriately evaluated. The maximum hold time for each process intermediate was justified based on analysis of an appropriate set of stability indicating assays and parameters, including a number of relevant performance attributes (HMW, degradation, bioburden, endotoxins, binding ELISA). The maximum hold times are considered adequate.

After PPQ was completed, the in-process control (IPC) ranges and classifications were reviewed and compared to manufacturing-scale data and small-scale process characterisation data. An overview of all adjusted IPC ranges was presented and each adjustment was justified. Furthermore, a number of process variables were upgraded to CPP.

The completed studies defined in the PPQ protocols demonstrated the effective and consistent production of nivolumab active substance. The upstream and downstream steps of the manufacturing process used to produce nivolumab active substance are considered qualified.

Control of critical steps / Control strategy

The manufacturing process development is based on an enhanced approach using several QbD elements, such as the use of available knowledge for each quality attribute, risk assessments to assign the criticality of quality attributes, Design of Experiments (DoE) studies to assign the criticality of process parameters and finally to define an integrated control strategy. It is noted that no Design Space is claimed.

Overall, the approach was deemed appropriate, and as requested the Applicant defined potency as an active substance CQA in line with the approach taken for the finished product. Extensive information was provided in support of the assigned process parameters and performance attributes based on an in-depth evaluation of each operation unit in terms of process parameter variability and impact on quality attributes. Process parameters and performance attributes and respective alert/actions limits were sufficiently supported by process characterisation studies.

Excursions from established alert, action or critical ranges were investigated to evaluate the possible root cause and possible impact on product quality. The Applicant justified that the investigation of these excursions entails sufficient rigour to ensure that the quality of the product is unaffected.

Multiple levels of control were established throughout the nivolumab manufacturing process to minimise the risk of contamination of the finished product with adventitious viruses. Cell banks were extensively tested for adventitious as well as endogenous agents. Cell culture is performed under serum free conditions using media that contain no components of animal or human origin. Pre-harvest samples from each nivolumab production batch are tested for bioburden, endotoxin, mycoplasma, in vitro adventitious agents and minute virus of mice.

Five process steps were investigated in virus validation studies to determine the manufacturing process capacity for virus inactivation/removal. In general, the strategy for virus validation was considered acceptable.

Characterisation

Overall, the analytical methods applied for characterisation of nivolumab were considered adequate and provide a complete characterisation of the molecule. The analytical methods applied were described in sufficient detail.

Control of active substance

The Applicant's approach for selection of the release and shelf life tests is based on a multistep process including definition of CQAs, consideration of clinical experience, patient safety limits, stability of the active substance and finished product and process and method variability. The approach is adequately documented in the application and is generally acceptable.

Specifications for the biological assays were also tightened, as requested during the procedure.

Discontinuation of the tests for impurities was based on consistent low results at the level of unformulated bulk active substance, often at or below the quantitation limit, in-process testing during process qualification, spiking studies at small scale and safety assessment. The results of the analysis of these studies indeed warrant discontinuation of these tests. The test for residual host-cell protein will be continued at the level of the unformulated active substance, which is endorsed. The Applicant agreed to develop and implement a process-specific HCP method, which should be available by the end of March 2016.

Analytical methods applied for batch release and stability testing were described in sufficient detail. Appropriate controls and relevant parameters ensuring the validity of the test were included. Methods were adequately validated and are suitable for their purpose.

Satisfactory active substance batch analysis data of have been provided. The data show, overall, that the process is running consistently.

Reference standards of materials

Adequate information was provided for reference standards. The current reference standard was prepared using Process B active substance; finished product manufactured from this lot was used in pivotal clinical trials for lung cancer and melanoma and thus provides a direct link to clinical experience. The Applicant plans to implement a two-tiered reference standard strategy, at which time the current Reference Standard will become the Primary Reference Standard (PRS) and will be used to qualify future Working Reference Standards.

Active substance stability

Since the active substance manufactured by Process C was shown to be comparable to active substance processed by Process B, the stability data available for Process B batches were regarded representative and could be considered supportive for the proposed active substance storage time of 24 months. In any case, updated finished product stability data was provided in the Applicant's responses supporting the acceptability of the proposed finished product shelf life of 24 months when stored at 2-8ºC and protected from light. The Applicant satisfactorily addressed a number of concerns raised about the analytical methods and the proposed active substance (release and) end-of-shelf-life specifications, with in some cases commitments.

<u>Finished product</u>

The manufacture of the finished product is a straightforward aseptic filling process for a liquid formulation. Satisfactory information with regard to manufacturing process development was provided.

Whilst the control strategy (i.e. the type and number of controls) was generally considered acceptable, a number of concerns was identified. The process to assign a criticality status to process parameters and controls was not sufficiently described for a number of parameters and some relevant parameters were lacking. These issues were considered resolved following the submission of the Applicant's responses.

The Applicant's approach for selection of the release and shelf life tests is based on a multi-step process including definition of critical quality attributes (CQAs), consideration of clinical experience, patient safety limits, stability of the active substance and finished product and process and method variability. The approach was adequately documented in the application and was generally acceptable.

The compatibility of the finished product with the container closure system was sufficiently studied. Compatibility was established for the recommended final infusion concentration range, diluents, containers, IV administration sets, in-line filters (0.2 and 1.2 µm polyethersulfone membranes) and infusion flow rates. An appropriate matrix design was used. The leachables studies did not give rise to any concerns. Given the relatively high concentration of citrate in the finished product, glass delamination was investigated with three compatibility studies and no risk of delamination was found.

Updated finished product stability data were provided in the Applicant's responses supporting the acceptability of the proposed finished product shelf life of 24 months at 2-8 °C and protected from light for both presentations.

Adventitious agents

The risk of TSE is considered very low in the nivolumab manufacturing process.

Viral elimination/inactivation studies were validated, but only a summary of the results was presented. The Applicant was asked to provide the full reports plus some other minor clarifications. All information submitted in the Applicant's responses was considered acceptable.

2.2.5. Conclusions on the chemical, pharmaceutical and biological aspects

Overall, the quality of Opdivo is considered to be in line with the quality of other approved monoclonal antibodies. The different aspects of the chemical, pharmaceutical and biological documentation comply with existing guidelines. The fermentation and purification of the active substance are adequately described, controlled and validated. The active substance is well characterised with regard to its physicochemical and biological characteristics, using state-of-the-art methods, and appropriate specifications are set. The manufacturing process of the finished product has been satisfactorily described and validated. The quality of the finished product is controlled by adequate test methods and specifications.

Viral safety and the safety concerning other adventitious agents including TSE have been sufficiently assured.

The overall Quality of Opdivo is considered acceptable. Several Recommendations on Quality aspects were agreed by the Applicant.

2.2.6. Recommendations for future quality development

In the context of the obligation of the MAHs to take due account of technical and scientific progress, the CHMP recommends several points for investigation.

2.3. Non-clinical aspects

2.3.1. Introduction

Pharmacology of PD-1 blockade was evaluated *in vitro* using primarily human and murine cells and *in vivo* in mice using a murine homolog product. Safety of nivolumab was evaluated exclusively in nonhuman primates, the only relevant and pharmacologically responsive model.

The pivotal toxicology studies supporting the safety of nivolumab were conducted in compliance with Good Laboratory Practice (GLP) regulations.

2.3.2. Pharmacology

Primary pharmacodynamic studies

Affinity study by surface plasmon resonance (MDX-1106-025-R930046580 and amendments)

Nivolumab binding to PD-1 on T cells and its affinity to human and cynomolgus PD-1 was assessed using surface plasmon resonance (SPR). Recombinant PD-1-Ig fusion protein was coated on a substrate chip and several concentrations of the antibody were flowed over the chip to obtain binding and dissociation kinetics. The results are shown in Table 5.

Table 5: Affinities of nivolumab by surface plasmon resonance

Antibody	antigen	K_D x 10^{-9} (M)	ka x 10^{4} (1/Ms)	kd x10^{-4} (1/s)
nivolumab	human PD-1	3.06	25.00	7.68
nivolumab	cynomolgus PD-1	3.92	11.8	4.65

PD-1 epitope sequence (BDX-1106-321 930072902)

Physical binding experiments were performed by immunoprecipitation with either intact PD-1 or PD-1 peptides generated from various enzymatic digests. As shown in the figure below, three peptides were identified, 1a (62-69), 1b (70-86) and 2 (118-136). Peptides 1a and 1b are adjacent to one another in the linear sequence of PD-1. Peptide 1a was found to have the strongest affinity to nivolumab. The binding of nivolumab to antigen was found to be dependent upon glycosylation of PD-1.

Figure 3: Amino acids 1-150 of human PD-1, peptide sequences precipitated by nivolumab are underlined

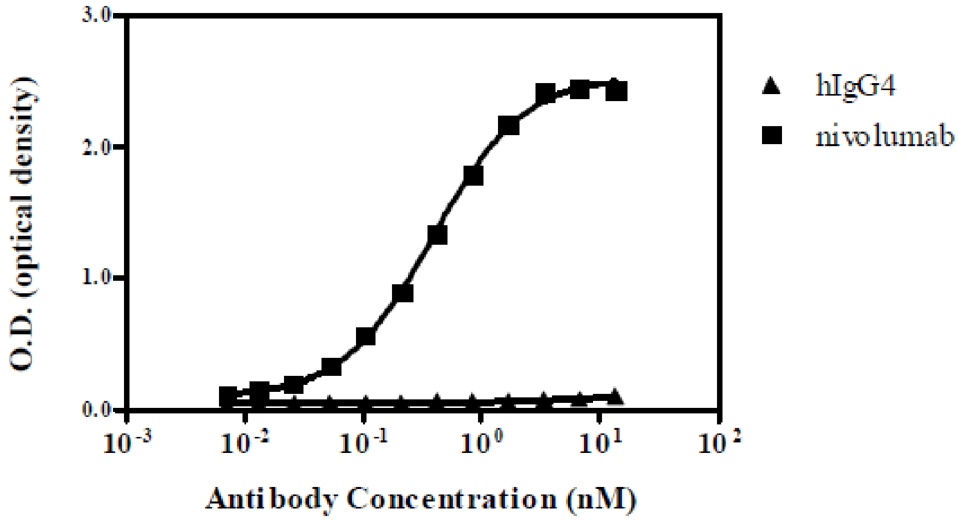
1 MQIPQAPWPVVWAVLQLGWRPGWFLDSPDRPWNPPTFSPALLVVTEGDNA 50
51 TFTCSFSNTSE<u>SFVLNWYRMSPSNQTDKLAAFPEDR</u>SQPGQDCRFRVTQL 100
101 PNGRDFHMSVVRARRND<u>SGTYLCGAISLAPKAQIKE</u>SLRAELRVTERRAE 150

The cynomolgus amino acid sequence is highly homologous to human PD-1 and within the peptides 1a, 1b and 2, all but one amino acid are identical.

Binding of nivolumab to recombinant human PD-1 and related proteins (MDX-1106-025-R 930046580 and amendments)

The ability of nivolumab to bind recombinant human PD-1-Ig fusion protein was demonstrated by ELISA. The results show an EC50 = 0.39 nM while demonstrating no binding to select other members of the related immunoglobulin supergene family i.e. CD28, ICOS, CTLA-4 and BTLA.

Figure 4: Binding of nivolumab to PD-1-Fc by enzyme-linked immunosorbent assay



Binding of nivolumab to activated human peripheral T cells (MDX-1106-025-R 930046580 and amendments)

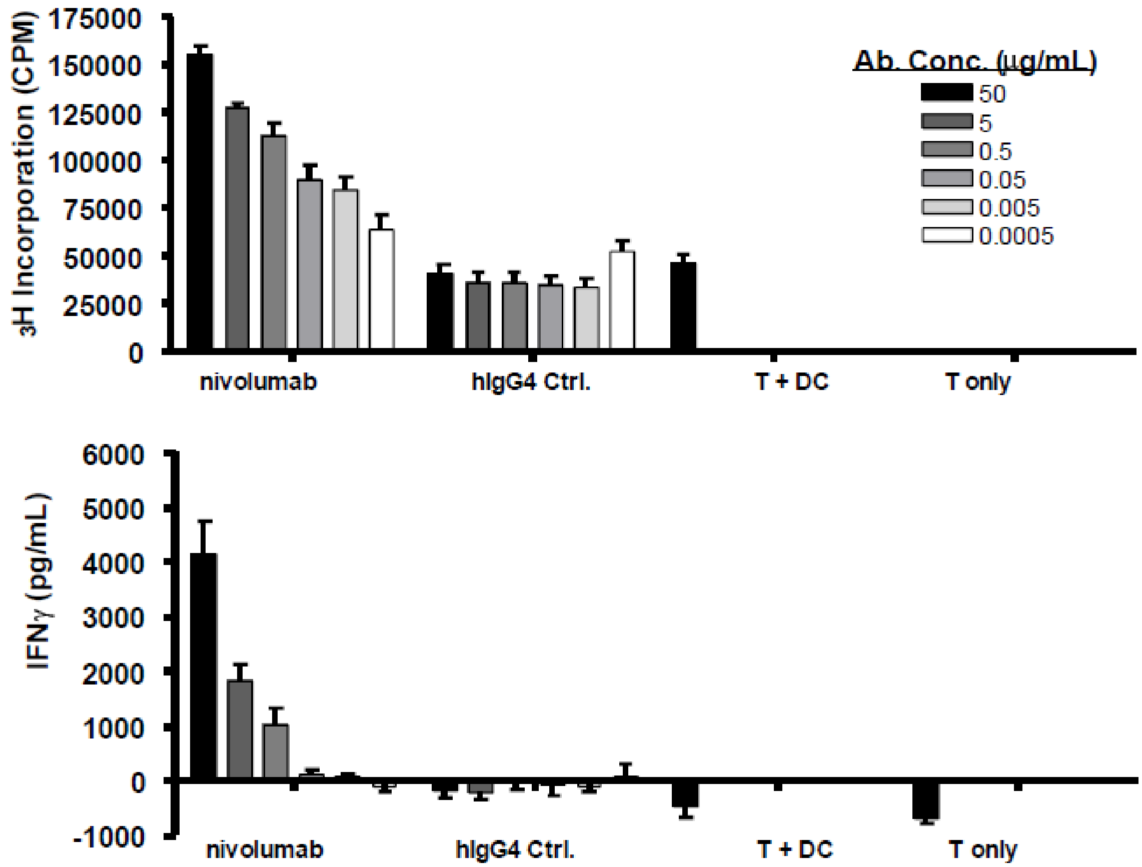
To confirm binding of nivolumab to native PD-1, activated human peripheral T cells which upregulate PD-1 on the cell surface were used to test binding of nivolumab by flow cytometry. Nivolumab binds to native PD-1 molecules expressed on activated T cells with an EC50 of 0.64 nM. No binding of the isotype control IgG4 antibody was observed. Notably, no binding of nivolumab was detected to activated rat or rabbit T cells.

Functional activity of nivolumab in vitro

Studies were undertaken to establish that nivolumab can inhibit binding between PD-1 and ligands PD-L1 and PD-L2. In vitro assays were performed for MLR, tumour antigen-specific CD8+ T cell restimulations and viral antigen-specific responses to evaluate the functional effect of nivolumab.

- *Activity of nivolumab in the mixed lymphocyte reaction (MDX-1106-026R 930046581):* The effect of PD-1 blockade by nivolumab on the immune response was studied using the MLR in which CD4+ T cells recognize allogeneic monocyte-derived dendritic cells and results in T cell proliferation and cytokine secretion. Results are shown in Figure 5.

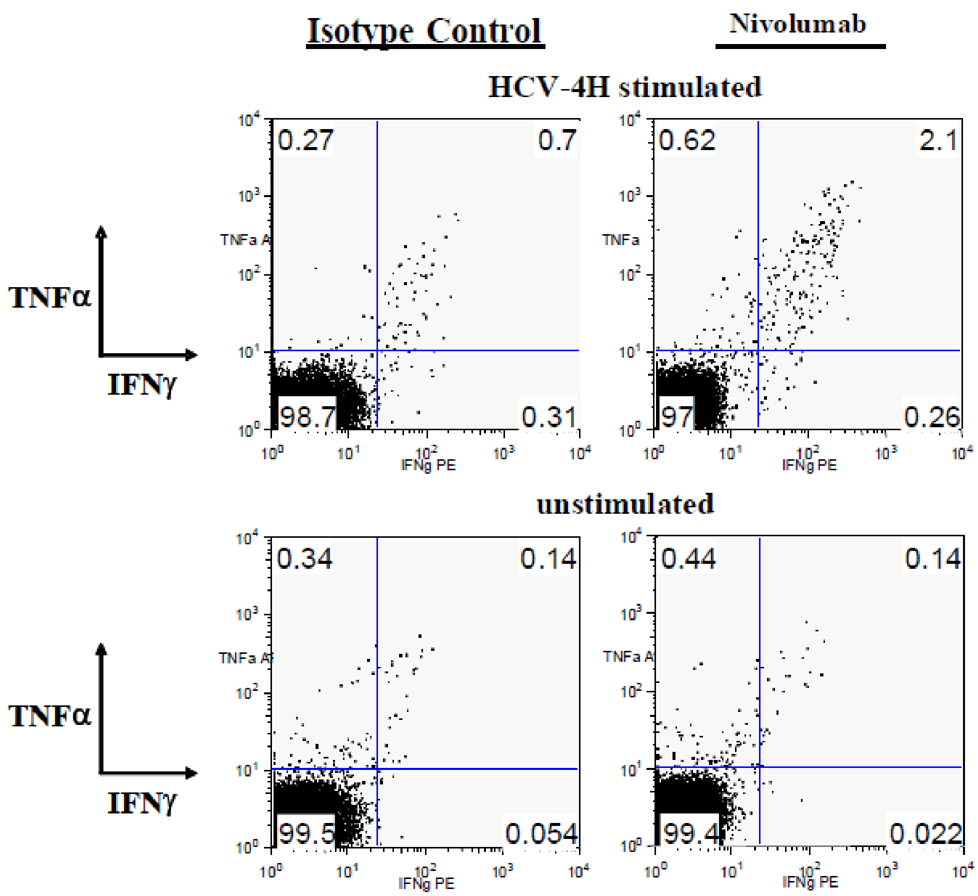
Figure 5: PD-1 blockade enhances CD4+ T cell proliferation and IFN-γ secretion in a dendritic cell-T cell MLR



T = T cells; DC = Dendritic cells; Ctrl. = control

- *Nivolumab enhances an antigen-specific recall response in vitro (MDX-1106-042R 930046579 and amendments):* The effect of nivolumab on an antigen-specific CD4+ T cell recall response was investigated using the response of human PBMC to a cytomegalovirus (CMV)-restimulation as assessed by secretion of IFN-γ. Nivolumab augmented IFN-γ secretion in response to stimulation by CMV antigen in a dose-dependent manner EC50 = 0.001 to 0.018 μg/mL (0.0067-0.12 nM). Maximum IFN-γ secretion was 2.5 to 5.3-fold more than CMV lysate stimulation alone.
- *PD-1 blockade and activation of HCV-specific T cells (MDX-1106-228R 930049318):* The effect of PD-1 blockade on the ability of HCV-specific CD8+ T cells to produce the cytokines IFN-γ and TNF-ɑ was measured using an intracellular cytokine stain (ICS). HCV-seropositive donor PBMCs were cultured in vitro with HCV-specific peptide (4H) and nivolumab or hIgG4 isotype antibody for 6 days. The PBMCs were harvested and restimulated with HCV-specific peptide and assayed for cytokines in the CD8+ T cell subset using flow cytometry. The results shown in Figure 6.

Figure 6: PD-1 blockade with nivolumab enhances IFN-γ and TNF-α production by HCV-specific CD8+ T cells in an ICS assay.



PD-1 blockade increases the frequency and absolute numbers of melanoma antigen-specific cytotoxic T lymphocytes

The applicant provided literature data to show that blocking PD-1 would result in an increase in antigen-specific cytotoxic T cells.[10]

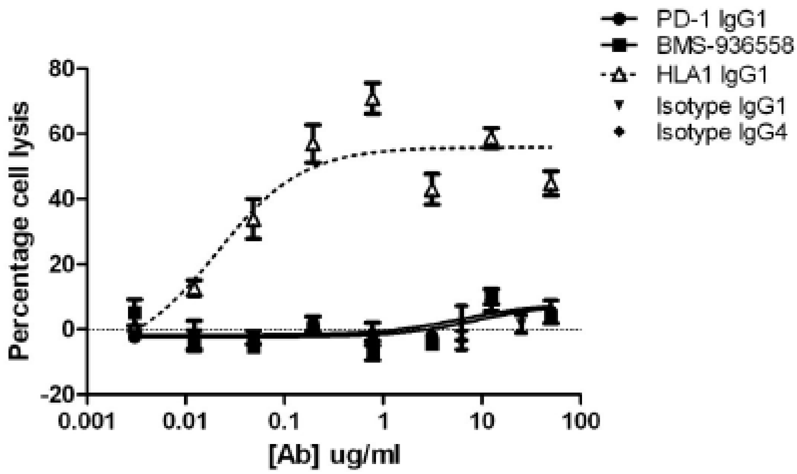
In vitro activity of nivolumab in assays of antibody effector function

Antibody-dependent cell-mediated cytotoxicity (BDX-1106-320 930073399)

An antibody-dependent cell-mediated cytotoxicity (ADCC) assay using IL-2 activated PBMC was conducted to evaluate whether nivolumab could induce ADCC of target cells. Among four donors, nivolumab did not mediate ADCC of activated human PD-1+ CD4+ T cells at most concentrations with only minor lytic activity at high concentrations. The mean values of percentage of cell lysis for nivolumab were 3.6 and 1.2% at 12.5 and 50 µg/mL, whereas for the positive control anti-HLA-1 antibody (HLA1 IgG1) the cell lysis were 33.8 and 26.1% at the same concentrations.

[10] Wong RM; Scotland RR, Lau RL, et al. Programmed death 1 blockade enhances expansion and functional capacity of human melanoma antigen-specific CTLs. Int Immunol 2007;19:1223-34.

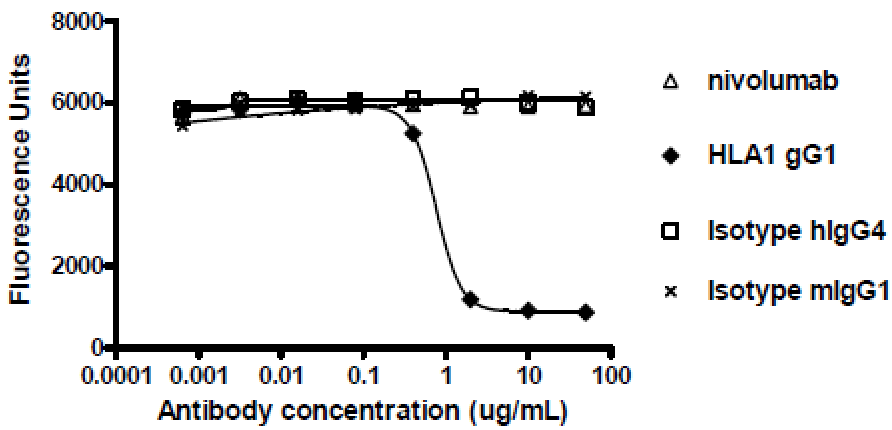
Figure 7: ADCC of Nivolumab on activated human CD4+ T cells



Complement-dependent cytotoxicity (MDX-1106-025R 930046580 and amendments)

Complement-dependent cytotoxicity (CDC) of nivolumab was examined using activated human PD-1+ CD4+ T cells. CDC studies demonstrated that, at concentrations up to 50 µg/mL, nivolumab did not elicit cytotoxicity of PD-1-expressing activated human CD4+ T.cells.

Figure 8: CDC of Nivolumab on Activated Human CD4+T cells



Effects of PD-1 blockade in tumour models (MDX-1106-028R 930046578)

The efficacy of PD-1 blockade on the growth of transplantable tumours in mice was tested. As nivolumab does not recognize mouse PD-1, a surrogate anti-mouse PD-1 antibody (clone 4H2; mouse IgG1 isotype) was derived and shown to bind to cells expressing murine PD-1 (EC50 = 2.9 nM) and block binding of mouse PD-L1 and PD-L2 to mouse PD-1, with an EC50 for blocking of 3.6 and 4.9 nM, respectively. The effects of 4H2 anti-PD-1 antibody blockade were tested in the following murine syngeneic tumor models: MC38 colon carcinoma, SA1/N fibrosarcoma, J558 melanoma, Renca kidney model, B16F10 melanoma, 4T1breast carcinoma, and CT26 colon carcinoma. The results are summarised in Table 6.

Table 6: Summary of key findings in in vivo studies with different tumour models

In Vitro Type of Study	Test System			Noteworthy Findings	Testing Facility	Report No.
Renca cells implanted into BALB/c mice using anti-PD-1 antibody; staged tumor study	Days 11, 14, 17, and 21	10 mg/kg of 4H2	Females 8 mice per group	No effect	Bristol-Myers Squibb	MDX-1106-036-R/ 930046562
4T1 cells implanted into BALB/c mice using anti-PD-1 antibody ; staged tumor study	Days 8, 11, 14 and 18 Saline 28 day study	10 mg/kg of 4H2	Females 9 mice per group	No effect	Bristol-Myers Squibb	MDX-1106-035-R/ 930046543
CT26 cells implanted into BALB/c mice using anti-PD-1 antibody; Staged	Days 10, 14, 17, and 21 Saline24 day study	10 mg/kg of 4H2	Females 10 mice per group	No effect	Bristol-Myers Squibb	MDX-1106-021-R/ 930046575 Amendment 01
B16F10 cells implanted into C57BL/6 mice using anti-PD-1 antibody; Staged	Days 8, 11, 14, and 17 Saline 28 day study	10 mg/kg of 4H2	Females 8 mice per group	No effect	Bristol-Myers Squibb	MDX-1106-020-R/ 930046576 Amendment 01
Sa1N cells implanted in A/J mice using anti-PD-1 antibody; staged tumor study	Days 7,10,13 and 16. Saline 41 day study	3 and 10 mg/kg of 4H2	Females 8 mice per group	Mean TGI on Day 20 was 31% for mice administered 10 mg/kg 4H2. Median TGI on Day 23 was 50% for mice administered 3 mg/kg 4H2 at and 16% for mice administered 10 mg/kg 4H2	Bristol-Myers Squibb	MDX-1106-013-R/ 930046563
Staged J558 cells implanted in BALB/c mice using anti-PD-1 antibody	Days 10, 13, and 17 Saline 38day study	10 mg/kg of 4H2	Females 8 mice per group	25% of the mice were tumor-free at the end of the study. Mean TGI on day 14 was 11% and median TGI on day 17 was 31%.	Bristol-Myers Squibb	MDX-1106-034-R/ 930046561 Amendment 01

The administration of anti-mouse PD-1 antibody 4H2 resulted in delayed tumor progression in the MC38, Sa1N, and J558 models, with complete tumor regressions observed in some individual mice in these studies. In these models, PD-1 blockade delivered at the time of tumor implantation (prophylactic or unstaged) or after tumors were established (therapeutic or staged) delayed or prevented the outgrowth of tumors. Four other syngeneic tumor models (Renca, 4T1, CT26, and B16F10) were refractory to anti-PD-1 treatment. In the Renca and 4T1 models, the prophylactic administration of an anti-PD-1 mAb did not impact tumor growth, and growth rates were nearly identical between the 4H2-treated and control groups. In the therapeutic B16F10 and CT26 models, the tumors grew very aggressively and 4H2 failed to significantly reduce the growth rate of established tumors.

MC38 model in C57BL/6 mice (MDX-1106-023R 930046542, MDX-1106-032R 930046566)

MC38 cells are colon adenocarcinoma cells derived from C57BL/6 mice that express PD-L1 *in vivo*, although *in vitro* expression of PD-L1 is very low. Two *in vivo* studies with MC38 tumours were conducted in mice. In both studies mice were transplanted with an unstaged MC38 tumour (subcutaneous implantation of 2 million cells) and treated with control IgG or anti-PD-1 antibody 4H2 (nivolumab murine surrogate) at 10 mg/kg IP. In the first study each animal received the first antibody or control dose the same day of the tumour implantation, and additional doses on days 3 and 6 post-implantation. At day 21, 83% tumour growth inhibition was observed, with 30% of mice tumour-free at the study endpoint. In the other study, treatment with control IgG or anti-PD-1 antibody 4H2 was conducted on days 7 post-tumour implantation and again on days 10 and 13, resulting in a mean tumour growth inhibition of 62% at day 13 and a median growth inhibition of 76% by day 20.

Dose titration of anti-PD-1 antibody in the MC38 model (MDX-1106-200R 930046571)

A dose titration was performed with anti-PD-1 antibody to determine whether there is a relationship between dose and efficacy in a staged MC38 tumour model. With dosing on days 6, 10 and 13 post-implantation, antitumor responses were most potent at 30 mg/kg and little difference was observed between animals treated with 3 and 10 mg/kg/dose. Results are shown in table 7.

Table 7: Median tumour volume over time in a staged MC38 model (dose titration)

Treatment	Day 6	Day 13		Day 20		Day 59
	Mean Tumor Volume (mm^3)	Mean Tumor Volume (mm^3)	Mean % TGI	Median Tumor Volume (mm^3)	Median % TGI	% Mice Tumor-Free
m-IgG1-10 mg/kg + r-IgG1-10 mg/kg	93.56	580.6	N/A	1566.0	N/A	0
4H2-10 mg/kg + r-IgG1-10 mg/kg	96.67	204.4	65	190.3	88	0
4H2-3 mg/kg	85.13	205.0	65	632.0	60	20
4H2-10 mg/kg	84.22	229.7	60	505.0	68	10
4H2-30 mg/kg	83.64	150.3	74	70.0	96	10

Secondary pharmacodynamic studies

The applicant did not submit secondary pharmacodynamic studies (see non-clinical discussion).

Safety pharmacology programme

In the single-dose cardiovascular safety study, groups of 6 telemetered Cynomolgus monkeys (3 per sex) were administered 0 (vehicle control), 10, or 50 mg/kg nivolumab intravenously. Both were well tolerated and there were no nivolumab-related effects on clinical signs, body weights, cardiovascular parameters, or body temperature. In repeat-dose toxicity studies, weekly (QW) or 2QW IV doses of nivolumab were administered for 1 or 3 months, respectively, at dose levels up to 50 mg/kg. No nivolumab-related clinical signs of toxicity or effects on body weight, food consumption, blood pressure, heart rate, respiration rate, or ophthalmic or electrocardiographic parameters were observed.

Pharmacodynamic drug interactions

The pharmacodynamics activity of nivolumab in cynomolgus monkeys was evaluated in combination with anti-CTLA (ipilimumab) and anti-LAG-3 (BMS-986016) monoclonal antibodies.

Combination study of nivolumab and ipilimumab (SUV00106 930036346)

The pharmacodynamic activity of nivolumab when administered in combination with ipilimumab was studied in a 1-month combination study in cynomolgus monkeys which included an assessment of the Immune function was evaluated by measurement of antibody response to keyhole limpet haemocyanin (KLH) (study SUV00106). Immunophenotypic analyses of peripheral blood were conducted at different intervals to identify changes in the composition of immune cells. This study included 5 animals/sex/group treated at 3 mg/kg ipilimumab and 10 mg/kg nivolumab or 10 mg/kg ipilimumab and 50 mg/kg nivolumab infused IV once weekly for 4 consecutive weeks.

The high dose combination showed variable but significant increases in the number and frequency of peripheral blood CD3+CD4+ T cells and CD3+CD8+ T cells on day 7, whereas no changes were observed in the number or frequency of monocytes or natural killer cells. Similar changes were observed in the 3-month toxicity study with nivolumab alone (study WIL-552003). In addition, an increased incidence of immune-mediated adverse effects (GI toxicity/colitis) was also observed in this combination study, which was consistent with the observed potentially enhanced T cell numbers and activity.

Combination study of nivolumab and BMS-986016 (DN12123 930070016)

In a 1-month combination study, nivolumab (50 mg/kg) or BMS-986016 alone (30 and 100 mg/kg) and in combination (50 mg/kg nivolumab and 100 mg/kg BMS-986016) were administered to monkeys (5/group/sex) for a total of 5 weekly doses. Serum nivolumab exposures when administered in combination with BMS-986016 were comparable to exposures observed when nivolumab was administered alone. There were no nivolumab-related changes in T-cell dependent antibody responses to KLH and hepatitis B surface antigen (HBsAg), ex vivo recall responses to KLH in CD8+CD4- T cells or HBsAg in CD4+CD8+ or CD8+CD4- T cells. However, nivolumab alone and in combination with BMS-986016 caused increases in immune cell parameters, consistent with pharmacological mechanisms of action of nivolumab and BMS-986016. In both cases, cytokine production was greater in the combination group than in the monotherapy groups.

2.3.3. Pharmacokinetics

The cynomolgus monkey was determined to be a relevant animal species because nivolumab binds to cynomolgus monkey PD-1 receptor. Therefore, all pharmacokinetic studies were conducted in cynomolgus monkeys. Since the intended route of administration in patients is intravenous, nivolumab was administered IV in all nonclinical studies as a solution in 20 nM sodium citrate, 50 nM sodium chloride, 20 μM diethylenetriamine pentacetic acid (DTPA), 3% mannitol, 0.01% polysorbate 80 (pH 6). This was similar to the clinical formulation.

Table 8: Pharmacokinetic data of nivolumab

Study ID	Species	N	Dose (mg/kg)	Route	Cmax (μ g/mL)	Tmax (h)	AUC∞ (μg·h/mL)	T½ el (h)
SUV00027	monkey	3/sex	1	IV	♂34.3±2.2	0.25-0.5	4470±423	124±20.3
					♀30.6 ±1.71	0.25	4050±616	139±12.7
		3 males	10		343±21.6	0.25-0.5	64200±27400	261±226
SUV00025	monkey	5/sex/group	1	IV	♂22.0±4.34 [a]	NC	1740±411 [b]	148±34.5
					♀23.1±3.37 [a]	NC	2040±370 [b]	146±25.5
			10		♂288±40.1 [a]	NC	23800±1650 [b]	267±104
					♀255±40.3 [a]	NC	21600±2950 [b]	223±37.5
			50		♂1120±133 [a]	NC	91700±9070 [b]	238±94.5
					♀1200±131 [a]	NC	109000±14200 [b]	260±133

a: concentration at 0.25 hours after dosing

b: AUC (0-168h). AUC∞ not reported because data was only collected for a period equivalent to approximately a single t½.

In study SUV00027, consistent with the long t½, total serum clearance (CLT) was low (0.172-0.250 mL/h/kg), with slightly more rapid clearance after 1 mg/kg doses than after the 10 mg/kg doses in males. The volume of distribution at steady state (Vss) was low, with values of 0.046-0.06 L/kg at all doses. For study SUV00025, consistent with the long t½, CTLs were 0.266-0.323 mL/h/kg at 1 mg/kg and 0.161-0.222 mL/h/kg at higher doses.

The applicant did not submit distribution, metabolism and excretion studies (see non-clinical discussion).

2.3.4. Toxicology

The nonclinical toxicity studies of nivolumab were performed in a battery of nonGLP studies, and GLP- and ICH-compliant *in vitro* tissue binding studies in multiple species and *in vivo* toxicology studies in cynomolgus monkeys.

Single dose toxicity

A single-dose IV of nivolumab at dose levels of 1 or 10 mg/kg was administered in cynomolgus monkeys.

Table 9: Single dose toxicity studies with nivolumab in Cynomolgus monkeys

Study ID	Species/ Sex/Number/ Group	Dose/Route	Approx. lethal dose / observed max non-lethal dose	Major findings
SUV00027/GLP	Cynomolgus monkey	1-10, 3/s/g low 3M high	NA/10	None

All animals survived the study, and no effect of nivolumab was observed on clinical observations, body-weight measurements, food consumption, or clinical pathology parameters.

Repeat dose toxicity

A 1- and 3-month IV repeat-dose toxicity studies of nivolumab were conducted in cynomolgus monkeys.

Table 10: Overview of repeat dose toxicity studies in Cynomolgus monkey including major study findings

Study ID	Species/Sex/ Number/Group	Dose/Route	Duration	NOEL/ NOAEL	Major findings
SUV00025 /GLP	Cynomolgus monkey, 5/s/g	0-1-10-50/ IV QW	30d	50 mg/kg	No major findings

WIL-552003/GLP	Cynomolgus monkey, 6/s/g	0-10-50/ IV /2QW	3 months	50 mg/kg	**Clinical observation** ≥10F:Slight increase in occurrence of soft feces, **Microscopy** ≥10M/F: Slight increase in incidence of mononuclear infiltrate in Heart, Kidney and Liver from minimal up to (on occasion) moderate severity, Slight increase in chronic inflammation of lungs minimal to mild ≥10M: Heart mononuclear infintrate minimal 50% both groups **Serum chemistry** =50M:↓Phos slight, =50F: ↓T3 **Cytometry** **=50:** ↑CD8 effector memory, ↑CD4+ effector memory, ↑CD8+ central memory **Other** Almost all male animals have immature epididymis, prostate, testes

In the 1-month toxicity study in cynomolgus monkeys, 5 weekly IV doses of nivolumab were administered at dose levels of 0 (vehicle control), 1, 10, or 50 mg/kg. There were no nivolumab-related clinical signs or changes in body weight; food consumption; blood pressure; heart and respiration rates; ophthalmic or electrocardiographic parameters; clinical pathology parameters (serum chemistry, hematology, coagulation, thyroid hormone, urinalysis, or urine chemistry); or gross or microscopic pathology findings. In addition, there were no nivolumab-related changes in immune cell parameters.

In the 3-month toxicity study in cynomolgus monkeys, IV doses of nivolumab were administered 2QW (for a total of 27 doses) at dose levels of 0 (vehicle control), 10, or 50 mg/kg. There were no nivolumab related effects on body weight, cardiovascular, neurologic, respiratory, ophthalmic, urinalysis, hematologic, or organ weight parameters, and no gross or microscopic pathology findings.

Genotoxicity

The applicant did not submit genotoxicity studies (see non-clinical discussion).

Carcinogenicity

The applicant did not submit carcinogenicity studies (see non-clinical discussion).

Reproduction Toxicity

In the ePPND study, nivolumab was administered 2QW at 10 or 50 mg/kg to pregnant monkeys from GD 20 to 22 until parturition. Nivolumab was well tolerated at both doses and there were no nivolumab-related effects on viability, clinical signs, food consumption, body weights, immunological endpoints, or clinical/anatomic pathology parameters in these females throughout the study. However, in the offspring, maternal nivolumab administration at both doses was associated with fetal/neonatal mortality characterized by: 1) dose-dependent increases in third trimester fetal losses (12.5% and 33.3% at 10 and 50 mg/kg, respectively, relative to 7.1% in controls), which occurred predominately after GD 120; and 2) increased neonatal mortality at 10 mg/kg, which was noted in 3 infants with extreme prematurity during the first 2 postnatal weeks. The cause(s) of these fetal losses and infant prematurity could not be determined. The remaining offspring of nivolumab-treated females survived to scheduled termination and there were no nivolumab-related effects on any of the parameters evaluated throughout the 6-month postnatal period.

Table 11: Overview of reproductive toxicity studies conducted with nivolumab in nonhuman primates

Study type/ Study ID / GLP	Species; Number Female/ group	Route & dose	Dosing period	Major findings
ePPND/DN12001	Cynomolgus; 16F/g	0-10-50 2QW	GD20-parturition	**Clinical observations** ≥10F: ↑Stillbirths on/after GD140 =50F: ↑Abortions before GD140 **Hematology** =10F: ↑RBC slight =50F: ↑LUC (NS high variance) **Serum chemistry** =50 pooled infants: ↑Phos d84 slight but not d182, ↑AST d182 slight, ↑Ca d182 minimal, **Cytometry** =50F: ↓cytotoxic activated lymphocytes NS high variance, ≥10 pooled infants: ↑T-lymphocytes NS high variance BD84, @50 elevated at BD182 NS, =50 pooled infants:↑T-helper regulatory cells at BD182, ↑Cytotoxic t-regulaotr lymphocytes and trend at 10mg/kg **Gross pathology** =50 male infants: ↑prostate **Mortalities** =10 infant M: 3xUnscheduled necropsy d1-13 (premature)

Table 12: Summary of pregnancy and infant losses

Nivolumab Dose	Nivolumab Dose (mg/kg)			Historical Controls[a]	
	0 (Control)	10	50	Overall Incidence (%)	Incidence Range (%)
First trimester (GD 20-50) loss	2/16 (12.5%) GD 33, 47	0	3/16 (18.8%)[b] GD 31, 32, 33	7.7	0-16.7
Second trimester (GD 51-100) loss	0	0	0	1.5	0-11.1
Third trimester (> GD 100) loss	1/14 (7.1%) GD 121	2/16 (12.5%) GD 124, 158	4/12 (33.3%) GD 113, 127, 161, 167	15.7	0-31.3
Infant loss	2/13 (15.4%) GD 138/BD 16 GD 153/BD 32	3/14 (21.4%)[c] GD 131/BD 1 GD 135/BD 1 GD 143/BD 13	0	11.2	0-20.0
Surviving infants	11	10	8	NA	NA

Abbreviation: NA = Not applicable.

For pregnancy loss, GD xx = Gestation day when pregnancy loss was first noted.

For infant loss, GD xx/BD xx = Gestation length/day of infant loss after birth.

a The first trimester results were based on control data from 6 embryo-fetal development studies (2006 to 2012) and 12 ePPND (2008 to present) studies conducted at the Testing Facility; results for all other periods/parameters were based on control data from12 ePPND studies.

b Excluded 1 embryonic loss caused by umbilical thrombus and was unrelated to nivolumab treatment.

c Excluded 1 infant loss, that occurred under ketamine sedation for blood sampling and was unrelated to nivolumab treatment.

Studies in juvenile animals

The potential developmental effects of nivolumab were examined in the ePPND study that included assessments in infant monkeys up to 6 month old. In addition, pivotal toxicity studies up to 3 months in duration included cynomolgus monkeys as young as 2 years of age, which is approximately equivalent to a 6-year-old human. No developmental effects were observed in the pivotal intermittent-dose toxicity studies, and no developmental effects were observed in the surviving infants in the ePPND study.

Toxicokinetic data

Table 13: TK data in nivolumab 1-month repeat dose study (SUV00025)

Parameter	Day	Nivolumab Dose					
		1 mg/kg		10 mg/kg		50 mg/kg	
		Males	Females	Males	Females	Males	Females
C(0.25h) (µg/mL)	1	22.0[a]	23.1[a]	288[a]	255[a]	1,120[a]	1,200[a]
	22	27.9/33.5[b]	21.4/27.8[b]	411/496[b]	496[a]	1,930/2,150[b]	2,230[a]
AUC(0-168h) (µg•h/mL)	1	1,740[a,c]	2,040[a,c]	23,800[a,c]	21,600[a,c]	91,700[a,c]	109,000[a,c]
	22	3,060[d]/ 3,820[b,c]	2,120[e,f]/ 2,850[b,e]	36,700[g]/ 45,700[b,h]	51,200[a,c]	196,000[h]/ 242,000[b,c]	224,000[a,c]

a No anti-nivolumab antibodies were detected.

b Means were calculated with the inclusion/exclusion of data from animals with detectable anti-nivolumab antibodies.

c Mean systemic exposures were averaged from individual AUC(0-168h) values.

d Mean systemic exposures were averaged from individual AUC(0-2h) and AUC(0-168h) values.

e Mean systemic exposures were averaged from individual AUC(0-24h) and AUC(0-168h) values.

f N = 4 because all of individual concentrations in Animal No. 2504 were less than the detection limit (1 µg/mL).

g Mean systemic exposures were averaged from individual AUC(0-24h), AUC(0-72h), and AUC(0-168h) values.

h Mean systemic exposures were averaged from individual AUC(0-72h) and AUC(0-168h) values.

Table 14: TK data in nivolumab 3-month repeat dose study (WIL-552003)

Parameter	Week	Nivolumab	
		10 mg/kg 2QW	50 mg/kg 2QW
Cmax[a] (µg/mL)	13	801	3,610
AUC(0-168h) (µg•h/mL)	13	117,000	531,000

a Cmax values were estimates of time 0 (i.e., initiation of dosing).

Local Tolerance

The local tolerance of nivolumab was assessed in single- and repeat-dose IV studies (QW, 2QW, and monthly) in monkeys. Nivolumab was administered at up to 50 mg/kg/dose. No irritation or local tolerance issues were observed.

Other toxicity studies

Immunogenicity

Nivolumab was immunogenic in study SUV00027, with 5 of 6 animals at 1 mg/kg and 2 of 3 animals at 10 mg/kg tested positive for anti-drug antibodies 27 days after dosing. There was no apparent effect of these antibodies on the PK of nivolumab.

Nivolumab binds cynomolgus monkey PD-1 and human PD-1 with similar affinities (KD of 3.06 nM to human PD-1 and 3.92 nM to cynomolgus monkey PD-1), and exhibits pharmacologic activity in humans and monkeys, but nivolumab does not bind homologous PD-1 in other traditional toxicology species (eg, rat, rabbit).

2.3.5. Ecotoxicity/environmental risk assessment

Nivolumab is a protein, which is expected to biodegrade in the environment and not be a significant risk to the environment. Thus, according to the "Guideline on the Environmental Risk Assessment of Medicinal Products for Human Use" (EMEA/CHMP/SWP/4447/00), nivolumab is exempt from preparation of an Environmental Risk Assessment as the product and excipients do not pose a significant risk to the environment.

2.3.6. Discussion on non-clinical aspects

The similarities in tissue binding profiles between cynomolgus monkeys and humans indicate that the cynomolgus monkey was an appropriate animal model for preclinical toxicity testing of nivolumab. In contrast, cross-reactivity studies conducted with rat and rabbit PD-1 showed no binding.

The applicant did not submit studies on distribution, metabolism and excretion. This is acceptable as in accordance with regulatory guidelines for biotechnology-derived pharmaceuticals (ICH S6), no tissues distribution studies, metabolism studies, mass balance are considered necessary. The single-dose PK of nivolumab in cynomolgus monkeys were characterized by a long t½, a low CLTs and a low distribution volume at the steady state (Vss) which are characteristics for a mAb. The metabolism and clearance of nivolumab *in vivo* is expected to follow the degradation route of Abs in general, via biochemical pathways that are independent of metabolising enzymes. Formal distribution studies with nivolumab were not conducted. The low volume of distribution ranging between 0.046-0.071 L/kg at steady state suggests that most of the monoclonal antibody remains in the bloodstream. While there was no accumulation of the mAb at 10mg/kg/month, accumulation was apparent at 1 to 50 mg/kg given weekly or bi-weekly in animals not positive for ADA. When ADA are not predominant, kinetics are typical for IgG/FcRn mediated elimination with long half-lives of 5.2-11.1 days and slow clearance 0.16-0.32mL/h/kg.

The apparent elimination half-life of nivolumab in cynomolgus monkeys following a single IV dose of 1 mg/kg was 124 to 148 hours and was longer with higher doses. Consistent with the long half-life, the total serum clearance after a single dose of nivolumab was low (0.16-0.32 mL/h/kg). The steady state volume of distribution was similar to the reported plasma volume of monkeys (0.046-0.071 L/kg), suggesting that nivolumab remains mainly in the vascular system.

Since nivolumab is a monoclonal antibody and does not belong to a class of medicinal products expected to cause cardiovascular effects, specific safety pharmacology studies were limited to a single-dose cardiovascular safety study in Cynomolgus monkeys. This limited safety pharmacology evaluation of nivolumab is supported by ICH guidelines S6(R1), S7A, and S9, which do not require specific safety pharmacology studies for biotechnology-derived products, such as monoclonal antibodies, or for anticancer pharmaceuticals. It is considered that for this type of product, the lack of studies for secondary pharmacodynamics are acceptable.

In the repeat toxicity studies, serum chemistry changes were limited to a 28% decrease in mean plasma triiodothyronine (T3) levels at 50 mg/kg in female monkeys at the end of the 3-month dosing phase of the study with complete recovery after a 4-week dose-free period. The relevance of the lower T3 levels in females, in the absence of any correlative changes in other hormones or in the thyroid or pituitary gland, is unknown but does not cause any particular concern.

The applicant did not submit studies for genotoxicity, carcinogenicity as well as fertility and early embryonic development. According to ICH guideline S6(R1) "Preclinical safety evaluation of biotechnology-derived pharmaceuticals" (EMA/CHMP/ICH/731268/1998), these studies are not required and are not warranted to support marketing for therapeutics intended to treat patients with advanced cancer (ICH guideline S9 "Nonclinical evaluation for anticancer pharmaceuticals" EMEA/CHMP/ICH/646107/2008), the lack of studies is acceptable.

In repeat-dose toxicity studies, no nivolumab-related gross or microscopic pathology changes were observed. Pivotal 1- and 3-month IV repeat-dose toxicity studies of nivolumab alone were conducted in cynomolgus monkeys. The product was well tolerated, with no adverse effects at doses ≤50 mg/kg administered up to one month once a week or 3 months twice a week. In the 1-month study, while PD-1 receptor occupancy was demonstrated throughout the dosing phase, no changes in peripheral blood immune cell subpopulations were observed. In the 3-month study, changes in immune cell parameters were observed, demonstrating that nivolumab elicited pharmacological responses in healthy monkeys.

Regarding reproductive toxicity, the ePPND study was conducted following the recommendations included in the PIP P/0064/2014 agreed with the EMA on March 2014 with the non clinical working group of the PDCO. Nivolumab was administered 2QW at 10 or 50 mg/kg to pregnant monkeys from GD 20 to 22 until parturition. Nivolumab was well tolerated and there were no nivolumab-related effects on viability, clinical signs, food consumption, body weights, immunological endpoints or clinical/anatomic pathology parameters in dams throughout the study. However, in the offspring, maternal nivolumab administration was associated with foetal/neonatal mortality, with dose-dependent increases in third trimester foetal loses and increased neonatal mortality at 10 mg/kg during the first 2 postnatal weeks. The cause of these foetal losses and infant prematurity could not be determined. However, a relationship between pregnancy failure and a break in tolerance by the abrogation of PD-1 signalling by nivolumab cannot be ruled out. This information is stated in section 5.3 and the potential risk to pregnant women is adequately addressed in section 4.6 of the SmPC. The risk of embryofoetal toxicity has been included in the RMP as an important potential risk.

No juvenile toxicity studies were conducted by the Applicant. In the ePPND study infant monkeys were assessed up to 6 month old, whereas in the pivotal toxicity studies some monkeys were 2 years old, approximately equivalent to a 6-year old human. No developmental effects were observed in any of these studies.

Considering the lack of irritation or local tolerance issues found in the repeat-dose toxicity studies in cynomolgus monkeys, no additional local tolerance studies are considered necessary.

As a selective immunomodulator, nivolumab is expected to elicit pharmacologically mediated effects on the immune system. Repeated administration of nivolumab alone for 3 months at ≥ 10 mg/kg produced pharmacologic effects on specific immune cell sub-populations, but no general nonspecific immunostimulatory or autoimmune-related toxicities were observed.

ADAs were detected in 4% of monkeys receiving 2QW doses of nivolumab, 20% to 53% of monkeys receiving weekly doses of nivolumab, and 67% of monkeys receiving monthly doses of nivolumab. Nivolumab concentrations were lower in animals with anti-nivolumab antibodies. ADA incidence was predominantly increased in the low dose groups, which may be a result of assay interference.

The applicant did not submit studies for the ERA. According to the guideline, in the case of products containing proteins as active pharmaceutical ingredient(s), an ERA justifying the lack of ERA studies is acceptable.

2.3.7. Conclusion on the non-clinical aspects

In conclusion, the non-clinical studies (pharmacology, pharmacokinetics and toxicology), submitted for the marketing authorisation application for nivolumab, were considered adequate and acceptable for the assessment of non-clinical aspects. The lack of carcinogenicity, mutagenicity, fertility and early embryonic development were well justified. Based on cynomolgus monkey studies, there is a potential risk for foetal loss in the third semester in humans. In addition, the cynomolgus monkey model showed development of ADAs after repeat administration. These risks are adequately addressed in the SmPC and RMP.

2.4. Clinical aspects

2.4.1. Introduction

GCP

The Clinical trials were performed in accordance with GCP as claimed by the applicant

The applicant has provided a statement to the effect that clinical trials conducted outside the community were carried out in accordance with the ethical standards of Directive 2001/20/EC.

- Tabular overview of clinical studies

Table 15: Summary of Study Designs of studies CA209037, MDX1106-03, and CA209066

	CA209037	MDX1106-03	CA209066
Study Design	Phase 3, randomised, open-label	Phase 1b (expansion), open-label	Phase 3, randomised, double-blind
Study Population	Progressed on or after anti-CTLA-4 therapy and for those with BRAF V600 mutations, progressed on or after a BRAF inhibitor in addition to anti CTLA-4-therapy	At least 1 prior systemic therapy; ipilimumab was not allowed	Previously untreated unresectable or metastatic BRAF WT melanoma
Number of Melanoma Subjects Treated with Nivolumab	N=268	N=107	N=206
Nivolumab Monotherapy Regimen	Nivolumab: 3 mg/kg Q2W vs Investigator's Choice Chemotherapy	Nivolumab: 0.1, 0.3, 1, 3, and 10 mg/kg Q2W	Nivolumab: 3 mg/kg Q2W vs Dacarbazine: 1000 mg/m^2 Q3W
Primary Efficacy Endpoint	<u>Co-Primary:</u> IRRC-assessed ORR (1st 120 nivolumab	Sponsor-assessed ORR	OS

Table 15: Summary of Study Designs of studies CA209037, MDX1106-03, and CA209066

	CA209037	MDX1106-03	CA209066
	treated subjects) and OS		
Secondary Efficacy Endpoints	IRRC-assessed PFS, efficacy by PD-L1 expression HRQoL: EORTC QLQ-C30	DCR, PFS, OS	PFS, ORR, OS by PD-L1 expression
Study Status	Enrollment closed; ORR primary analysis completed; OS follow-up remains ongoing	Analysis of primary endpoint completed; 3-year OS is available; OS follow-up remains ongoing	Primary endpoint analysis of OS completed, DBL planned Aug 2014; OS follow-up remains ongoing

- **Table 16: Summary of clinical studies contributing to pharmacology profiling of nivolumab**

Study number	Treatment	Number of treated subjects	Pharmacology component
MDX1106-01 (CA209001) Phase 1	0.3, 1, 3, 10 mg/kg	39	Single dose PK popPK
MDX1106-03 (CA209003) Phase 1	0.1, 0.3, 1, 3, 10 mg/kg Q2W	306 NSCLC=129 Melanoma=107 RCC=34 CRC=19 mCPRC=17	Multiple dose PK Dose selection popPK Receptor binding T cell distribution ALC PD-L1 tumour tissue Exposure response immunogenicity
CA209009 *study ongoing*	0.3, 2, 10 mg/kg Q3W	91 RCC	PBMC Cytokine concentrations
CA209010 Phase 2	0.3, 2, 10 mg/kg Q3W	167 RCC	Sparse PK – popPK QTc prolongation
CA209063 **Phase 2**	**3 mg/kg Q2W**	**117 NSCLC**	Sparse PK – popPK immunogenicity

(pivotal)			
CA209037 **Phase 3** **(pivotal)**	**3 mg/kg Q2W**	**268 melanoma**	Sparse PK – popPK Exposure response immunogenicity
ONO-4538-01 Phase 1 Japanese	1, 3, 10, 20 mg/kg Q2W	17	Single dose PK and sparse PK for multiple dose - popPK
ONO-4538-02 Japanese	2 mg/kg Q3W	35 melanoma	Sparse PK - popPK

2.4.2. Pharmacokinetics

Absorption

Nivolumab is dosed via the IV route and therefore is completely bioavailable.

Distribution

Single Dose Pharmacokinetics: Study MDX1106-01

The single-dose PK of nivolumab was described by non-compartmental analysis (NCA) of data from 39 subjects in MDX1106-01 (also known as CA209001), which was a Phase 1 study in patients with selected refractory or relapsed malignancies. The single-dose PK of nivolumab was determined from serum concentrations collected up to 85 days following single doses of 0.3, 1, 3 and 10 mg/kg, given as 1-hour IV infusion in MDX1106-01.

Following a single-dose IV administration of nivolumab ranging from 0.3 mg/kg to 10 mg/kg mean volume of distribution (Vz) varied between 83 to 113 mL/kg across doses. Mean clearance and mean elimination half-life ranged from 0.13 to 0.19 ml/h/kg and between 17 and 25 days, across the range of 0.3 to 10 mg/kg dose.

Table 17: Summary of nivolumab single-dose pharmacokinetic parameters – Study MDX1106-01

Dose (mg/kg)	Cmax (µg/mL) Geo. Mean [N] (%CV)	Tmax (h) Median [N] (Min-Max)	AUC(0-T) (µg*h/mL) Geo. Mean [N] (%CV)	AUC(INF) (µg*h/mL) Geo. Mean [N] (%CV)	T-HALF (day) Mean [N] (SD)	CLT (mL/h/kg) Geo. Mean [N] (%CV)	Vz (mL/kg) Mean [N] (SD)
0.3	6.7 [6]	3.0[6]	970 [6]	2343 [3]	18.9 [3]	0.13 [3]	82.8 [3]
	(21.6)	(1.0-6.8)	(47)	(16)	(7.05)	(16.93)	(27.19)
1	16.0 [6]	1.9 [6]	3244 [6]	6014 [4]	17.0 [4]	0.17 [4]	99.6 [4]
	(32.1)	(1.0-7.0)	(62)	(30)	(2.36)	(29.80)	(23.04)
3	60.0 [5]	3.1 [5]	13909 [5]	15813 [5]	17.0 [5]	0.19 [5]	112.7 [5]
	(27.6)	(1.0-5.0)	(44)	(44)	(4.70)	(42.66)	(39.50)
10	196.3 [21]	1.6 [21]	55324 [21]	76541 [19]	24.8 [19]	0.13 [19]	109.4 [19]
	(19.5)	(0.9-7.0)	(39)	(27)	(7.22)	(28.42)	(26.70)

Abbreviations: Geo mean=geometric mean; CV=coefficient of variation; SD=standard deviation

Volume of distribution as estimated by popPK analysis was 8.00 L (35.3%).

Multiple-dose Administration: Study MDX1106-03

The multiple-dose PK of nivolumab given Q2W was assessed by NCA in MDX1106-03. Intensive PK serum concentration samples were collected from all subjects enrolled in 0.1, 0.3, and 1 mg/kg melanoma cohorts and first 16 subjects each from 3 and 10 mg/kg NSCLC cohorts over 336 hours (15 days) after first dose (Cycle 1) and ninth dose (Cycle 3). Limited PK samples were collected from all other pre-amendment 4 subjects and from all subjects enrolled in 1 mg/kg RCC cohort, 1 mg/kg NSCLC and remaining 16 subjects each from 3 and 10 mg/kg NSCLC in this study. Single samples were collected to evaluate serum concentrations of nivolumab at all follow-up visits.

The results from study MDX1106-03 is shown in Table 18.

Table 18: Summary of nivolumab multiple dose pharmacokinetic parameters – Study MDX1106-03

Nivolumab Dose	Dose Number	Cmax (µg/mL) GEO.MEAN[N] (%CV)	Tmax (h) MEDIAN[N] (MIN-MAX)	AUC(TAU) (µg*h/mL) GEO.MEAN[N] (%CV)	AI_Cmax GEO.MEAN[N] (%CV)	AI_AUC GEO.MEAN[N] (%CV)	CLT (mL/h) GEO.MEAN[N] (%CV)	Effective T-HALF (h) Mean [N] (SD)
0.1 mg/kg	First	1.9[15] (23.6)	1.1[15] (0.3-51.0)	279.4[13] (32.5)				
	Ninth	3.7[5] (42.2)	8.0[5] (0.6-24.0)	1104.4[4] (26.6)	2.3[4] (13.0)	3.1[4] (31.0)	8.3[4] (40.0)	622 [4] (235)
0.3 mg/kg	First	7.0[17] (32.3)	1.2[17] (0.9-24.3)	954.7[15] (26.9)				
	Ninth	17.8[2] (26.6)	24.7[2] (1.3-48.0)	3406.1[2] (12.8)	2.0[2] (26.7)	2.9[2] (6.1)	6.9[2] (17.8)	555 [2] (42)
1 mg/kg	First	19.6[17] (29.5)	1.2[17] (0.9-48.0)	3589.6[10] (23.8)				
	Ninth	46.9[10] (26.1)	1.0[10] (0.9-24.1)	10190.4[9] (25.8)	2.4[9] (21.4)	3.1[9] (34.7)	8.0[9] (31.1)	636 [9] (267)
3 mg/kg	First	61.3[13] (26.4)	2.1[13] (0.8-8.0)	8785.8[13] (22.7)				
	Ninth	132.0[7] (19.8)	4.0[7] (1.0-8.0)	30640.3[5] (17.5)	2.4[5] (13.6)	3.3[5] (25.5)	10.3[5] (18.1)	661 [5] (202)
10 mg/kg	First	191.2[14] (40.0)	3.9[14] (1.0-48.2)	31095.1[12] (25.4)				
	Ninth	475.0[5] (24.6)	22.3[5] (1.0-24.5)	99621.7[3] (26.0)	2.4[3] (12.6)	3.1[3] (11.0)	8.5[3] (6.4)	595 [3] (80)

The population PK analysis showed a 2-compartment PK model with first order elimination. Both the clearance and the volume in the central compartment on nivolumab increase with body weight (ranged from 34 to 162 kg with a mean weight of 81 kg). This fact has been addressed by means of weight normalized dosing.

Elimination

The mean terminal elimination half-life of nivolumab ranged between 17 and 27.5 days following single dose (study MDX1106-01) and Q2W administration (MDX1106-03) across the range of 0.1 to 10 mg/kg dose.

The geometric mean (%CV) of PopPK model-based estimates of individual nivolumab CL, volume of distribution at steady state (VSS), and terminal half life were 9.5 mL/h (49.7%), 8.0 L (30.4%), and 26.7 days (101.0%), respectively. The typical clearance was 8.7 mL/h.

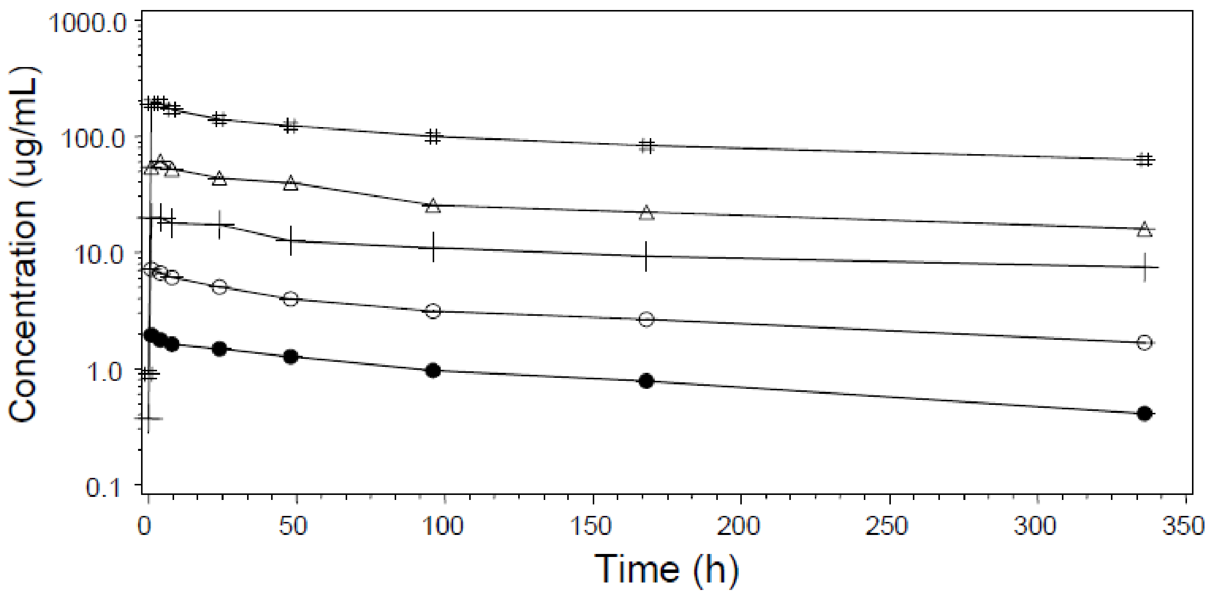
Metabolism

No formal studies were conducted as nivolumab is a human monoclonal immunoglobulin and not metabolized by cytochrome P450 enzymes, it is degraded to small peptides and individual amino acids.

Dose proportionality and time dependencies

The proportionality of the pharmacokinetics of nivolumab over the dose range 0.1 mg/kg-10 mg/kg was investigated in study MDX1106-03 and is presented in Figure 9. Following a one hour IV infusion, maximum concentrations of nivolumab were reached at median Tmax of 1.1 to 3.9h after Cycle 1/Day 1 dose.

Figure 9: Plot of Mean Nivolumab Serum Concentration Profile versus Time After First Nivolumab Dose - Study MDX1106-03



0.1 mg/kg nivolumab (●); 0.3 mg/kg nivolumab (o); 1 mg/kg nivolumab (+); 3 mg/kg nivolumab (Δ); 10 mg/kg nivolumab (#)

- **Time dependency**

Following Q2W administration, accumulation of nivolumab Cmin from first to ninth dose was in the range of 3.1 to 4.8, whereas accumulation of Ceoinf was in the range of 1.5 to 2.2.

Table 19: Summary of trough and end of infusion concentration values of nivolumab administered every two weeks – Study MDX1106-03

Nivolumab Dose	Dose Number	Cmin (µg/mL) GEO.MEAN[N] (%CV)	Ceoinf (µg/mL) GEO.MEAN[N] (%CV)	AI_Cmin GEO.MEAN[N] (%CV)	AI_Ceoinf GEO.MEAN[N] (%CV)
0.1 mg/kg	First	0.3[16] (56.9)	1.9[16] (27.7)		
	Ninth	2.5[7 (27.7)	2.8[4] (53.1)	4.8[7] (26.2)	1.5[4] (58.9)
0.3 mg/kg	First	1.4[15] (47.6)	6.9[18] (32.8)		
	Ninth	6.4[5] (47.1)	17.2[2] (31.3)	4.7[5] (102.3)	1.9[2] (31.5)
1 mg/kg	First	5.5[72] (42.8)	19.7[82] (31.3)		
	Ninth	19[35] (38.8)	39.7[38] (30.1)	3.1[35] (34.5)	1.9[36] (32.6)
3 mg/kg	First	16.6[46] (34.4)	58.6[50] (28.3)		
	Ninth	57[21] (35.9)	121.5[23] (20.7)	3.2[20] (25.3)	2.2[23] (49.4)
10 mg/kg	First	56.5[116] (30.6)	179.6[120] (26.3)		
	Ninth	188.8[44] (36.9)	331.4[43] (33.6)	3.2[44] (34.6)	1.8[42] (39.4)

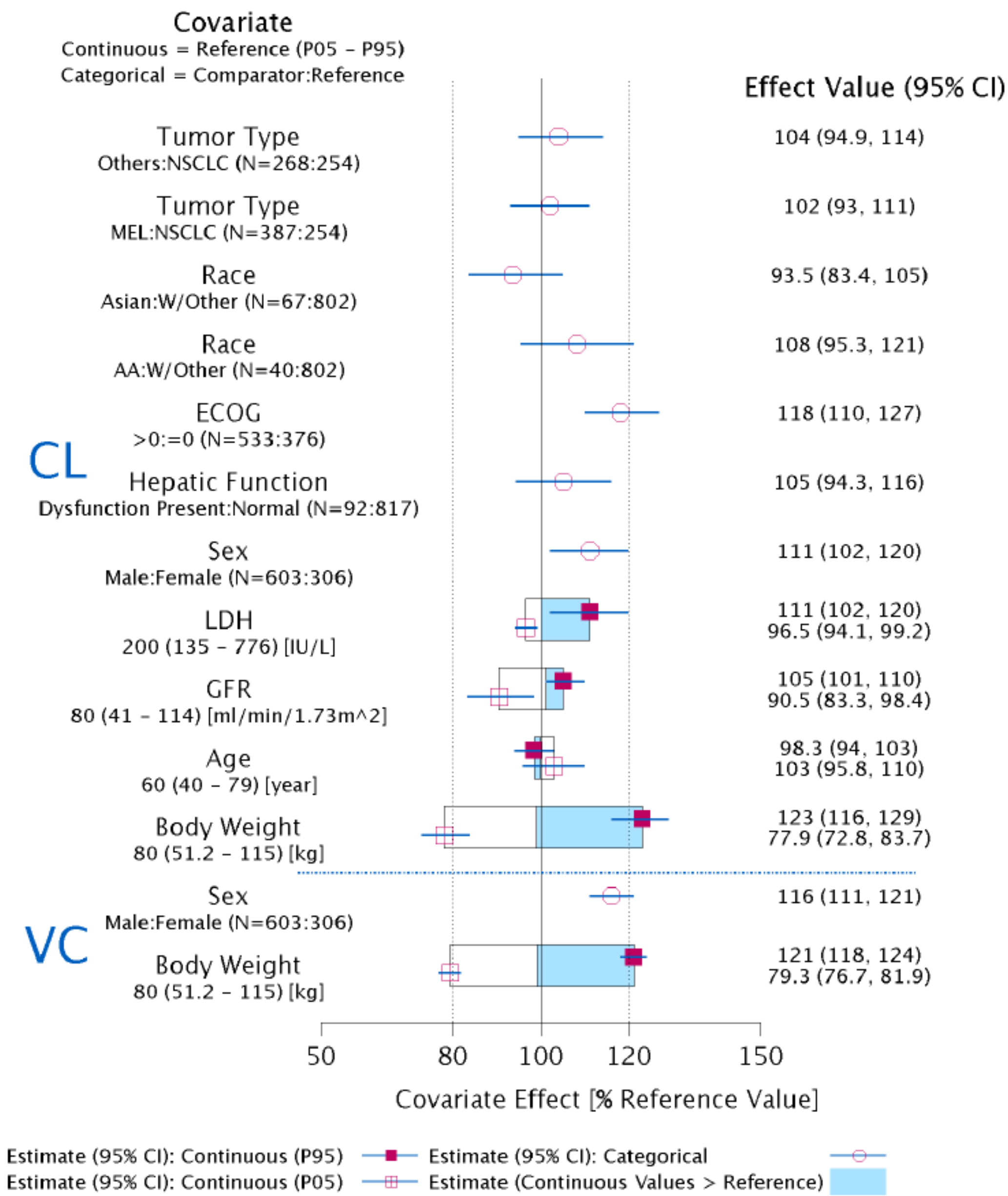
Cmin and Ceoinf data shown are based on Day 1 samples, except for the Cycle1 Cmin which is based on Day 15 samples

Special populations

PopPK analyses

Population PK (popPK) analysis was based on intensive and sparse PK sampling mainly between Day 1 (Cycle 1) and Day 99 (Cycle 8) from 909 patients with solid tumors who received 3 mg/kg or 10 mg/kg Q2W during the dosing period (MDX1106-01, ONO-4538-01, ONO- 4538-02, MDX1106-03, CA209010, CA209063 and CA209037). The PopPK model parameters were estimated with precision, and the model evaluation demonstrated that there was good agreement between model predictions and observations (Figure 10).

Figure 10: Covariate effect on PK model parameters (full PPK model)



Note 1: Categorical covariate effects (95% CI) are represented by open symbols (horizontal lines).

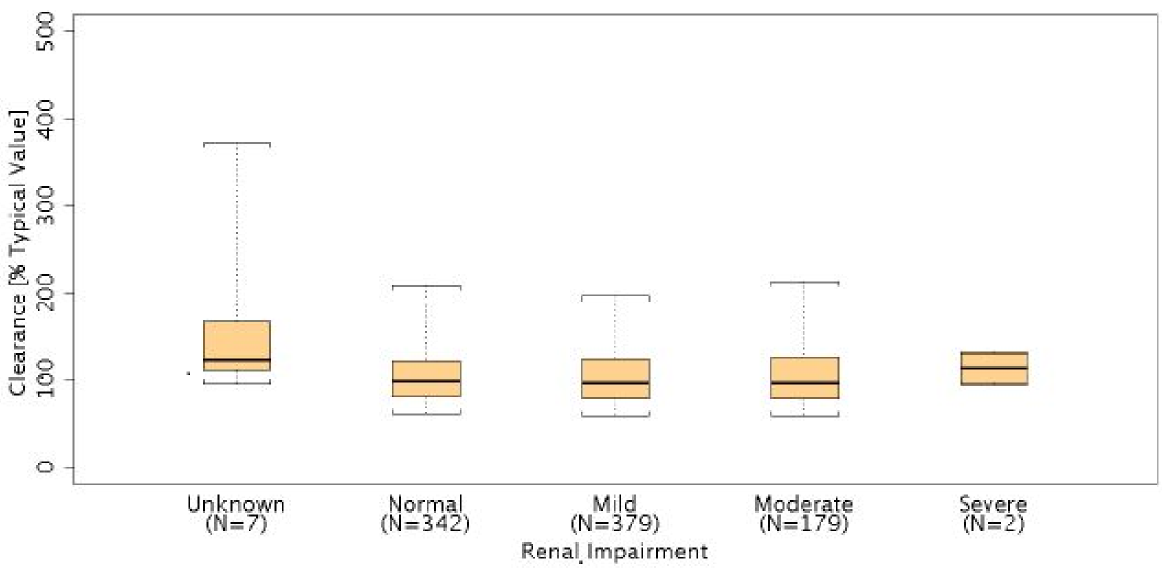
Note 2: Continuous covariate effects (95% CI) at the 5th/95th percentiles of the covariate are represented by the end of horizontal boxes (horizontal lines). Open/shaded area of boxes represents the range of covariate effects from the median to the 5th/95th percentile of the covariate.

Note 3: Reference subject is CA/others female age=60 yr, ECOG=0, LDH=200 IU/L, GFR=80 ml/min/1.73m^2 and body weight=80kg, subject with normal hepatic function. Parameter estimate in reference subject is considered as 100% (vertical solid line) and dashed vertical lines are at 80% and 120% of this value.

Impaired renal function

Specific PK studies in patients with renal impairment were not conducted. Lack of effect of renal function (normal, mild or moderate) on the PK of nivolumab was obtained from the PopPK analysis. Limited data (n=2) were available for severe renal impairment assessment.

Figure 11: Boxplots showing the lack of relationship between renal fundction status and nivolumab clearance



- **Impaired hepatic function**

No formal studies were conducted. The assessment of hepatic status is limited to the normal and mild abnormalities integrated in the PopPK analysis. PopPK analysis showed mild hepatic impairment had similar CL and exposures relative to normal subjects.

- **Body Weight**

PopPK analysis with CDA209037 showed that both clearance (CL) and volume of central compartment (VC) increase with body weight. However, nivolumab exposures (dose normalized Cminss and Cavgss) are comparable across the range of body weight (34-162 kg) when administered based on mg/kg.

- **Elderly**

PopPK analysis with CDA209037 showed that age was not a significant covariate on nivolumab CL. Nivolumab exposure (dose normalized Cavgss) was similar across the age ranging from 29 to 87 years.

Pharmacokinetic interaction studies

The applicant did not submit drug-drug interaction studies (see pharmacology discussion).

Pharmacokinetics using human biomaterials

The applicant did not submit PK using biomaterials studies (see pharmacology discussion).

2.4.3. Pharmacodynamics

The PD effects of nivolumab were studied by assessing receptor occupancy (RO), peripheral immune cell population modulation, systemic cytokine modulation, and change in absolute lymphocyte count (ALC) in studies MDX1106-03 and/or CA209009.

Mechanism of action

PD-1 Receptor Occupancy by Nivolumab

PD-1 receptor occupancy by nivolumab was investigated in studies MDX1106-03 and CA209009. In study MDX1106-03, RO was determined in using frozen peripheral CD3+ T-cells from 65 melanoma subjects treated with one cycle (4 doses Q2W) of nivolumab at doses of 0.1 to 10.0 mg/kg. The median PD-1-receptor occupancy by nivolumab was 64 to 70% across all dose levels (Table 20).

Table 20: Receptor occupancy prior to fifth dose administration (all treated subjects with receptor occupancy) - Study MDX1106-03

Nivolumab Dose (mg/kg)	N	MEAN	SD	MEDIAN	MIN	MAX
0.1	11	61.12	11.761	64.1	29.8	73.8
0.3	11	63.81	8.338	63.9	52.0	80.6
1.0	21	66.00	8.919	65.0	48.9	85.0
3.0	12	67.39	10.792	67.8	51.1	84.6
10.0	10	69.52	9.045	70.1	57.1	81.9

In the nivolumab metastatic RCC study CA209009, receptor occupancy was assessed using fresh whole blood specimens. RO of >d90% was achieved at one hour post nivolumab treatment at all dose levels, and remained near this level through Dose 8 Day 1.

Lymphocyte phenotype, absolute lymphocyte count, cytokine and chemokine modulation by nivolumab

Peripheral immune cell populations as measured by flow cytometry and change in ALC from baseline was evaluated in study MDX1106-03. The effect of nivolumab on cytokine modulation was assessed in CA209009 by measuring cytokine levels during the course of nivolumab treatment. The activated CD8+ T-cell mean changes by nivolumab dose were on average 3.8%, 0.4%, 2.3%, 5.8%, and 0.1% for 0.1, 0.3, 1, 3, and 10 mg/kg, respectively. No meaningful rise over baseline was observed in mean ALC.

The effect of nivolumab on cytokine modulation was assessed in CA209009 by measuring cytokine levels during the course of nivolumab treatment. Nivolumab was administered at 0.3, 2 and 10 mg/kg dose levels every three weeks. Cytokine levels were measured at Dose 1 (0, 3, 7, 24 hr), Dose 2 (0, 168 hr), Dose 4 (0 hr), Dose 7 (0,168 hr) and Dose 8 (0 hr). The study showed that IL-1A, IL-1B, INF-γ, TNF-α, IL-12P and IL-23M were found to be below the lower limit of quantification. The levels of IL-6, IL-10 and IL-2 soluble receptor α showed transient changes but these were not consistent between individuals. The levels of CXCL-9 and CXCL-10 increased from baseline across any treatment dose group.

Primary and Secondary pharmacology

PD-1 Receptor Occupancy by Nivolumab

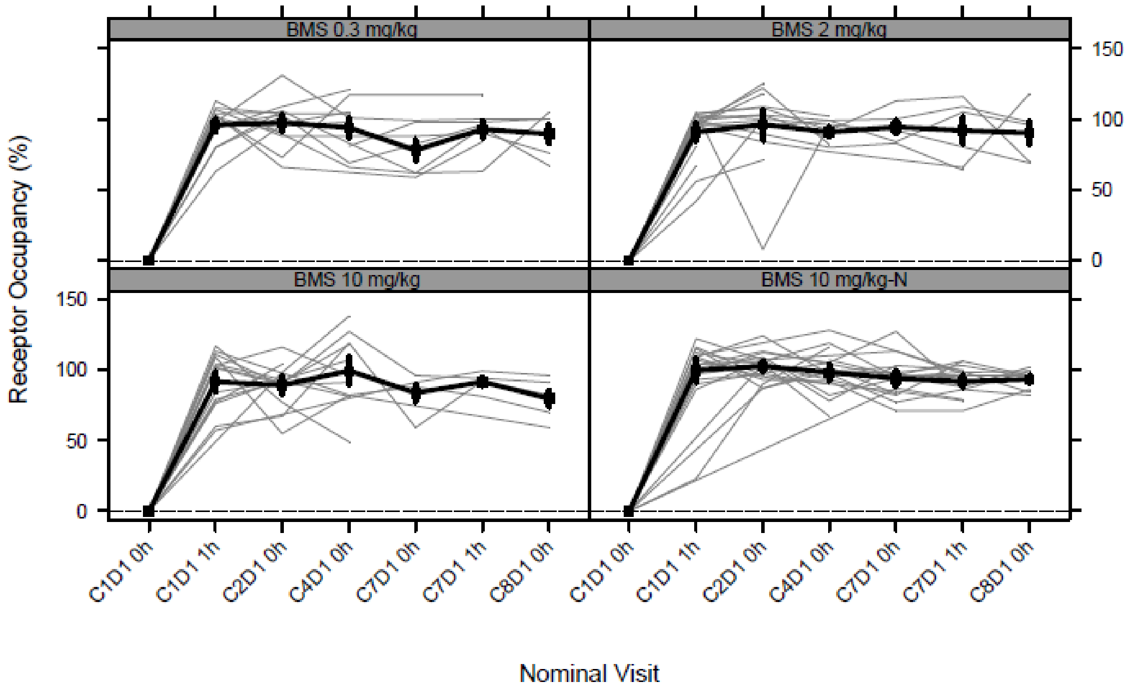
PD-1 RO by nivolumab was investigated in studies MDX1106-03 and CA209009. In study MDX1106-03, RO was determined in using frozen peripheral CD3+ T-cells from 65 melanoma subjects treated with one cycle (4 doses Q2W) of nivolumab at doses of 0.1 to 10.0 mg/kg. The median PD-1-receptor occupancy by nivolumab was 64 to 70% across all dose levels (Table 21). These results demonstrate that the majority of PD-1 receptors are bound by nivolumab at the lowest dose tested (0.1 mg/kg). Increasing doses up to 10.0 mg/kg did not substantially increase PD-1 receptor occupancy at the time point tested.

Table 21: Receptor Occupancy Prior to Fifth Dose Administration - All Treated Subjects with Receptor Occupancy - Study MDX1106-03

Nivolumab Dose (mg/kg)	N	MEAN	SD	MEDIAN	MIN	MAX
0.1	11	61.12	11.761	64.1	29.8	73.8
0.3	11	63.81	8.338	63.9	52.0	80.6
1.0	21	66.00	8.919	65.0	48.9	85.0
3.0	12	67.39	10.792	67.8	51.1	84.6
10.0	10	69.52	9.045	70.1	57.1	81.9

In the nivolumab metastatic RCC study CA209009, RO was assessed using fresh whole blood specimens. RO of peripheral CD3+ T cells (and CD4+ or CD8+ subsets) was measured at baseline and at six timepoints following initiation of nivolumab treatment (Dose 1-1H, Dose 2-0H, Dose 4-0H, Dose 7-0H, Dose 7-1H, Dose 8-0H). Kinetics of RO were similar across all dose cohorts (0.3 mg/kg, 2 mg/kg, 10 mg/kg, 10 mg/kg-treatment naive) (Figure 12) Receptor occupancy of >d90% was achieved at one hour post nivolumab treatment at all dose levels, and remained near this level through Dose 8 Day 1.

Figure 12: Time course of PD-1 occupancy by nivolumab by dose level – Study CA209009



PD-L1 Expression as a Potential Biomarker

In study MDX1106-03, evaluable archival tumour tissue was available from 36% (38/107) of melanoma and 49% (63/129) of NSCLC subjects indicating a low ascertainment rate. For PD-L1 expression data analysis, 1% and 5% thresholds were utilized to assess PD-L1 positivity. The proportion of PD-L1 positive tumour samples were determined in both melanoma and NSCLC tumours (Table 22).

PD-L1 expression in one or more immune cells was analysed. Using either tumour cell PD-L1 positivity (1% and 5%) or any immune cell PD-L1 expression as an indicator of a positive sample, the proportion of PD-L1 positive subjects was around 90% in both melanoma and NSCLC.

Table 22: PD-L1 expression in melanoma and NSCLC tissue samples – Study MDX1106-03

	n (%)	
PD-L1 Expression	**Melanoma N = 107**	**NSCLC N = 129**
No. of evaluable subjects	38	63
1% - tumor only	25 (66)	35 (56)
5% - tumor only	17 (45)	31 (49)
5% - tumor + immune cells[a]	35 (92)	56 (89)

aIncludes 5% PD-L1 expression on tumour cells or immune cells

2.4.4. Discussion on clinical pharmacology

The clinical pharmacology profile of nivolumab has been characterized based on data from 8 Phase 1/2/3 clinical studies conducted in the clinical program. Pharmacokinetics has mainly been documented in patients with different type of solid tumors (NSCLC, Melanoma, RCC, CRC, CRPC, others) and not in healthy volunteers.

The dose proposed for nivolumab montherapy is 3 mg/kg administered intravenously over 60 minutes every 2 weeks.

Study MDX1106-01 pharmacokinetics of nivolumab was analysed by non-compartmental analysis (NCA) of data from 39 subjects dose ranging between 0.3 and 10 mg/kg. The single-dose PK of nivolumab was dose proportional in the range of 0.3 to 10 mg/kg: there was no correlation between dose and clearance, volume of distribution or elimination half-life. The mean volume of distribution after a single dose administration was small (between 83 to 113 mL/kg) and consistent with localization in the extracellular fluid, as observed for other IgG mAbs with large molecular weight. Pharmacokinetics of nivolumab seemed dose proportional over the dose range 0.1 mg/kg-10 mg/kg. No signs of time dependent PK parameters were observed over the period studied. According to popPK analysis nivolumab accumulation index of approximately 3-fold was consistent with the estimated half-life of 26.7 days for 3 mg/kg dosing every 2 weeks. Steady state was achieved approximately at the 6th dose (12 weeks).

The geometric mean (%CV) of PopPK model-based estimates of individual nivolumab CL, volume of distribution at steady state (VSS), and terminal half life were 9.5 mL/h (49.7%), 8.0 L (30.4%), and 26.7 days (101.0%), respectively. The typical clearance was 8.7 mL/h.Nivolumab, is expected to be cleared through receptor mediated endocytosis or non-specific endocytosis followed by proteolytic degradation mainly in hepatic or reticuloendothelial cells. Therefore, no renal elimination is expected given the large molecular weight of monoclonal antibodies. As nivolumab is not subject of metabolism by CYP450 enzymes no classical studies regarding metabolism or elimination were deemed necessary. The estimated terminal half-life ranged between 17 and 25 days and was consistent with a fully human mAb. The proposed dosing interval of 2 weeks is shorter than the observed terminal half-life.

No formal studies have been conducted in special populations, such as renal and hepatic impaired patients as only patients with mild hepatic abnormalities were included in the clinical studies. PopPk analysis has shown a lack of effect on the PK of nivolumab in patients with renal impairment. PopPK analysis showed that eGFR did not have a clinically relevant effect on nivolumab CL. Results indicate that no dose adjustment is needed for subject with mild and moderate renal impairment. PopPk analysis has shown a lack of effect on the PK of nivolumab in patients with hepatic impairment. Subjects with mild hepatic impairment had similar CL and exposures relative to normal subjects, suggesting that no dose adjustment is needed for subjects with mild hepatic impairment. PopPK analysis did not have any subject with moderate and severe hepatic impairment, thus the assessment of effect of moderate and severe hepatic was not available. This range is reflected in the SmPC in section 4.2. PopPK analysis indicated a higher clearance of nivolumab with increasing body weight. With dosing of nivolumab on an mg/kg basis, no additional dose adjustment is needed for body weight. Thus, dosing per body weight is acceptable across the range of 34-162 kg. The PopPK model seems to provide an adequate description of nivolumab concentration-time data in solid tumours. The values of the PopPK analysis are consistent with the corresponding values estimated by non-compartmental analysis and they can be considered in line with data from other fully human IgG antibodies.

No formal drug-drug interaction studies have been conducted to support the use of nivolumab as monotherapy. As nivolumab is not expected to be metabolized by liver cytochrome P450 or other drug metabolizing enzymes, it is unlikely to have an effect on CYPs or other drug metabolizing enzymes in terms of inhibition or induction.

However, recent literature reports suggest that therapeutic proteins that modulators of cytokines may indirectly affect expression of cytochrome (CYP) enzymes. The indirect drug-drug interaction potential of nivolumab was assessed using systemic cytokine modulation data for cytokines known to modulate CYP enzymes, at single and multiple doses of 0.3 to 10 mg/kg Q3W from CA209009. This dose range covers the exposure of nivolumab at proposed dosing regimen of 3 mg/kg Q2W. There was no meaningful change in cytokines across all dose levels of nivolumab (0.3, 2 and 10 mg/kg) during the course of treatment. This lack of cytokine modulation suggests that nivolumab has no or low potential for modulating CYP enzymes, thereby indicating a low risk of therapeutic protein-drug interaction.

No thorough QT/QTc study with nivolumab was submitted, which is considered acceptable. The nivolumab exposure obtained with the 10 mg/kg Q3W in the QT study was considered sufficient for obtaining a relevant outcome as 10 mg/kg Q3W provides higher Cmin and AUC compared to proposed 3 mg/kg Q2W dosing regimen. Nivolumab, within the range of doses studied up to 10 mg/kg Q3W did not meaningfully affect the QTc interval. There was no discernible relationship between nivolumab serum concentration and change in QTcF.

Peripheral RO of PD-1 was saturated at doses ≥ 0.3 mg/kg dose levels, which was lower than the proposed dose of 3 mg/kg. In peripheral blood, neither clinically relevant changes in the count of activated T cells nor changes in the absolute lymphocyte counts during treatment with nivolumab were observed. Demonstration of peripheral immunomodulatory activity of nivolumab was limited to chemokines CXCL9 and CXCL10. Relation between baseline absolute lymphocyte count and response to treatment could not be established. PD-L1 expression on the tumour did not appear to be related to efficacy response in study MXD1106-03. The value of PD-L1 and PD-L2 as biomarker to predict the efficacy of nivolumab should be further investigated (see clinical discussion).

2.4.5. Conclusions on clinical pharmacology

In conclusion, pharmacokinetics of nivolumab has been mainly characterized by means of a PopPK model which is considered acceptable. The dose is considered to be appropriately investigated and well defined. The CHMP is of the opinion that the relevance of PD-L1 and PD-L2 expression as biomarkers, in the tumour microenvironment as well as in the peripheral compartment, should be further explored (see clinical conclusions).

2.5. Clinical efficacy

2.5.1. Dose response study

Dose rationale

In study MDX1106-03, increasing doses of nivolumab were tested in order to evaluate efficacy response in patients with different type of tumours. There was a greater percent of objective responses observed in NSCLC subjects treated with 3 mg/kg (24.3%) and 10 mg/kg (20.3%) nivolumab than with 1 mg/kg (3%) nivolumab. There was no apparent relationship between nivolumab dose and ORR in melanoma and RCC (Table 23).

Table 23: Overview of objective response rates of nivolumab across tumour types and dose levels – Study MDX1106-03

Nivolumab Dose(mg/kg)	% Objective Response Rate (95% Confidence Interval)					
	0.1	0.3	1	3	10	Total
All NSCLC	NA	NA	3.0 (0.1, 15.8) N=33	24.3 (11.8, 41.2) N=37	20.3 (11.0, 32.8) N=59	17.1 (11.0, 24.7) N=129
Melanoma	35.3 (14.2, 61.7) N=17	27.8 (9.7, 53.5) N=18	31.4 (16.9, 49.3) N=35	41.2 (18.4, 67.1) N=17	20.0 (5.7, 43.7) N= 20	30.8 (22.3, 40.5) N=107
RCC	NA	NA	27.8 (9.7, 53.5) N=18	NA	31.3 (11.0, 58.7) N=16	29.4 (15.1, 47.5) N=34

In study MDX1106-03, increasing doses of nivolumab were tested in order to evaluate efficacy response in patients with different type of tumours. The nature, frequency, and severity of adverse events (AEs) were similar across the dose range 0.1 to 10 mg/kg and across tumour types (Table 24).

Table 24: Frequency of Drug-related Adverse Events across Dose Groups – Study MDX1106-03

Total Number (%) subjects with AE	Dose (mg/kg Q2W)					
	0.1 (N=17)	0.3 (N=18)	1 (N=86)	3 (N=54)	10 (N=131)	Total (N=306)
Any Grade DR-AE	13 (77)	14 (78)	70 (81)	40 (74)	93 (71)	230 (75)
Gr 3-4 DR-AE	5 (29)	3 (17)	12 (14)	11 (20)	21(16)	52 (17)
Gr 3-4 DR-SAE	1 (6)	0	4 (5)	5 (9)	14 (11)	24 (8)
DR-AE leading to DC	3 (18)	0	9 (11)	4 (7)	16 (12)	32 (11)
DR-AE deaths	0	0	1 (1)	0	1 (1)	2 (1)

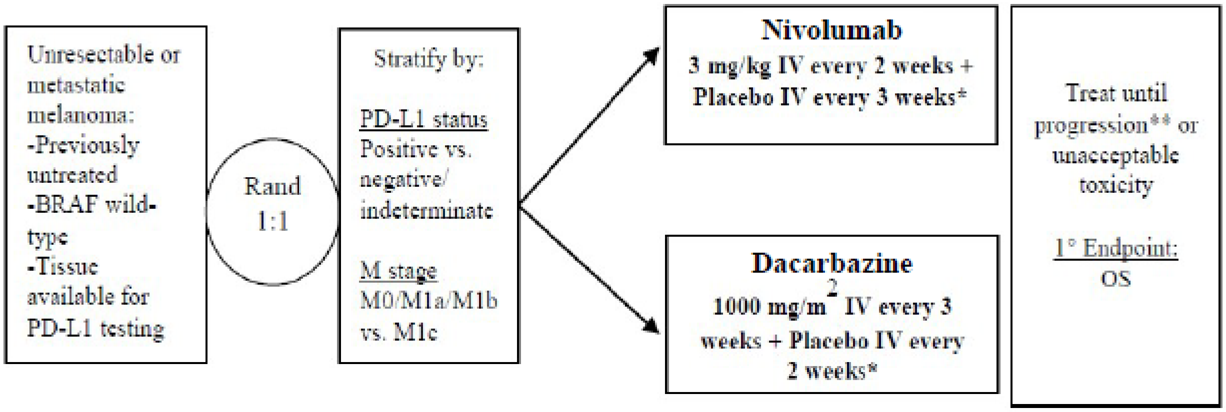
Based on above data and analyses across tumour types, 3 mg/kg IV Q2W was selected as the nivolumab monotherapy dose and schedule for all indications.

2.5.2. Main studies

Study CA209066: Phase 3, randomised, double-blind study of nivolumab vs dacarbazine in subjects with previously untreated, unresectable or metastatic melanoma

Methods

Figure 13: Study design schematic – Study CA209066



* Subjects who received nivolumab also received a dacarbazine-matched placebo, while subjects who received dacarbazine also received a nivolumab-matched placebo.

** Treatment beyond initial investigator-assessed RECIST v1.1-defined progression was considered in subjects experiencing investigator-assessed clinical benefit and tolerating study therapy. These subjects discontinued therapy when further progression was documented.

Study Participants

<u>Key Inclusion Criteria:</u>

- Men and women ≥ 18 years of age.
- Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1.
- Untreated, histologically confirmed unresectable Stage III or Stage IV melanoma, as per AJCC staging system.
- Note that prior adjuvant or neoadjuvant melanoma therapy is permitted if it was completed at least 6 weeks prior to randomisation and all related adverse events have either returned to baseline or stabilised.
- Measurable disease as per RECIST 1.1.
- Tumour tissue from an unresectable or metastatic site of disease must be provided for biomarker analyses. In order to be randomised, a subject must be classified as PD-L1 positive, PD-L1 negative, or PD-L1 indeterminate.
- Known BRAF wild-type as per regionally acceptable V600 mutational status testing. BRAF mutant subjects and those with indeterminate or unknown BRAF status are not permitted to randomise.

<u>Key Exclusion Criteria:</u>

- Active brain metastases or leptomeningeal metastases. Subjects with brain metastases were eligible if these had been treated and there was no magnetic resonance imaging (MRI) evidence of progression for at least 8 weeks after treatment was complete and within 28 days prior to first dose. There was also the requirement for no administration of immunosuppressive doses of systemic corticosteroids (> 10 mg/day prednisone equivalents) for at least 2 weeks prior to study drug administration.

- Ocular melanoma.
- Subjects with active, known or suspected autoimmune disease. Subjects with vitiligo, type I diabetes mellitus, residual hypothyroidism due to autoimmune condition only requiring hormone replacement therapy, psoriasis not requiring systemic treatment, or conditions not expected to recur in the absence of an external trigger were permitted to enroll.
- Prior malignancy active within the previous 3 years except for locally curable cancers that have been apparently cured.
- Subjects with a condition requiring systemic treatment with either corticosteroids (> 10 mg daily prednisone equivalents) or other immunosuppressive medications within 14 days of study drug administration.
- Prior treatment with an anti-PD-1, anti-PD-L1, anti-PD-L2, anti-CD137, or anti-CTLA-4
- Antibody, or any other antibody or drug specifically targeting T-cell co-stimulation or checkpoint pathways.

Subjects had to also meet other study criteria including exclusions for medical history, positive Hep B/C, HIV, and pregnancy tests, and other laboratory criteria.

Treatments

Subjects received one of the following treatments:

Nivolumab group: nivolumab at 3 mg/kg Q2W by IV infusion plus a dacarbazine-matched placebo Q3W by IV infusion.

Dacarbazine group: dacarbazine at 1000 mg/m^2 by IV infusion Q3W plus a nivolumab matched placebo Q2W by IV infusion.

Objectives

<u>Primary Objective</u>: To compare the clinical benefit, as measured by the duration of OS, provided by BMS-936558 (Nivolumab) vs. dacarbazine in subjects with previously untreated, unresectable or metastatic melanoma.

<u>Secondary Objectives</u>:

1. To compare the duration of investigator-assessed progression-free survival (PFS) of BMS-936558 (Nivolumab) vs dacarbazine in subjects with previously untreated, unresectable or metastatic melanoma.

2. To compare the investigator-assessed objective response rate (ORR) of BMS-936558 (Nivolumab) vs. dacarbazine in subjects with previously untreated, unresectable or metastatic melanoma.

3. To evaluate whether PD-L1 expression is a predictive biomarker for OS

4. To evaluate the Health Related Quality of Life (HRQoL) as assessed by European Organisation for Research and Treatment of Care (EORTC) QLQ-C30.

<u>Key Exploratory Objectives</u>

To evaluate duration of and time to objective response of nivolumab and dacarbazine.

To assess the overall safety and tolerability of nivolumab vs dacarbazine.

To characterize the immunogenicity of nivolumab.

Outcomes/endpoints

The primary endpoint was OS in all randomised subjects.

The secondary endpoints were investigator-assessed PFS and ORR per RECIST v1.1 criteria, and OS based on PD-L1 expression level.

Exploratory efficacy endpoints were time to response (TTR) and duration of response (DOR).

OS was defined as the time between the date of randomisation and the date of death. For subjects without documentation of death, OS was censored on the last date the subject was known to be alive.

Investigator-assessed PFS was defined as the time from randomisation to the date of the first documented progression or death due to any cause, whichever occurred first. Subjects who died without a reported progression were considered to have progressed on the date of their death. Subjects who did not progress or die were censored on the date of their last evaluable tumour assessment. Subjects who did not have any on study tumour assessments and did not die were censored on their date of randomisation. Subjects who started any subsequent anticancer therapy without a prior reported progression were censored on the date of their last evaluable tumor assessment prior to initiation of subsequent anti-cancer therapy.

Investigator-assessed ORR was defined as the number of subjects with a BOR of CR or PR divided by the number of randomised subjects for each treatment arm. The BOR is defined as the best response designation, as determined by the investigator using RECIST v1.1 without requirement of confirmation, recorded between the date of randomisation and the date of objectively documented progression per RECIST v1.1 or the date of subsequent therapy, whichever occurs first.

OS based on PD-L1 expression level defined as the percent of tumour cells demonstrating plasma membrane PD-L1 staining in a minimum of 100 evaluable tumour cells per a Dako PD-L1 IHC assay (this is referred to as quantifiable PD-L1 expression). In this CSR, OS was assessed based on PD-L1 status using a verified assay with ≥5% tumour cell membrane expression cutoff.

Tumour tissues (from metastatic or unresectable site) were collected at screening and assessed for PD-L1 status prior to randomisation for use as a stratification factor.

Evaluation of Target Lesions

- Complete Response (CR): Disappearance of all target lesions. Any pathological lymph nodes (whether target or non-target) must have reduction in short axis to < 10 mm.
- Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters.
- Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression).

- Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum diameters while on study.

Tumor assessments should first be performed at 9 weeks (± 1 wk) following randomisation then every 6 weeks (± 1 wk) thereafter for the first 12 months and then every 12 weeks (± 1 wk) until disease progression or treatment discontinuation.

Sample size

The study design required at least 312 deaths to ensure approximately 90% power to detect a hazard ratio (HR) of 0.69 with an overall type I error of 0.05 (two-sided). The HR of 0.69 corresponds to a 45% increase in the median OS, assuming a median OS of 10 months for dacarbazine and 14.49 months for nivolumab. One formal interim analysis was planned to be conducted. The stopping boundaries at the interim and final analyses will be derived based on the exact number of deaths using Lan-DeMets alpha spending function with O'Brien-Fleming boundaries. At the time of the database lock for analysis, 146 deaths were observed. The boundary for statistical significance was calculated based on the Lan-DeMets alpha spending function with O'Brien-Fleming type boundaries and required the unadjusted log rank p-value to be less than 0.0021 to conclude OS benefit for nivolumab relative to dacarbazine. The corresponding confidence interval is the 99.79% [(1-.0021)*100%] confidence interval presented in this analysis.

Randomisation

Patients who met all eligibility criteria were randomised by IVRS in a 1:1 ratio to the nivolumab group or the dacarbazine group, with stratification by:

- PD-L1 status
    - PD-L1 positive (≥ 5% tumour cell membrane staining in a minimum of a hundred evaluable tumour cells) vs
    - PD-L1 negative (≤ 5% tumour cell membrane staining in a minimum of a hundred evaluable tumour cells)/indeterminate (tumour cell membrane scoring hampered by high cytoplasmic staining or melanin content)
- M stage
    - M0/M1a/M1b vs
    - M1c

Blinding (masking)

The applicant, patients, investigators, and site staff were blinded to the study drug administered (nivolumab or dacarbazine).

Statistical methods

Standard methods of survival analysis were used. The primary analysis of OS in all randomised subjects was conducted using a two-sided logrank test stratified by PD-L1 status and M stage. The hazard ratio and corresponding two-sided (1-adjusted α)% confidence interval (CI) were estimated using a Cox proportional hazards model, with treatment arm as a single covariate, stratified by the above factors.

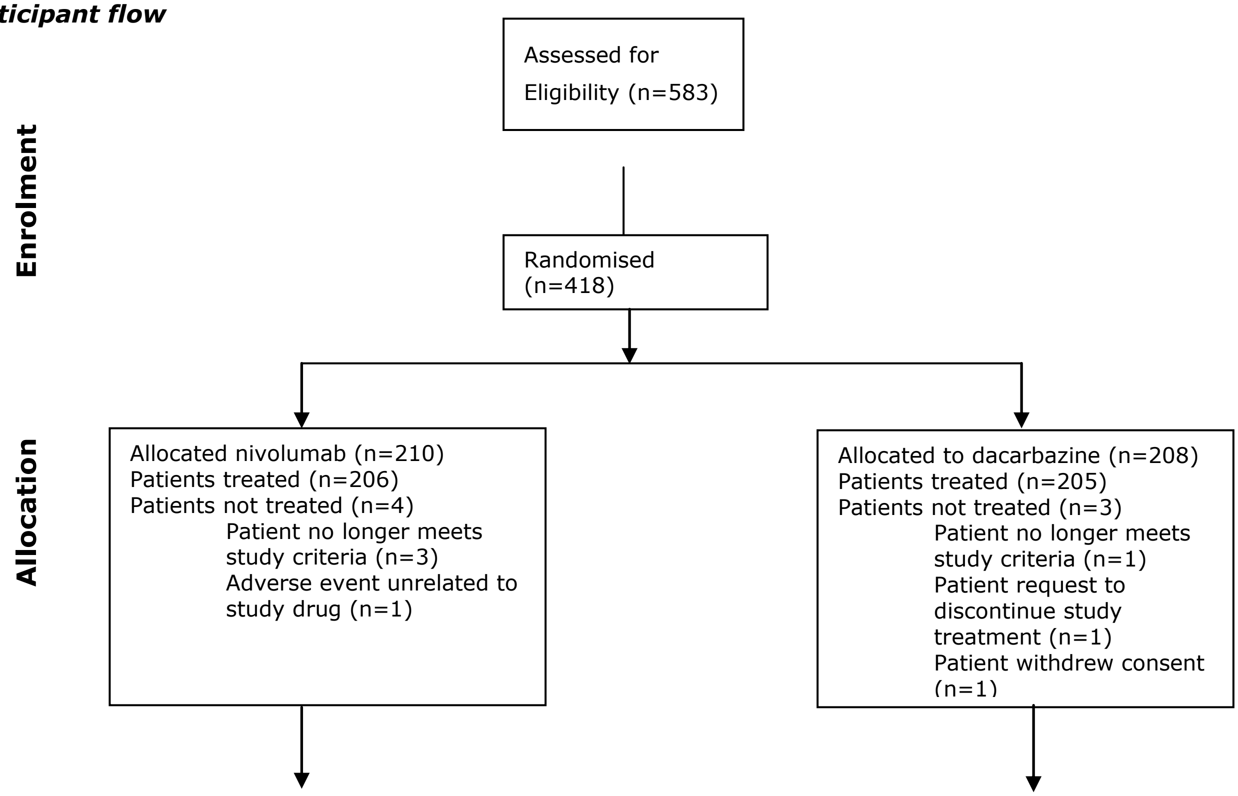
P-values other than those provided for the OS primary analysis and the hierarchical analysis of key secondary endpoints were for descriptive purpose only and not adjusted for multiplicity:

- If OS superiority in all randomised subjects was demonstrated, a gatekeeping testing approach for key secondary endpoints would be applied as described in the SAP. Key secondary endpoints included PFS in all subjects and ORR in all randomised subjects.
- Analysis of PFS in all randomised subjects was conducted using a two-sided log-rank test stratified by PD-L1 status and M stage. The hazard ratio and corresponding two-sided (1- adjusted α)% confidence interval was estimated using a Cox proportional hazards model, with treatment arm as a single covariate, stratified by the above factors. PFS curves, PFS medians with 95% CIs, and PFS rates at 6 and 12 months with 95% CIs was estimated using Kaplan- Meier methodology.
- Analysis of ORR was conducted using a Cochran-Mantel Haenszel (CMH) test stratified by the corresponding factors above for all randomised subjects. ORR estimates and corresponding 95%CIs, calculated using the Clopper-Pearson method, would be provided by treatment arm.

To evaluate PD-L1 expression as a predictive biomarker, a Cox proportional hazards model was to be used to test the interaction between PD-L1 expression (positive vs negative) and treatment arm for the OS endpoint. Additionally, OS was analysed within each PD-L1 expression subgroup (positive and negative) including log-rank tests and hazard ratios with corresponding confidence intervals. OS curves and medians was estimated using Kaplan-Meier methodology. These analyses will be descriptive and not adjusted for multiplicity. Other exploratory analyses, such as associations between PD-L1 expression and other efficacy endpoints and evaluations of different thresholds for PD-L1 positivity, were also planned.

Results

Participant flow



Follow-up

Discontinued nivolumab (n=100)	Discontinued dacarbazine (n=180)
n=81 progression	n=159 progression
n=0 deaths	n=2 deaths
n=4 study drug toxicity	n= 8 study drug toxicity
n=3 AE unrelated to study drug	n=2 AE unrelated to study drug
n=5 patient request to discontinue study treatment	n=1 patient request to discontinue study treatment
n=2 withdrew consent	n=4 withdrew consent
n=1 maximum clinical benefit	n=0 maximum clinical benefit
n=4 other	n=4 other
Still on therapy (n=110)	Still on therapy (n=28)

Recruitment

Patients were enrolled from January 2013 until February 2014. This study was conducted at 76 sites in 16 countries (Argentina, Australia, Canada, Chile, Denmark, Finland, France, Germany, Greece, Israel, Italy, Mexico, Norway, Poland, Spain, and Sweden).

Conduct of the study

An independent DMC was established to provide oversight of safety and efficacy considerations, study conduct, and risk-benefit ratio. Following review, the DMC was to recommend continuation, modification, or discontinuation of this study based on reported safety and efficacy data.

During the first DMC safety review on 05-Nov-2013, the committee noted a potential difference in OS and requested to receive monthly death summaries starting in December 2013 and Kaplan- Meier curves of OS starting in Jan-2014. With the 5-May-2014 data transfer, the report included a table with median survival times along with the hazard ratio and its corresponding 95% confidence interval. At the 10-Jun-2014 meeting, the DMC requested the O'Brien-Fleming boundary be given for the fraction of events presented in the June summary report.

The plan for the study as described in the DMC charter included only a single interim review of OS scheduled to occur when approximately 218 deaths had occurred (70% of the total 312 deaths). This was expected to occur in Mar-2015. The transfer used for the Jun-2014 report included a total of 110 deaths (35% of the 312 total expected deaths). The O'Brien-Fleming boundary based on a Lan-DeMets spending function calculated in East 6.2 software for 110 deaths would have a p-value of 0.00032 (hazard ratio of 0.50). The log-rank p-value based on the 110 deaths is 0.000085. Assuming two interim analyses (eg, one at 110 deaths and one at 218 deaths), the nominal significance level would remain approximately unchanged at 0.0455 (hazard ratio of 0.80) for the final OS analysis at 312 deaths. (The plan in the protocol with a single interim look at 218 deaths had a nominal significance level of 0.0455 for the final OS analysis at 312 deaths.)

Based on the 10-Jun-2014 review of data from the DMC-requested 23-May-2014 database lock, the DMC informed BMS that there was clear evidence of a survival benefit in subjects receiving nivolumab compared to dacarbazine. As a result, the DMC recommended that the study be unblinded and subjects randomised to dacarbazine who had ended study treatment be allowed to receive treatment with nivolumab. They did allow for those subjects deriving benefit from dacarbazine to remain on current therapy. The efficacy data from CA09066 that is presented is based on a clinical cutoff date of 24-Jun-2014, prior to any dacarbazine subjects being treated with nivolumab.

Changes to the study protocol was based on 6 amendments.

Table 25: Protocol amendments – Study CA209066

Document	Site(s)	Date of Issue	No of Subjects Treated at Time of Amendment
Amendment No. 01	All sites	20-Sep-2012	0
	• Permit the collection and storage of blood samples for use in future exploratory pharmacogenetic research. Bristol-Myers Squibb will use DNA obtained from the blood sample and health information collected to study the association between genetic variation and drug response.		
Amendment No. 02	Country-specific: All sites in France	25-Jan-2013	1
	• Yellow fever vaccine, live attenuated vaccines, and prophylactic phenytoin, added as exclusion criteria (if received within 28 days of first dose of study drug) and as prohibited and/or restricted medications. • A urinalysis procedure (to include protein and urine sediment) added at the baseline/screening visit to ensure appropriate monitoring of renal function of subjects participating in the study. • Appendix 4 was developed to include recommended management algorithms for suspected pulmonary toxicity, diarrhea/colitis, suspected hepatotoxicity, suspected endocrinopathy and nephrotoxicity previously located only in the Investigator Brochure.		
Amendment No. 03	All sites	07-Mar-2013	8
	• Update the Summary of Safety section to include new preliminary reproductive toxicology data that was distributed as a Non-clinical Expedited Safety Report and to include changes to the guidance on contraception.		
Amendment No. 04	Country-specific: All sites in Spain	20-Mar-2013	14
	• Matching placebo IV solution (0.9% sodium chloride or 5% glucose) is to be considered as investigational product.		
Amendment No. 05	Country-specific: All sites in Germany	09-Apr-2013	18
	• Adjust the exclusion criteria to ensure that HIV positive subjects are excluded from the study. • Added HIV test to Laboratory Tests at screening for subjects enrolled in Germany.		
Amendment No. 06	All sites	08-May-2013	49
	• Facilitate accrual by removing the specified timeframe around the timing of tumor tissue collection.		

After the clinical cutoff date (24-Jun-2014), the protocol was amended on 09-Jul-2014 per DMC recommendations to allow subjects in the follow-up phase of the study who had received and discontinued from dacarbazine to receive nivolumab, as well as to provide a mechanism for subjects who were receiving dacarbazine to receive nivolumab upon discontinuation of dacarbazine.

Baseline data

Among all randomised subjects, the mean age was 62.7 (range: 18, 87) years and the majority of subjects were White (99.5%) and male (58.9%). The median duration of time from initial diagnosis was 1.9 (range: 0.1-32.6) years in the nivolumab group and 1.7 (range: 0.1-22.2) years in the dacarbazine group.

All subjects have BRAF wild type melanoma and advanced disease (as specified by the protocol). A high percentage of subjects had poor prognostic factors, including 55.7% M1c stage. Across treatment groups, the most common visceral sites of disease included lung (60.5% of subjects) and liver (30.9% of subjects). The majority of subjects should have a baseline ECOG performance status of 0 or 1 (as specified in the protocol). Prior adjuvant therapy, neo-adjuvant therapy, surgery related to cancer, and radiotherapy were balanced across treatment groups.

Table 25: Baseline Disease Characteristics and Tumour Assessments (All Randomised Subjects) – Study CA209066

Demographics	*Group A* *N=210*	*Group B* *N=208*	*Total* *N=418*
Age (years)			
Mean (SD)	61.6 (13.0)	63.7 (12.6)	62.7 (12.8)
Range (min, max)	18, 86	25, 87	18, 87
Quartiles (25th, median, 75th)	52, 64, 70	57, 66, 73	55, 65, 72
Age category, n (%)			
< 65	106 (50.5)	94 (45.2)	200 (47.8)
65-85	103 (49.0)	111 (53.4)	214 (51.2)
> 85	1 (0.5)	3 (1.4)	4 (1.0)
Sex, n (%)			
Male	122 (58.1)	124 (59.6)	246 (58.9)
Female	88 (41.9)	84 (40.4)	172 (41.1)
Race, n (%)			
White	209 (99.5)	207 (99.5)	416 (99.5)
Asian	0	1 (0.5)	1 (0.2)
Other	1 (0.5)	0	1 (0.2)

Column header counts and denominators are the number of subjects randomized.
Denominators are the number of subjects with non-missing data for each characteristic.

	Group A N=210 n (%)	Group B N=208 n (%)	Total N=418 n (%)
Baseline disease stage			
Stage III	30 (14.3)	24 (11.5)	54 (12.9)
Stage IV	180 (85.7)	184 (88.5)	364 (87.1)
Baseline M status			
M0	21 (10.0)	24 (11.5)	45 (10.8)
M1a	22 (10.5)	22 (10.6)	44 (10.5)
M1b	47 (22.4)	49 (23.6)	96 (23.0)
M1c	120 (57.1)	113 (54.3)	233 (55.7)
Years from initial diagnosis to randomization			
N	210	208	418
Median	1.9	1.7	1.8
Min, Max	0.1, 32.6	0.1, 22.2	0.1, 32.6
<1	69 (32.9)	61 (29.3)	130 (31.1)
1 - <2	37 (17.6)	51 (24.5)	88 (21.1)
2 - <3	27 (12.9)	24 (11.5)	51 (12.2)
3 - <4	24 (11.4)	18 (8.7)	42 (10.0)
4 - <5	13 (6.2)	10 (4.8)	23 (5.5)
>5	40 (19.0)	44 (21.2)	84 (20.1)

Column header counts and denominators are the number of subjects randomized.

	Group A N=210	Group B N=208	Total N=418
Subjects with at least one lesion, n (%)	**208 (99.0)**	**208 (100.0)**	**416 (99.5)**
Site of target and non-target lesions			
Visceral, lung	127 (60.5)	126 (60.6)	253 (60.5)
Lymph node	117 (55.7)	121 (58.2)	238 (56.9)
Visceral, liver	69 (32.9)	60 (28.8)	129 (30.9)
Soft tissue	40 (19.0)	57 (27.4)	97 (23.2)
Visceral, other	31 (14.8)	51 (24.5)	82 (19.6)
Skin	23 (11.0)	37 (17.8)	60 (14.4)
Bone	25 (11.9)	18 (8.7)	43 (10.3)
Intestine	7 (3.3)	3 (1.4)	10 (2.4)
Other	56 (26.7)	47 (22.6)	103 (24.6)
Total number of target and non-target lesion sites, n (%)			
1	47 (22.4)	49 (23.6)	96 (23.0)
2	77 (36.7)	63 (30.3)	140 (33.5)
3	48 (22.9)	54 (26.0)	102 (24.4)
4	29 (13.8)	31 (14.9)	60 (14.4)
≥5	7 (3.3)	11 (5.3)	18 (4.3)
Sum of longest diameter of target lesions (mm)			
N	208	207	415
Median	54	63	60
Min, Max	10, 237	10, 321	10, 321

Column header counts and denominators are the number of subjects randomized. Subjects are counted at most once in each row.

Table 26: Prior medications – Study CA209066

Setting /prior systemic regimen (verbatim)	*Group A N=210 n (%)*	*Group B N=208 n (%)*	*Total N=418 n (%)*
Subjects with prior regimen in the adjuvant setting	**32 (15.2)**	**37 (17.8)**	**69 (16.5)**
Interferon	17 (8.1)	22 (10.6)	39 (9.3)
Interferon alfa 2b	3 (1.4)	6 (2.9)	9 (2.2)
Interferon alfa	1 (0.5)	2 (1.0)	3 (0.7)
Interferon alfa 2a	3 (1.4)	0	3 (0.7)
Cisplatin	2 (1.0)	0	2 (0.5)
Investigational immunotherapy	1 (0.5)	1 (0.5)	2 (0.5)
Ipilimumab	1 (0.5)	1 (0.5)	2 (0.5)
Anastrozole/Cyphos/epirub/flur/Letrozole	0	1 (0.5)	1 (0.2)
Bleomycin/Cisplatin	0	1 (0.5)	1 (0.2)
Dacarbazine	0	1 (0.5)	1 (0.2)
Dactinomycin/Melphalan	1 (0.5)	0	1 (0.2)
Imatinib	0	1 (0.5)	1 (0.2)
Interferon alfa 2a/Interferon alfa 2b	1 (0.5)	0	1 (0.2)
Investigational drug	1 (0.5)	0	1 (0.2)
Levamisol	1 (0.5)	0	1 (0.2)
Melphalan/Tumor necrosis factor	0	1 (0.5)	1 (0.2)
Mitotane	1 (0.5)	0	1 (0.2)
Subjects with prior regimen in the neo-adjuvant setting	**1 (0.5)**	**1 (0.5)**	**2 (0.5)**
Dacarbazine/Interferon	1 (0.5)	0	1 (0.2)
Investigational antineoplastic	0	1 (0.5)	1 (0.2)
Subjects with prior regimen in the metastatic setting	**0**	**0**	**0**

Column header counts and denominators are the number of subjects randomized. A subject is counted at most once in each row.

Numbers analysed

Of the 583 subjects who enrolled, 418 subjects were randomized (210 nivolumab group and 208 dacarbazine group) and 411 subjects were treated (206 nivolumab group and 205 dacarbazine group. Seven (1.7%) subjects were not treated because they failed to meet study eligibility criteria, withdrew consent, were not compliant, or had an AE preventing them from starting study treatment.

Table 27: Analysis Population – Study CA209066

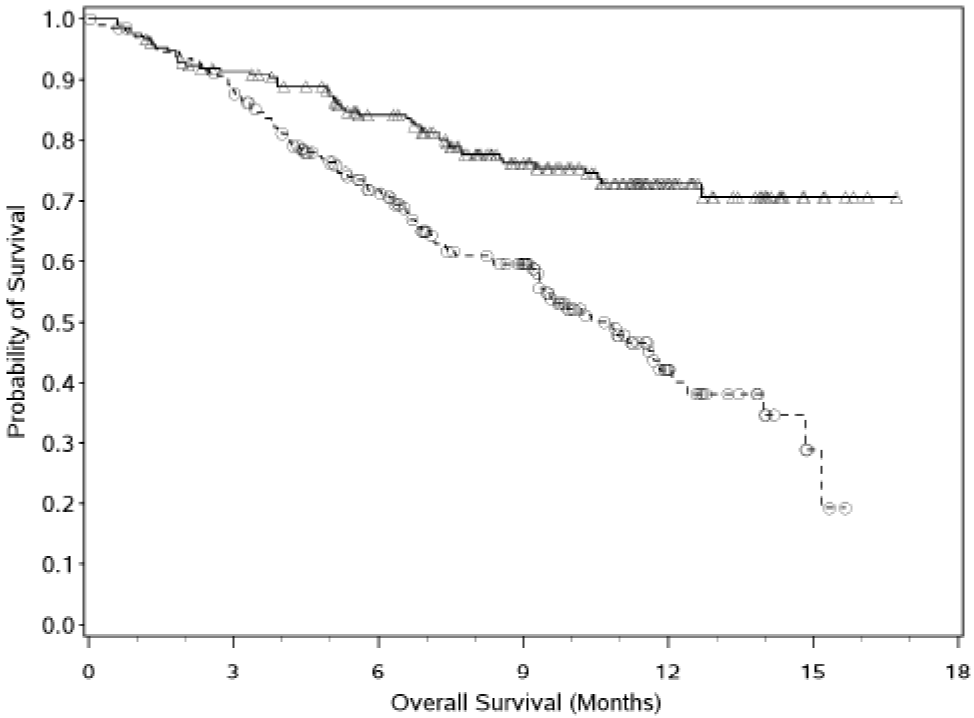
Population	Nivolumab Group N	Dacarbazine Group N	Total N
All Enrolled Subjects: All subjects who signed an informed consent form and were registered into the IVRS.	NA	NA	583
All Randomized Population: All subjects who were randomized to any treatment arm in the study. This is the primary dataset for analyses of study conduct, study population and efficacy.	210	208	418
All Treated Population: All subjects who received at least one dose of nivolumab, nivolumab-placebo, dacarbazine, or dacarbazine-placebo. This is the primary dataset for analyses of exposure and safety.	206	205	411
Response-Evaluable Subjects: All randomized subjects with measurable disease at a baseline tumor assessment and at least one on-treatment tumor assessment.	180	163	343
Immunogenicity Subjects: All nivolumab-treated subjects with baseline and at least one post-baseline assessment for ADA	107	NA	NA

Outcomes and estimation

<u>Overall survival</u>

The OS data are presented in Figure 14. Median OS was not reached in the nivolumab group and was 10.84 months in the dacarbazine group.

Figure 14: Kaplan-Meier Overall survival plot (all randomised patients) - Study CA209066



Number of Subjects at Risk

Nivolumab							
	210	185	150	105	45	8	0
Dacarbazine							
	208	177	123	82	22	3	0

—△— Nivolumab (events : 50/210), median and 95% CI : N.A.

- -⊖- - Dacarbazine (events : 96/208), median and 95% CI : 10.84 (9.33, 12.09)

Hazard Ratio (Nivolumab over Dacarbazine) and 95% CI (1): 0.42 (0.30, 0.60)
Hazard Ratio (Nivolumab over Dacarbazine) and 99.79% CI (1): 0.42 (0.25, 0.73)
Stratified log-rank p-value: <0.0001

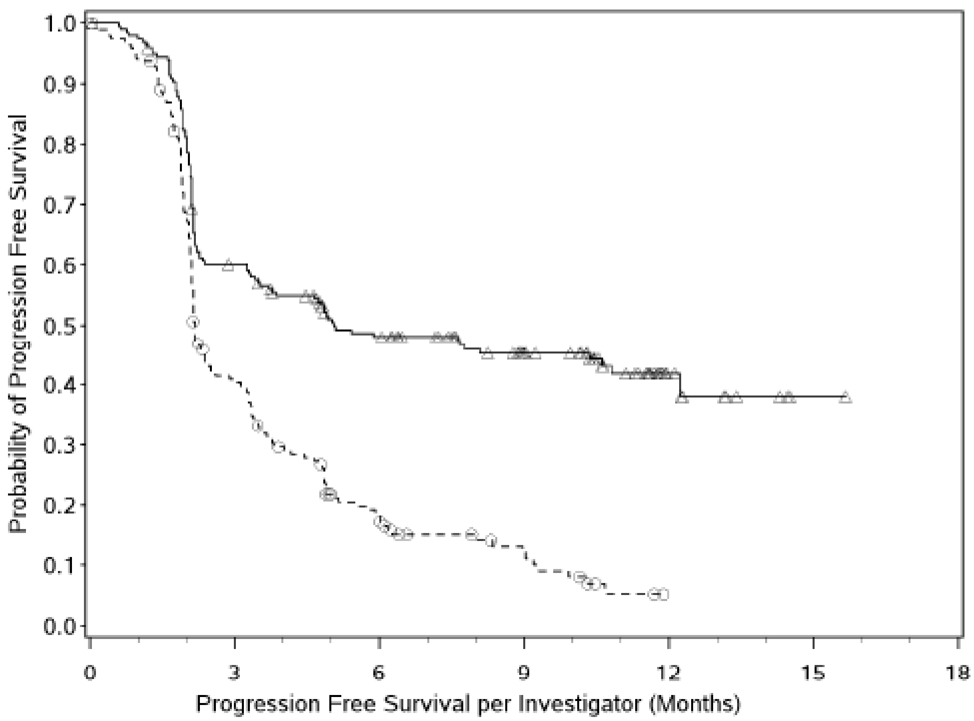
The boundary for statistical significance requires the p (1) from stratified Cox proportional hazard model using randomized arm as a single covariate. Symbols represent censored observations.

NOTE: Two-sided, 95% CI for median OS were computed by Brookmeyer and Crowley method (log log transformation). The HR and the corresponding CI were estimated in a stratified Cox proportional hazard model using randomized arm as a single covariate.

<u>Progression free survival</u>

The results of PFS are presented in Figure 15. The median PFS was 5.06 months (95% CI: 3.48, 10.81) in the nivolumab group and 2.17 months (95% CI: 2.10, 2.40) in the dacarbazine group. PFS rates for the nivolumab and dacarbazine groups at 6 months were 48.0% vs 18.5%, respectively, and at 12 months were 41.8% vs not produced (as all PFS times were less than 12 months for the dacarbazine group), respectively . There were 102 (48.6%) subjects in the nivolumab (21.6%) subjects in the dacarbazine group censored, with the most common reasons for censoring listed as 'still on treatment' for the nivolumab group (37.6%) and 'received subsequent therapy' for the dacarbazine group (12.0%).

Figure 15: Kaplan-Meier plot of progression free survival (all randomised patients) - Study CA209066



Number of Subjects at Risk

Nivolumab							
	210	116	82	57	12	1	0
Dacarbazine							
	208	74	28	12	0	0	0

—△— Nivolumab (events : 108/210), median and 95% CI : 5.06 (3.48, 10.81)

- -○- - Dacarbazine (events : 163/208), median and 95% CI : 2.17 (2.10, 2.40)

Hazard Ratio (Nivolumab over Dacarbazine) and 95% CI (1): 0.43 (0.34, 0.56)

Stratified log-rank p-value: <0.0001

(1) from stratified Cox proportional hazard model using randomized arm as a single covariate.
Symbols represent censored observations.

Overall response

The ORR data are presented in Table 28.

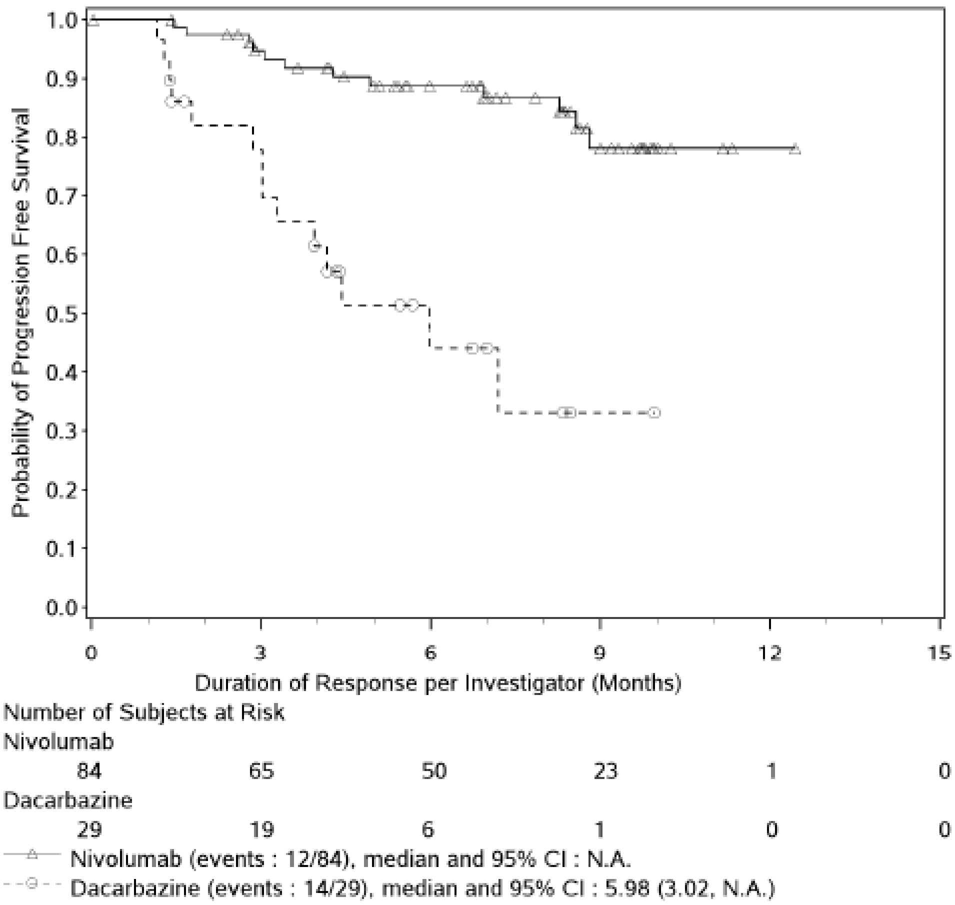
Table 28: Best overall response (all randomised subjects) - Study CA209066

	Nivolumab N = 210	Dacarbazine N = 208
BEST OVERALL RESPONSE (A):		
COMPLETE RESPONSE (CR)	16 (7.6)	2 (1.0)
PARTIAL RESPONSE (PR)	68 (32.4)	27 (13.0)
STABLE DISEASE (SD)	35 (16.7)	46 (22.1)
PROGRESSIVE DISEASE (PD)	69 (32.9)	101 (48.6)
UNABLE TO DETERMINE (UTD)	22 (10.5)	32 (15.4)
OBJECTIVE RESPONSE RATE (1) (95% CI)	84/210 (40.0%) (33.3, 47.0)	29/208 (13.9%) (9.5, 19.4)
DIFFERENCE OF OBJECTIVE RESPONSE RATES (2) (95% CI)	26.1% (18.0, 34.1)	
ESTIMATE OF ODDS RATIO (3) (4) (95% CI)	4.06 (2.52, 6.54)	
P-VALUE (5)	<0.0001	

Duration of response

The duration of response is presented in Figure 16.

Figure 16: Kaplan-Meier plot of duration of response (all randomised patients) - Study CA209066



Ancillary analyses

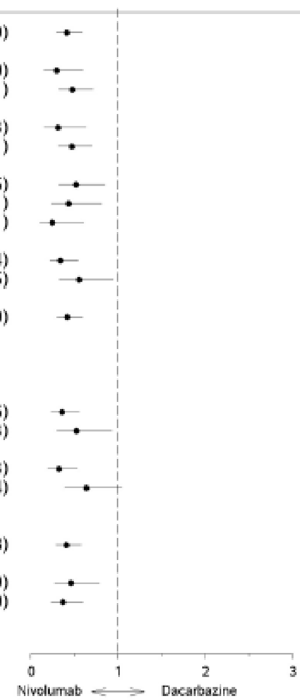
Subpopulations analyses

Overall survival

Subgroup analyses assessing the impact of age, gender, race, region, baseline ECOG performance status, PD-L1 expression status, M stage at study entry, history of brain metastases, smoking status, baseline LDH, AJCC stage, or prior neo-adjuvant or adjuvant therapy were also conducted and are shown in Figures 17, 18 and19.

Figure 17: Forest plot of treatment effect on OS in pre-defined subsets (all randomised patients) - Study CA209066

	N	Nivolumab N of Events (N of subjects)	Nivolumab mOS	Nivolumab (95% CI)	Dacarbazine N of Events (N of subjects)	Dacarbazine mOS	Dacarbazine (95% CI)	Unstratified Hazard Ratio	(95% CI)
Overall	418	50(210)	N.A.		96(208)	10.84	(9.33, 12.09)	0.42	(0.30, 0.59)
PD-L1 Status									
Positive	148	11(74)	N.A.		29(74)	12.39	(9.17, N.A.)	0.30	(0.15, 0.60)
Negative/Indeterminate	270	39(136)	N.A.		67(134)	10.22	(7.59, 11.83)	0.48	(0.32, 0.71)
M Stage at Study Entry									
M0/M1A/M1B	163	11(82)	N.A.		29(81)	12.39	(9.56, N.A.)	0.31	(0.16, 0.63)
M1C	255	39(128)	N.A.		67(127)	9.33	(6.64, 11.17)	0.47	(0.32, 0.71)
Age Category									
< 65	200	28(106)	N.A.		40(94)	11.17	(9.17, N.A.)	0.52	(0.32, 0.85)
>= 65 and < 75	151	16(77)	N.A.		30(74)	11.70	(9.33, 15.18)	0.44	(0.24, 0.81)
>= 75	67	6(27)	N.A.		26(40)	6.64	(4.86, 9.33)	0.25	(0.10, 0.61)
Gender									
Male	246	27(121)	N.A.		62(125)	9.92	(8.38, 11.70)	0.34	(0.22, 0.54)
Female	172	23(89)	N.A.		34(83)	12.39	(7.59, N.A.)	0.56	(0.33, 0.95)
Race									
White	416	50(209)	N.A.		95(207)	10.41	(9.26, 12.39)	0.42	(0.30, 0.60)
Black	0	0(0)	N.A.		0(0)	N.A.			
Asian	1	0(0)	N.A.		1(1)	12.09	(N.A., N.A.)		
Other	1	0(1)	N.A.		0(0)	N.A.			
Region									
W.Europe+Canada	290	29(145)	N.A.		66(145)	9.92	(8.34, 13.96)	0.36	(0.23, 0.56)
Rest of World	128	21(65)	N.A.	(12.68, N.A.)	30(63)	11.17	(7.59, 14.85)	0.53	(0.30, 0.93)
Baseline ECOG Performance Status									
0	269	23(148)	N.A.		48(121)	11.83	(9.59, 15.18)	0.32	(0.20, 0.53)
1	144	26(60)	12.68	(7.46, N.A.)	46(84)	7.43	(5.16, 11.70)	0.64	(0.40, 1.04)
History of Brain Metastases									
Yes	15	1(7)	N.A.	(0.59, N.A.)	2(8)	N.A.	(3.35, N.A.)		
No	403	49(203)	N.A.		94(200)	10.41	(9.26, 12.09)	0.41	(0.29, 0.58)
Smoking Status									
Yes	163	24(90)	N.A.	(12.68, N.A.)	33(73)	10.22	(9.33, 15.18)	0.46	(0.27, 0.79)
No	232	25(111)	N.A.		58(121)	10.41	(7.39, 12.09)	0.37	(0.23, 0.60)



	N	Nivolumab N of Events (N of subjects)	Nivolumab mOS	Nivolumab (95% CI)	Dacarbazine N of Events (N of subjects)	Dacarbazine mOS	Dacarbazine (95% CI)	Unstratified Hazard Ratio	(95% CI)
Baseline LDH 1									
<= ULN	245	16(120)	N.A.		38(125)	14.85	(11.17, N.A.)	0.35	(0.19, 0.62)
> ULN	153	29(79)	N.A.	(8.51, N.A.)	53(74)	6.31	(4.21, 8.38)	0.41	(0.26, 0.65)
Baseline LDH 2									
<= 2*ULN	355	35(178)	N.A.		73(177)	11.17	(9.56, 14.85)	0.38	(0.25, 0.57)
> 2*ULN	43	10(21)	10.28	(1.84, N.A.)	18(22)	2.58	(0.95, 4.21)	0.44	(0.20, 0.96)
Prior Neo-adjuvant or Adjuvant Therapy									
Neo-adjuvant	2	0(1)	N.A.		1(1)	7.43	(N.A., N.A.)		
Adjuvant	68	4(32)	N.A.		13(36)	12.09	(8.34, N.A.)	0.29	(0.10, 0.91)
Time from Completion of Prior Adjuvant Therapy to Randomization									
< 6 Months	19	1(7)	N.A.	(7.26, N.A.)	3(12)	11.60	(4.17, N.A.)		
>= 6 Months	49	3(25)	N.A.		10(24)	12.09	(5.16, N.A.)	0.24	(0.07, 0.88)
AJCC Stage at Study Entry									
Stage III	49	4(27)	N.A.		9(22)	11.60	(7.13, 15.18)	0.35	(0.10, 1.16)
Stage IV	369	46(183)	N.A.		87(186)	10.22	(8.38, 12.39)	0.44	(0.30, 0.62)

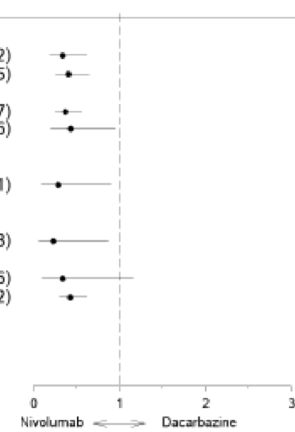
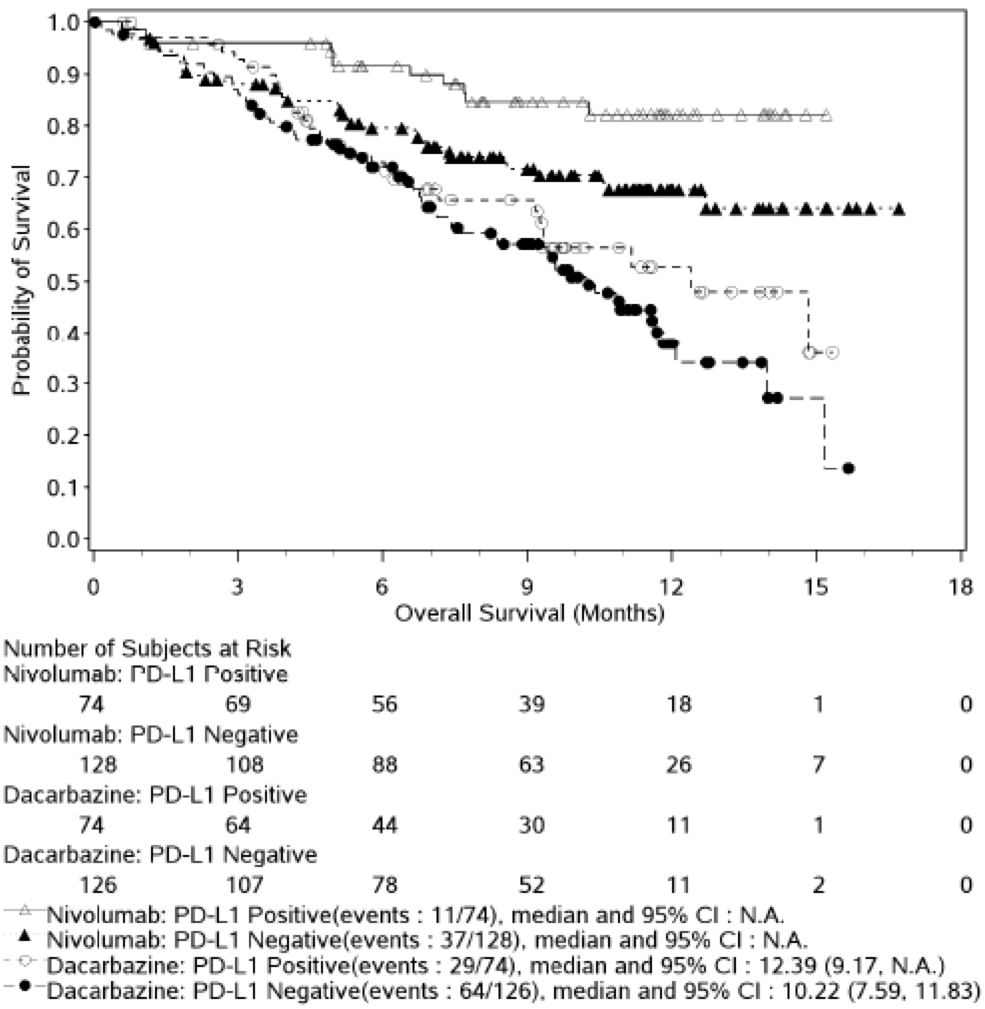


Figure 18: Kaplan-Meier plot of OS by PD-L1 expression status (all randomised patients) - Study CA209066

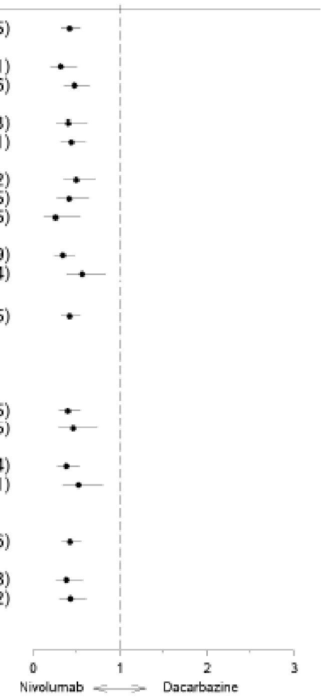


<u>Progression free survival</u>

Progression-free survival benefit was observed across subgroups with nivolumab vs dacarbazine and in both PD-L1 positive and negative (HR=0.33; 95% CI: 0.21, 0.51) and PD-L1 negative/indeterminate subjects (HR=0.49; 95% CI: 0.36, 0.66).

Figure 19: Forest plot of treatment effect on PFS in pre-defined subsets (all randomised patients) - Study CA209066

	N	Nivolumab N of Events (N of subjects)	Nivolumab mPFS	Nivolumab mPFS (95% CI)	Dacarbazine N of Events (N of subjects)	Dacarbazine mPFS	Dacarbazine mPFS (95% CI)	Unstratified Hazard Ratio	Unstratified Hazard Ratio (95% CI)
Overall	418	108(210)	5.06	(3.48, 10.81)	163(208)	2.17	(2.10, 2.40)	0.43	(0.33, 0.55)
PD-L1 Status									
Positive	148	33(74)	10.61	(4.80, N.A.)	55(74)	2.17	(2.10, 2.50)	0.33	(0.21, 0.51)
Negative/Indeterminate	270	75(136)	4.67	(2.17, 7.75)	108(134)	2.14	(2.10, 3.22)	0.49	(0.36, 0.66)
M Stage at Study Entry									
M0/M1A/M1B	163	40(82)	8.08	(4.80, N.A.)	57(81)	2.17	(2.07, 3.81)	0.41	(0.27, 0.63)
M1C	255	68(128)	3.88	(2.17, N.A.)	106(127)	2.14	(2.10, 2.40)	0.45	(0.33, 0.61)
Age Category									
< 65	200	59(106)	4.67	(2.20, 10.61)	75(94)	2.17	(2.10, 3.52)	0.51	(0.36, 0.72)
>= 65 and < 75	151	37(77)	7.62	(2.33, N.A.)	59(74)	2.17	(2.10, 3.06)	0.43	(0.28, 0.65)
>= 75	67	12(27)	12.22	(2.10, N.A.)	29(40)	2.10	(1.87, 2.89)	0.27	(0.13, 0.55)
Gender									
Male	246	55(121)	7.75	(4.73, N.A.)	102(125)	2.14	(2.07, 2.37)	0.35	(0.25, 0.49)
Female	172	53(89)	3.42	(2.14, 10.35)	61(83)	2.17	(2.10, 3.38)	0.58	(0.40, 0.84)
Race									
White	416	108(209)	5.06	(3.48, 10.81)	163(207)	2.17	(2.10, 2.40)	0.43	(0.33, 0.55)
Black	0	0(0)	N.A.		0(0)	N.A.			
Asian	1	0(0)	N.A.		0(1)	N.A.			
Other	1	0(1)	N.A.		0(0)	N.A.			
Region									
W.Europe+Canada	290	76(145)	5.03	(3.25, 10.81)	117(145)	2.14	(2.10, 2.17)	0.41	(0.30, 0.55)
Rest of World	128	32(65)	5.42	(2.14, N.A.)	46(63)	3.35	(2.00, 4.17)	0.47	(0.30, 0.75)
Baseline ECOG Performance Status									
0	269	71(148)	7.75	(3.88, N.A.)	99(121)	2.17	(2.14, 3.25)	0.39	(0.29, 0.54)
1	144	36(60)	3.52	(2.10, 5.42)	62(84)	2.10	(1.91, 2.60)	0.53	(0.35, 0.81)
History of Brain Metastases									
Yes	15	3(7)	N.A.	(0.59, N.A.)	7(8)	3.52	(1.87, 4.90)		
No	403	105(203)	5.06	(3.42, 10.81)	156(200)	2.14	(2.10, 2.37)	0.43	(0.34, 0.56)
Smoking Status									
Yes	163	49(90)	4.90	(2.14, 7.75)	63(73)	2.10	(1.94, 2.14)	0.39	(0.27, 0.58)
No	232	56(111)	5.09	(3.32, N.A.)	92(121)	2.30	(2.14, 3.29)	0.44	(0.31, 0.62)



	N	Nivolumab N of Events (N of subjects)	Nivolumab mPFS	Nivolumab mPFS (95% CI)	Dacarbazine N of Events (N of subjects)	Dacarbazine mPFS	Dacarbazine mPFS (95% CI)	Unstratified Hazard Ratio	Unstratified Hazard Ratio (95% CI)
Baseline LDH 1									
<= ULN	245	56(120)	10.61	(5.03, N.A.)	98(125)	2.37	(2.14, 3.35)	0.35	(0.25, 0.49)
> ULN	153	49(79)	2.14	(2.04, 3.32)	60(74)	2.10	(1.91, 2.37)	0.60	(0.41, 0.87)
Baseline LDH 2									
<= 2*ULN	355	89(178)	7.66	(4.73, N.A.)	140(177)	2.17	(2.14, 3.06)	0.39	(0.30, 0.52)
> 2*ULN	43	16(21)	1.95	(1.68, 2.00)	18(22)	1.54	(0.92, 2.50)	0.89	(0.45, 1.76)
Prior Neo-adjuvant or Adjuvant Therapy									
Neo-adjuvant	2	1(1)	5.03	(N.A., N.A.)	1(1)	4.86	(N.A., N.A.)		
Adjuvant	68	17(32)	4.90	(2.14, N.A.)	30(36)	2.07	(1.87, 2.17)	0.36	(0.19, 0.67)
Time from Completion of Prior Adjuvant Therapy to Randomization									
< 6 Months	19	4(7)	1.97	(1.77, N.A.)	10(12)	2.04	(1.41, 4.44)		
>= 6 Months	49	13(25)	8.08	(2.17, N.A.)	20(24)	2.09	(1.87, 2.37)	0.32	(0.15, 0.66)
AJCC Stage at Study Entry									
Stage III	49	13(27)	12.22	(2.10, 12.22)	13(22)	3.29	(2.07, 9.20)	0.61	(0.28, 1.36)
Stage IV	369	95(183)	5.06	(3.48, 10.61)	150(186)	2.14	(2.10, 2.37)	0.41	(0.32, 0.53)

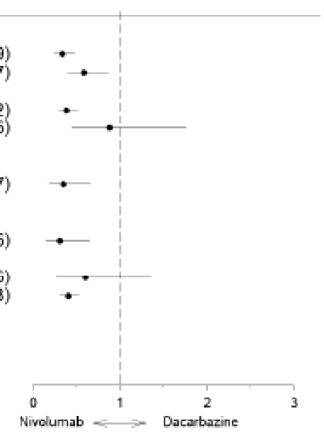
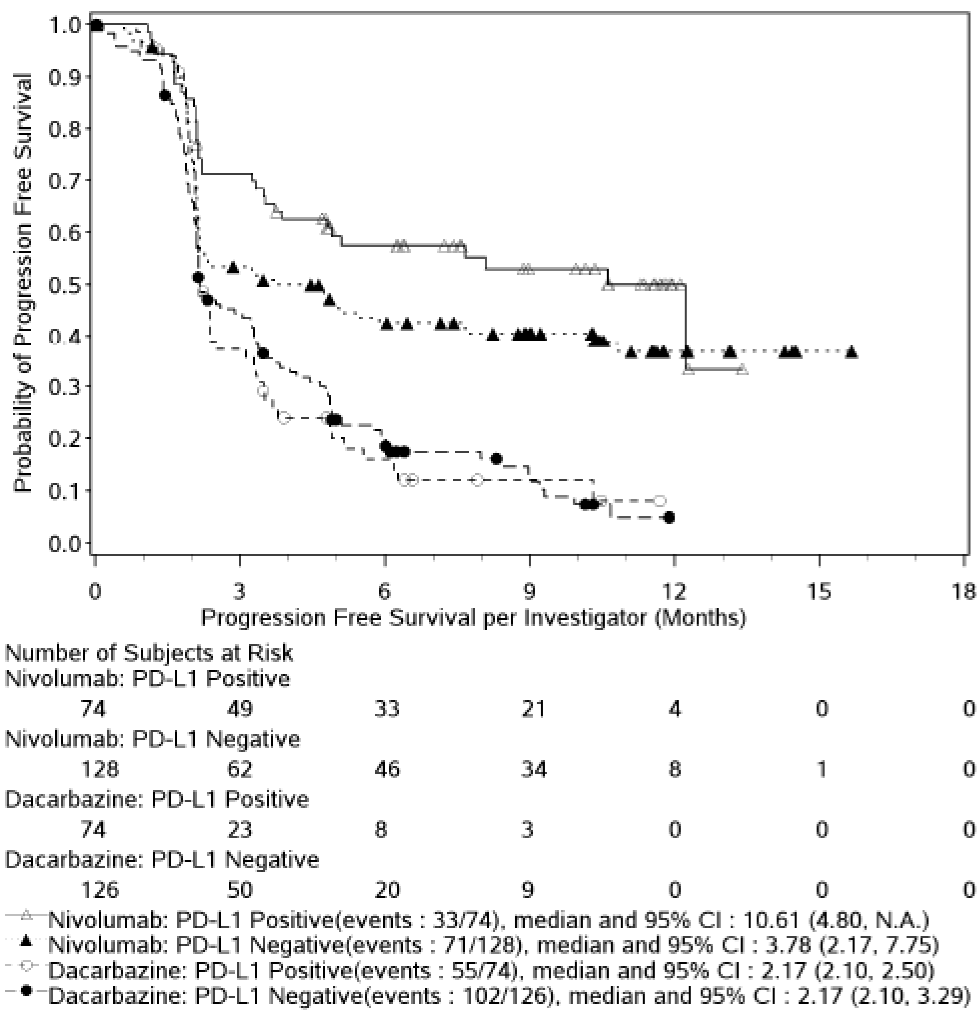


Figure 20: Kaplan-Meier plot of PFS by PD-L1 expression status (all randomised patients) - Study CA209066



Summary of main study

The following tables summarise the efficacy results from the main studies supporting the present application. These summaries should be read in conjunction with the discussion on clinical efficacy as well as the benefit risk assessment (see later sections).

The results of the primary, secondary, and a subset of exploratory efficacy endpoints are summarised in Table 29.

Table 29: Summary of Efficacy for Study CA209066

Title: Study CA209066: Phase 3, randomised, double-blind study of nivolumab vs dacarbazine in subjects with previously untreated, unresectable or metastatic melanoma

Study identifier	CA209066		
Design	Phase 3, randomized, open-label, multicenter study of nivolumab monotherapy (3 mg/kg Q2W) versus dacarbazine (1000 mg/m^2) in patients with previously untreated advanced (unresectable or metastatic) melanoma		
	Duration of main phase:		FPFV: 18 Jan-2013 LPLV: 24 Jan-2014
	Duration of Run-in phase:		n/a
	Duration of Extension phase:		n/a
Hypothesis	improve OS by nivolumab treatment in comparison to dacarbazine treatment.		
Treatments groups	Nivolumab 3 mg/kg		Nivolumab was administered as an IV infusion of 60 minutes on day 1 of each 2-week cycle. 206 patients has received at least 1 infusion of nivolumab plus placebo IV every 3 weeks
	Dacarbazine		DTIC at 1000 mg/m^2 as a 30 to 60 minute IV infusion on day 1 of a treatment cycle every 3 weeks plus placebo IV every 3 weeks plus placebo IV every 2 weeks 205 patients received at least one dose of dacarbazine
Endpoints and definitions	Primary endpoint	OS	The time from the first dose of study drug until the date of death. Patients who did not die were censored at the last known alive date
	Secondary endpoint	PFS	The time from randomization to the date of the first documented progression as assessed by investigator or death due to any cause whichever occurred first. Patients who did not progress or die were censored on the date of their last evaluable tumour assessment.

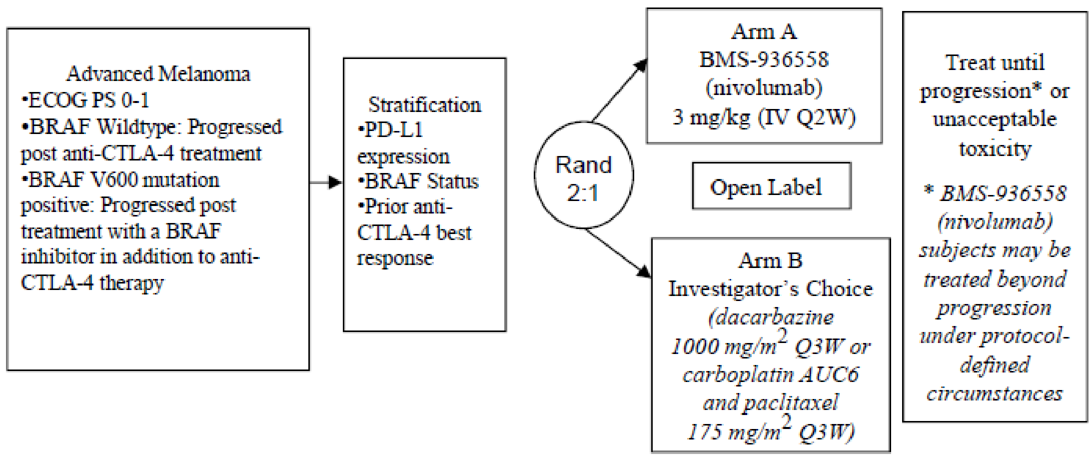
	Secondary endpoint	ORR	ORR: there percentage of patients in the analysis population with a confirmed best overall response (BOR) of complete response (CR) or partial response (PR) per investigator using RECIST v 1.1
	Secondary endpoint	OS based on PD-L1 expression	OS sub analysis based on PD-L1 expression levels
Database lock	27 May-2014		
Results and Analysis			

Analysis description	**Primary Analysis**		
Analysis population and time point description	All randomized patients		
Descriptive statistics and estimate variability	Treatment group	Nivolumab 3 mg/kg	dacarbazine
	Number of subject	210	208
	OS Median (months)	>14	11.8
	ORR	13.6%	5.9%
Effect estimate per comparison	Hazard ratio OS (nivolumab/dacarbazine)	0.46	
	95% CI	0.31, 0.69	
	P value	0.000085	

Study CA209037: A Randomized, Open-Label, Phase 3 Trial of BMS-936558 (Nivolumab) Versus Investigator's Choice in Advanced (Unresectable or Metastatic) Melanoma Patients Progressing Post Anti-CTLA-4 Therapy

Methods

Figure 21: Study Design schematic – Study CA209037



Study Participants

Main inclusion criteria

- Histologically confirmed Stage III (unresectable) or Stage IV melanoma.
- Classified as PD-L1 positive, PD-L1 negative, or PD-L1 indeterminate by a pre-treatment recent core, excision or punch biopsy from an unresectable or metastatic site. No intervening systemic therapy should be administered between the time of biopsy and randomization.
- Objective evidence of disease progression (e.g., clinical or radiological) during or after at least 1 (V600 wildtype) or at least 2 (V600 mutation positive) prior treatment regimens for advanced melanoma.
    - Subjects BRAF wildtype: evidence of progression of disease (PD) during, or following treatment with anti-CTLA-4 containing therapy for advanced melanoma and those subjects who have received another treatment regimen must have objective evidence of PD during or following at least 1 cycle of treatment for advanced melanoma.
    - Subjects BRAF V600 mutation positive: objective evidence of PD post-treatment with anti-CTLA-4 containing therapy and BRAF inhibitor therapy for advanced melanoma, in any sequence or in any combination.
- Prior chemotherapy or immunotherapy (tumor vaccine, cytokine, or growth factor given to control the cancer) was completed at least 4 weeks before study drug administration, and all AEs have either returned to baseline or stabilized.

- Prior anti-CTLA-4 therapy was completed at least 6 weeks before study drug administration.
- Eastern Cooperative Oncology Arm (ECOG) performance status of 0 or 1.

Main exclusion criteria

- Active brain metastasis or leptomeningeal metastasis: Subjects with brain metastases were eligible if these were treated, and had no evidence of progression for at least 8 weeks prior to start of study treatment by MRI (except where contraindicated in which case CT scan was acceptable) and did not require immunosuppressive doses of systemic corticosteroids (>10mg/day prednisone or equivalent) for at least 2 weeks prior to study drug administration.
- Ocular melanoma.
- Subjects whose melanoma BRAF status was unknown.
- Subjects with a condition requiring systemic treatment with either corticosteroids (> 10 mg daily prednisone equivalent) or other immunosuppressive medications within 14 days of first dose of study drug. Corticosteroids with minimal systemic absorption (for example topical or inhalational), and adrenal replacement steroid doses > 10 mg daily prednisone equivalent, were permitted in the absence of active autoimmune disease.
- Prior systemic melanoma therapy with both dacarbazine and carboplatin and paclitaxel.

Subjects with a known history of the following anti-CTLA-4 therapy related adverse reactions based on the CTCAE v4.0 criteria:

- Grade 4 anti-CTLA-4 therapy related adverse reaction except resolved nausea, fatigue, infusion reactions or endocrinopathies where clinical symptoms were able to be controlled with appropriate hormone replacement therapy. Grade 3 anti-CTLA-4 therapy related adverse reactions must have resolved or been controlled within 12 weeks.
- Any ≥ Grade 2 eye pain or reduction of visual acuity that did not respond to topical therapy and did not improve to ≤ Grade 1 severity within 2 weeks of starting topical therapy or that required systemic treatment.
- Any ≥ Grade 3 sensory neurologic toxicity.
- Any Grade 4 laboratory abnormalities, except aspartase aminotransferase (AST), alanine aminotransferase (ALT) or total (T.) bilirubin;

  1) ASTorALT > 10 x upper limit of normal (ULN)

  2) T.bilirubin > 5 x ULN
- Subjects who required infliximab or other immune suppressants including mycophenolic acid for management of drug-related toxicities.

Treatments

Nivolumab

Nivolumab at 3 mg/kg, on Day 1 of each 2 week cycle, was administered as an IV infusion over 60 minutes using a volumetric pump with a 0.2/0.22 micron in-line filter. Dosing calculations were based on the subject's body weight. No premedications were recommended for initiation of dosing.

Subjects received nivolumab until disease progression (or discontinuation of study therapy in patients receiving nivolumab beyond progression), discontinuation due to toxicity, or other protocol-specified reasons.

Dose escalation or reduction was not permitted

Subjects were permitted to continue nivolumab treatment beyond initial RECIST v1.1 defined PD as long as they met the following criteria:

- Investigator-assessed clinical benefit; and
- Subject was tolerating study drug.

Investigator's choice

All subjects were treated with an option not previously received as a prior anti-cancer therapy. A subject was only treated with one of the following options:

1) Dacarbazine: administered as an open-label dose of 1000 mg/m^2 as a 30 to 60 minute IV infusion on Day 1 of a treatment cycle every 3 weeks (21 days).

2) Carboplatin and paclitaxel: paclitaxel administered as an open-label dose of 175mg/m^2 as a 180-minute IV infusion on Day 1 of a treatment cycle every 3 weeks (21 days). Carboplatin administered as an open-label dose of AUC 6 as a 30 minute IV infusion on Day 1 of a treatment cycle every 3 weeks (21 days).

There was no dose escalation permitted for the investigator's choice group. Dose reduction for the investigator's choice group, was permitted due to hematologic and non-hematologic toxicities. Subjects who received the investigator's choice therapy were not permitted to cross-over to nivolumab therapy, nor to be treated beyond progression.

Objectives

Primary Objective

- To estimate the ORR in nivolumab treatment group and to compare OS of nivolumab to investigator's choice in subjects with advanced melanoma.

Secondary Objectives

- To compare the PFS of nivolumab to investigator's choice in subjects with advanced melanoma.
- To evaluate whether PD-L1 expression is a predictive biomarker for ORR and OS.
- To evaluate HRQoL as assessed by European Organisation for Research and Treatment of Care (EORTC) QLQ-C30.

Exploratory Objectives

- To assess the overall safety and tolerability of nivolumab and investigator's choice.
- To characterize the PK of nivolumab and explore exposure-response relationships with respect to safety and efficacy.

- To characterize the immunogenicity of nivolumab.
- To explore potential biomarkers associated with clinical response to nivolumab by analyzing tumor tissue specimens and serum for proteins.
- To assess the effects of natural genetic variation (single nucleotide polymorphism [SNPs]) in select genes including but not limited to PD-1, PD-L1, PD-L2 and CTLA-4 on clinical endpoints and/or on the occurrence of adverse events (AEs).
- To assess changes in health status in treatment groups by the EuroQoL EQ-5D both on treatment and during the survival follow-up period.

Outcomes/endpoints

The primary efficacy endpoint were ORR (as determined per IRRC) in nivolumab treated patients and OS. ORR was defined as the percentage of patients with a confirmed best overall response (BOR) of complete response (CR) or partial response (PR).

Secondary efficacy endpoints included:

- PFS
- Best overall response rate (BOR) and Duration of Response (DOR; time between the date of first confirmed response to the date of the first documented progression)
- Time to Objective Response (TTR)
- Potential association between PD-L1 expression and efficacy endpoints of nivolumab and investigator's choice. Biomarkers were measured in tumour tissues at screening and during the study from all patients. For determination of the PD-L1 status for analyses of the efficacy the Validated Dako PD-L1 IHC assay was used. The PD-L1 expression status as determined by the (verified) Dako PD-L IHC assay also determined by a second (validated) Dako PD-L1 IHC assay. PD-L1 expression analysis was performed centrally (LabCorp Clinical trials, Los Angeles, CA).
- Health related Quality of Life (HR QoL) as measured by the EORTC QLQ-C30

<u>Primary endpoints</u>

The primary ORR analysis was performed on treated subjects among the ORR population (all randomized subjects to any treatment group with at least 6 months of follow-up at the time of ORR analysis timepoint) using treatment group 'as treated'.

The IRRC-determined ORR in the nivolumab group was estimated and its corresponding 95% exact two-sided CI was calculated using the Clopper-Pearson method.

The IRRC-determined ORR in subjects treated with investigator's choice is also presented, along with exact 95% two-sided CI.

The unweighted differences in ORR between the two treatment groups and corresponding 95% two-sided CI using the method of Newcombe is provided.

The BOR was summarized by response category for each treatment group.

To support the primary analyses, similar analyses of ORR were performed with the following modifications:

- ORR as determined per investigator among the ORR population for each treatment group 'as treated'. For subjects who continued nivolumab beyond progression, the BOR was determined based on response designations recorded up to the time of the initial RECISTv 1.1-defined progression as assessed by the investigator or the date of subsequent anticancer therapy.
- The IRRC-determined ORR on the entire ORR population (i.e., including randomized nontreated subjects among ORR population). The analysis was then performed by treatment group 'as randomized'.

To assess concordance between IRRC and investigator assessments, BOR was cross-tabulated by treatment group as randomized and assessment type (investigator vs. IRRC). Concordance Rate of Responders was computed as the frequency with which investigator and IRRC agree on classification of a subject as responder/non responder as a proportion of the total number of subjects assessed.

Tumour responses and progression were assessed using RECIST v 1.1. Tumour assessments were conducted at 9 weeks (± 1 week) following randomization then every 6 weeks (±1 week) for the first 12 months, thereafter every 12 weeks (± 1 week) until disease progression or treatment discontinuation, whichever occurred later. Immune related non-conventional responses were not included in calculations for response based analyses.

Secondary Endpoint

The criteria to analyze PFS were not reached at the time; therefore, only descriptive analyses of PFS for the ORR population were produced.

The PFS (based on IRRC assessment) curves for each treatment group were estimated using the K-M product-limit method on the ORR population. Median PFS (with 95% CI) and PFS rate at 6 months (with 95% CI) were computed.

The hazard ratio (HR) of nivolumab to investigator's choice and corresponding 99.99% CI were estimated using a Cox-proportional hazards model, with treatment group as a single covariate, stratified by BRAF status, prior anti-CTLA-4 benefit, and PD-L1 status (IVRS source).

The source of PFS event (death vs. progression) was summarised by treatment group.

Exploratory endpoints included safety and tolerability of nivolumab and investigator's choice, PK, evaluation of immunogenicity of nivolumab (serum ADA) and HRQoL measured by EuroQol EQ-5D.

Sample size

Approximately 390 subjects were to be randomised to the two treatment groups in a 2:1 ratio. The sample size of the study accounts for the co-primary efficacy endpoints: ORR (per IRRC) and OS with an alpha allocation of 0.1% and 4.9% respectively. The formal analyses of ORR and OS are planned to be conducted at different times: with ORR being analyzed first (reported in this assessment report) and subsequent interim and final OS analyses to follow.

Randomisation

After eligibility was established and informed consent was obtained, subjects were enrolled into the study via an IVRS. Subjects meeting all eligibility criteria were randomized in a 2:1 ratio to Arm A [BMS-936558 (nivolumab)] or Arm B (investigator's choice) respectively. Subjects were stratified and balanced according to the following factors:

- PD-L1 status
  - PD-L1 positive (≥5% tumor cell membrane staining in a minimum of a hundred evaluable tumor cells) vs.
  - PD-L1 negative (< 5% tumor cell membrane staining in a minimum of a hundred evaluable tumor cells)/indeterminate (tumor cell membrane scoring hampered by high cytoplasmic staining or melanin content)
- BRAF status: wildtype vs V600 mutation positive
- Prior anti-CTLA-4 best response: Prior anti-CTLA-4 therapy clinical benefit (Best Overall Response of Stable Disease, Partial Response or Complete Response) vs. No Prior anti-CTLA-4 therapy clinical benefit (Best Overall Response of Progressive Disease)

Blinding (masking)

This study was an open label study, hence the patients and treating physicians were not blinded to the treatment received.

Statistical methods

<u>ORR</u>

The primary analysis of ORR in the nivolumab treatment group was performed when the first 120 subjects treated with nivolumab (i.e., approximately 180 treated subjects in total: 120 in the nivolumab group and 60 in the investigator's choice group) had a minimum follow-up of 6 months and formal analysis was restricted to the subjects treated with nivolumab with at least 6 months of follow up. This allowed sufficient follow up for ORR to have a stable estimate, information on duration of response, as well as adequate safety follow up.

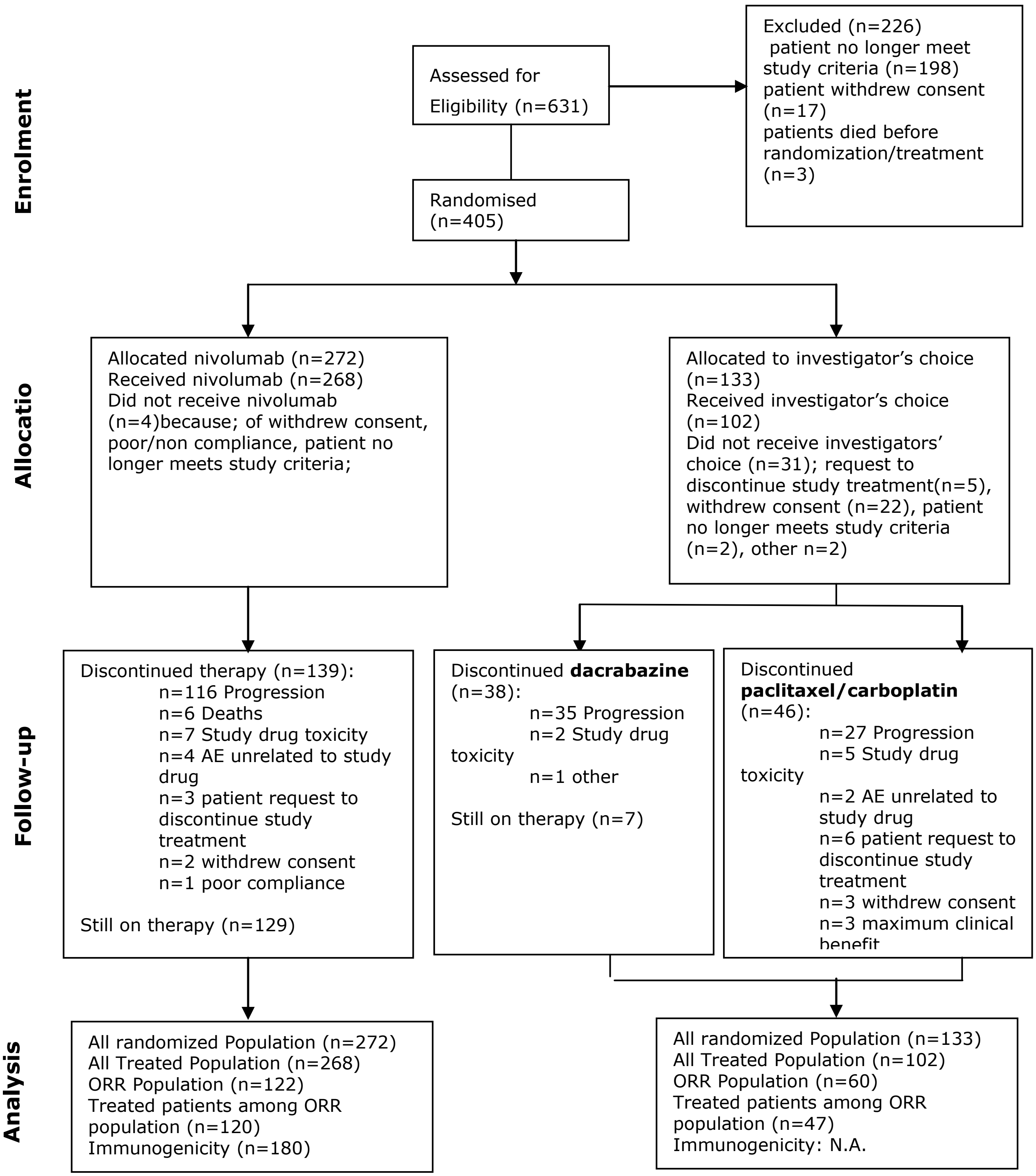
The maximum width of the exact two-sided 95% CI was 17.1% when the ORR was expected to be in the 5% to 30% range.

<u>OS</u>

At least 260 deaths are required to provide approximately 90% power to detect a HR of 0.65, corresponding to a median OS of 8 vs. 12.3 months (4.3 month difference) for the investigator's choice and nivolumab groups, respectively, with an overall two-sided type I error of 4.9%. One OS interim analysis will be conducted when at least 169 deaths (i.e., 65% of total events) have been observed. The stopping boundaries at the interim and final OS analyses will be derived based on the exact number of deaths using Lan-DeMets alpha spending function with O'Brien-Fleming boundaries. If the interim analysis of OS is performed at exactly 169 events, the study could be stopped for efficacy if the p-value is less than 0.0105. The nominal significance level for the final OS after 260 events would then be 0.0457.

Results

Participant flow



Recruitment

The enrollment period lasted from 21-Dec-2012 until 10-Jan-2014 and included 90 sites in 14 countries (US, Austria, Belgium, Brazil, Canada, Denmark, France, Germany, Israel, Italy, Netherlands, Spain, Switzerland, and United Kingdom).

A total of 631 subjects were enrolled. Of these, 405 (272 in the nivolumab and 133 in the investigator's choice groups) were randomized into the study. The most common reasons for screening failure were:

- 198 (31.4%) subjects no longer met study criteria.
- 17 (2.7%) subjects withdrew consent.

In addition, 3 subjects died before randomisation/treatment.

Conduct of the study

Protocol deviations were low in frequency, with 5 (1.8%) subjects in the nivolumab group and 5 (3.8%) subjects in the investigator's choice group. The original protocol was amended 11 times. Main amendments consist of:

1. Date 19-Mar-2013: the summary of safety section was updated to include new preliminary reproductive toxicology data that was distributed as a non-clinical expedited safety report and to include change to the guidance on contraception.
2. Date 08-May-2013: the eligibility criteria were modified to expand the number of prior therapies allowed. Confirmation of objective response required and clarification that patients receiving palliative radiotherapy would be classified as having unequivocal progression.
3. Date 24-Dec-2013: the study design was updated to allow an adequately powered statistical comparison of the co-primary endpoint of ORR at an earlier time point while maintaining the power for statistical comparison of the other co-primary endpoint of OS. Formal analyses of ORR and OS will be conducted at different time points with ORR being analysed first followed by interim and final OS analyses.
4. Date 21-Apr-2014: The protocol was modified to include the updated statistical plans for a non-comparative estimation of the ORR.

Baseline data

The majority of patients were males (64.4%) and white (98.3%), with a median age of 60.0 years (range 23- 88 yr), and an ECOG PS 0 (above 60%). The median duration of time since first diagnosis was 3.57 years (range: 0.4-25.3) in the nivolumab group and 3.73 years (range 0.3-31.1). Most patients in both treatment groups had stage IV (96.0% vs 98.5%) and M1c disease (74.6% vs 76.7%) of which 51.1% of patients in the nivolumab and 34.6% in the control arm had high LDH. Most patients had extensive disease with ≥2 sites of disease (78.7% vs. 78.9%). The most common visceral sites included lung and liver. A higher percentage of patients in the nivolumab group than in the investigator's choice group had a history of brain metastasis (19.5% vs. 13.5%, respectively).

Table 30: Baseline characteristic – Study CA209037

	ORR Population			All Randomized Subjects		
	NIVOLUMAB N = 122	INVESTIGATOR'S CHOICE N = 60	Total N = 182	NIVOLUMAB N = 272	INVESTIGATOR'S CHOICE N = 133	Total N = 405
AGE (YEARS)						
N	122	60	182	272	133	405
MEAN	57.9	61.1	59.0	58.7	60.3	59.2
MEDIAN	58.0	63.5	60.0	59.0	62.0	60.0
MIN , MAX	25 , 88	29 , 84	25 , 88	23 , 88	29 , 85	23 , 88
STANDARD DEVIATION	13.81	13.08	13.62	14.12	12.43	13.59
AGE CATEGORIZATION (%)						
< 65	83 (68.0)	33 (55.0)	116 (63.7)	177 (65.1)	80 (60.2)	257 (63.5)
>= 65 AND < 75	25 (20.5)	18 (30.0)	43 (23.6)	55 (20.2)	37 (27.8)	92 (22.7)
>= 75	14 (11.5)	9 (15.0)	23 (12.6)	40 (14.7)	16 (12.0)	56 (13.8)
GENDER (%)						
MALE	79 (64.8)	36 (60.0)	115 (63.2)	176 (64.7)	85 (63.9)	261 (64.4)
FEMALE	43 (35.2)	24 (40.0)	67 (36.8)	96 (35.3)	48 (36.1)	144 (35.6)
RACE (%)						
WHITE	120 (98.4)	57 (95.0)	177 (97.3)	269 (98.9)	129 (97.0)	398 (98.3)
BLACK OR AFRICAN AMERICAN	0	2 (3.3)	2 (1.1)	1 (0.4)	2 (1.5)	3 (0.7)
ASIAN	2 (1.6)	0	2 (1.1)	2 (0.7)	0	2 (0.5)
AMERICAN INDIAN OR ALASKA NATIVE	0	0	0	0	0	0
NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER	0	0	0	0	0	0
OTHER	0	1 (1.7)	1 (0.5)	0	2 (1.5)	2 (0.5)

	ORR Population			All Randomized Subjects		
	NIVOLUMAB N = 122	INVESTIGATOR'S CHOICE N = 60	Total N = 182	NIVOLUMAB N = 272	INVESTIGATOR'S CHOICE N = 133	Total N = 405
REGION						
WESTERN EUROPE+CANADA+ROW	70 (57.4)	31 (51.7)	101 (55.5)	166 (61.0)	75 (56.4)	241 (59.5)
USA	52 (42.6)	29 (48.3)	81 (44.5)	106 (39.0)	58 (43.6)	164 (40.5)
BASELINE LDH						
<= ULN	53 (43.4)	36 (60.0)	89 (48.9)	129 (47.4)	77 (57.9)	206 (50.9)
> ULN	69 (56.6)	23 (38.3)	92 (50.5)	139 (51.1)	46 (34.6)	185 (45.7)
<= 2*ULN	102 (83.6)	49 (81.7)	151 (83.0)	222 (81.6)	105 (78.9)	327 (80.7)
> 2*ULN	20 (16.4)	10 (16.7)	30 (16.5)	46 (16.9)	18 (13.5)	64 (15.8)
NOT REPORTED	0	1 (1.7)	1 (0.5)	4 (1.5)	10 (7.5)	14 (3.5)
HISTORY OF BRAIN METASTASES						
YES	22 (18.0)	8 (13.3)	30 (16.5)	53 (19.5)	18 (13.5)	71 (17.5)
NO	100 (82.0)	52 (86.7)	152 (83.5)	219 (80.5)	115 (86.5)	334 (82.5)
BRAF MUTATION TEST						
COBAS/THXID	42 (34.4)	17 (28.3)	59 (32.4)	74 (27.2)	31 (23.3)	105 (25.9)
OTHER	63 (51.6)	29 (48.3)	92 (50.5)	145 (53.3)	79 (59.4)	224 (55.3)
UNKNOWN	17 (13.9)	14 (23.3)	31 (17.0)	53 (19.5)	23 (17.3)	76 (18.8)
SMOKING STATUS						
YES	54 (44.3)	28 (46.7)	82 (45.1)	109 (40.1)	60 (45.1)	169 (41.7)
NO	64 (52.5)	30 (50.0)	94 (51.6)	148 (54.4)	65 (48.9)	213 (52.6)
UNKNOWN	4 (3.3)	2 (3.3)	6 (3.3)	15 (5.5)	8 (6.0)	23 (5.7)
PERFORMANCE STATUS (ECOG) [%]						
0	72 (59.0)	37 (61.7)	109 (59.9)	162 (59.6)	84 (63.2)	246 (60.7)
1	50 (41.0)	22 (36.7)	72 (39.6)	110 (40.4)	48 (36.1)	158 (39.0)
2	0	1 (1.7)	1 (0.5)	0	1 (0.8)	1 (0.2)

SUBTYPE						
MUCOSAL	10 (8.2)	4 (6.7)	14 (7.7)	27 (9.9)	14 (10.5)	41 (10.1)
CUTANEOUS	88 (72.1)	48 (80.0)	136 (74.7)	198 (72.8)	98 (73.7)	296 (73.1)
ACRAL	6 (4.9)	3 (5.0)	9 (4.9)	17 (6.3)	7 (5.3)	24 (5.9)
OCULAR/UVEAL	0	1 (1.7)	1 (0.5)	0	1 (0.8)	1 (0.2)
OTHER	18 (14.8)	4 (6.7)	22 (12.1)	30 (11.0)	13 (9.8)	43 (10.6)
M STAGE AT STUDY ENTRY						
M0	5 (4.1)	0	5 (2.7)	10 (3.7)	2 (1.5)	12 (3.0)
M1A	8 (6.6)	3 (5.0)	11 (6.0)	15 (5.5)	11 (8.3)	26 (6.4)
M1B	17 (13.9)	11 (18.3)	28 (15.4)	44 (16.2)	18 (13.5)	62 (15.3)
M1C	92 (75.4)	46 (76.7)	138 (75.8)	203 (74.6)	102 (76.7)	305 (75.3)
AJCC STAGE AT STUDY ENTRY						
STAGE III	5 (4.1)	0	5 (2.7)	11 (4.0)	2 (1.5)	13 (3.2)
STAGE IV	117 (95.9)	60 (100.0)	177 (97.3)	261 (96.0)	131 (98.5)	392 (96.8)
SUBJECTS WITH AT LEAST ONE LESION (%)	122 (100.0)	60 (100.0)	182 (100.0)	271 (99.6)	132 (99.2)	403 (99.5)
SITE OF LESION (A) (B) (%)						
BONE	19 (15.6)	10 (16.7)	29 (15.9)	36 (13.2)	15 (11.3)	51 (12.6)
CENTRAL NERVOUS SYSTEM	11 (9.0)	1 (1.7)	12 (6.6)	17 (6.3)	6 (4.5)	23 (5.7)
INTESTINE	14 (11.5)	9 (15.0)	23 (12.6)	35 (12.9)	18 (13.5)	53 (13.1)
LIVER	53 (43.4)	20 (33.3)	73 (40.1)	102 (37.5)	38 (28.6)	140 (34.6)
LUNG	68 (55.7)	35 (58.3)	103 (56.6)	151 (55.5)	73 (54.9)	224 (55.3)
LYMPH NODE	79 (64.8)	33 (55.0)	112 (61.5)	159 (58.5)	74 (55.6)	233 (57.5)
OTHER	16 (13.1)	7 (11.7)	23 (12.6)	52 (19.1)	28 (21.1)	80 (19.8)
SKIN	19 (15.6)	14 (23.3)	33 (18.1)	39 (14.3)	25 (18.8)	64 (15.8)
SOFT TISSUE	39 (32.0)	22 (36.7)	61 (33.5)	81 (29.8)	44 (33.1)	125 (30.9)
VISCERAL, OTHER	44 (36.1)	20 (33.3)	64 (35.2)	92 (33.8)	41 (30.8)	133 (32.8)
NUMBER OF SITES WITH AT LEAST ONE LESION (B) (%)						
1	22 (18.0)	10 (16.7)	32 (17.6)	57 (21.0)	27 (20.3)	84 (20.7)
2	27 (22.1)	19 (31.7)	46 (25.3)	58 (21.3)	37 (27.8)	95 (23.5)
3	28 (23.0)	13 (21.7)	41 (22.5)	77 (28.3)	34 (25.6)	111 (27.4)
4	30 (24.6)	10 (16.7)	40 (22.0)	53 (19.5)	20 (15.0)	73 (18.0)
>=5	15 (12.3)	8 (13.3)	23 (12.6)	26 (9.6)	14 (10.5)	40 (9.9)

ORR population consists of randomized subjects with at least 6 months between randomization date and clinical cutoff date.
(A) Subjects may have lesions at more than one site.
(B) Includes both target and non-target lesions.

Table 31: Prior Cancer therapy – Study CA209037

	ORR Population			All Randomized Subjects		
	NIVOLUMAB N = 122	INVESTIGATOR'S CHOICE N = 60	Total N = 182	NIVOLUMAB N = 272	INVESTIGATOR'S CHOICE N = 133	Total N = 405
PRIOR NEO-ADJUVANT THERAPY						
YES	0	0	0	1 (0.4)	2 (1.5)	3 (0.7)
NO	122 (100.0)	60 (100.0)	182 (100.0)	271 (99.6)	131 (98.5)	402 (99.3)
PRIOR ADJUVANT THERAPY						
YES	18 (14.8)	8 (13.3)	26 (14.3)	58 (21.3)	25 (18.8)	83 (20.5)
NO	104 (85.2)	52 (86.7)	156 (85.7)	214 (78.7)	108 (81.2)	322 (79.5)
NUMBER OF PRIOR SYSTEMIC REGIMEN RECEIVED IN METASTATIC SETTING						
1	39 (32.0)	17 (28.3)	56 (30.8)	77 (28.3)	34 (25.6)	111 (27.4)
2	61 (50.0)	34 (56.7)	95 (52.2)	139 (51.1)	68 (51.1)	207 (51.1)
>2	22 (18.0)	9 (15.0)	31 (17.0)	56 (20.6)	31 (23.3)	87 (21.5)
PRIOR CHEMOTHERAPY IN METASTATIC SETTING (A)						
YES	60 (49.2)	29 (48.3)	89 (48.9)	145 (53.3)	72 (54.1)	217 (53.6)
NO	62 (50.8)	31 (51.7)	93 (51.1)	127 (46.7)	61 (45.9)	188 (46.4)
PRIOR IMMUNOTHERAPY IN METASTATIC SETTING (B)						
YES	17 (13.9)	12 (20.0)	29 (15.9)	37 (13.6)	35 (26.3)	72 (17.8)
NO	105 (86.1)	48 (80.0)	153 (84.1)	235 (86.4)	98 (73.7)	333 (82.2)
PRIOR SURGERY RELATED TO CANCER						
YES	121 (99.2)	60 (100.0)	181 (99.5)	271 (99.6)	133 (100.0)	404 (99.8)
NO	1 (0.8)	0	1 (0.5)	1 (0.4)	0	1 (0.2)
PRIOR RADIOTHERAPY						
YES	54 (44.3)	29 (48.3)	83 (45.6)	124 (45.6)	65 (48.9)	189 (46.7)
NO	68 (55.7)	31 (51.7)	99 (54.4)	148 (54.4)	68 (51.1)	216 (53.3)
SUBJECTS WITH PRIOR ANTI-CTLA4 THERAPY (A)	122 (100.0)	60 (100.0)	182 (100.0)	272 (100.0)	133 (100.0)	405 (100.0)
SUBJECTS WITH PRIOR ANTI-CTLA4 THERAPY IN ADJUVANT SETTING ONLY	0	0	0	1 (0.4)	0	1 (0.2)
DURATION OF LAST PRIOR ANTI-CTLA4 THERAPY (A) (MONTH)						
N (B)	122 (100.0)	60 (100.0)	182 (100.0)	271 (99.6)	133 (100.0)	404 (99.8)
MEDIAN (MIN - MAX)	2.10 (0.3 - 16.4)	2.14 (0.7 - 23.3)	2.10 (0.3 - 23.3)	2.10 (0.0 - 76.9)	2.14 (0.0 - 28.6)	2.10 (0.0 - 76.9)
PRIMARY DISCONTINUATION REASON OF LAST PRIOR ANTI-CTLA4 THERAPY REGIMEN (A)						
DISEASE PROGRESSION	64 (52.5)	32 (53.3)	96 (52.7)	145 (53.3)	78 (58.6)	223 (55.1)
MAXIMUM CLINICAL BENEFIT	1 (0.8)	0	1 (0.5)	1 (0.4)	0	1 (0.2)
TOXICITY	7 (5.7)	2 (3.3)	9 (4.9)	16 (5.9)	6 (4.5)	22 (5.4)
COMPLETED TREATMENT	49 (40.2)	24 (40.0)	73 (40.1)	105 (38.6)	46 (34.6)	151 (37.3)
OTHER	1 (0.8)	2 (3.3)	3 (1.6)	5 (1.8)	3 (2.3)	8 (2.0)

ORR population consists of randomized subjects with at least 6 months between randomization date and clinical cutoff date.
(A) Any setting.
(B) Duration of therapy is not derived for subject with missing start or stop date.

Table 32: Prior BRAF inhibitor therapy for patients with mutated BRAF – Study CA209037

	NUMBER OF SUBJECTS (%)					
	ORR Population			All Randomized Subjects		
	NIVOLUMAB N = 27	INVESTIGATOR'S CHOICE N = 13	TOTAL N = 40	NIVOLUMAB N = 59	INVESTIGATOR'S CHOICE N = 27	TOTAL N = 86
SUBJECTS WITH PRIOR BRAF INHIBITOR THERAPY (A)	27 (100.0)	13 (100.0)	40 (100.0)	59 (100.0)	27 (100.0)	86 (100.0)
SUBJECTS WITH PRIOR BRAF INHIBITOR THERAPY IN ADJUVANT SETTING ONLY	0	0	0	0	0	0
SUBJECTS WITH PRIOR INVESTIGATIONAL BRAF INHIBITOR THERAPY ONLY (B)	0	0	0	1 (1.7)	1 (3.7)	2 (2.3)
DURATION OF LAST PRIOR BRAF INHIBITOR THERAPY (A) (MONTH)						
N (C)	27 (100.0)	13 (100.0)	40 (100.0)	59 (100.0)	27 (100.0)	86 (100.0)
MEDIAN (MIN - MAX)	7.49 (0.3 - 18.1)	6.18 (1.7 - 18.5)	6.77 (0.3 - 18.5)	7.98 (0.1 - 30.1)	6.01 (0.0 - 22.9)	6.83 (0.0 - 30.1)
PRIMARY DISCONTINUATION REASON OF LAST PRIOR BRAF INHIBITOR THERAPY REGIMEN (A)						
DISEASE PROGRESSION	21 (77.8)	9 (69.2)	30 (75.0)	47 (79.7)	21 (77.8)	68 (79.1)
MAXIMUM CLINICAL BENEFIT	1 (3.7)	1 (7.7)	2 (5.0)	3 (5.1)	1 (3.7)	4 (4.7)
TOXICITY	4 (14.8)	2 (15.4)	6 (15.0)	7 (11.9)	4 (14.8)	11 (12.8)
COMPLETED TREATMENT	1 (3.7)	0	1 (2.5)	2 (3.4)	0	2 (2.3)
OTHER	0	1 (7.7)	1 (2.5)	0	1 (3.7)	1 (1.2)
SUBJECTS WHO PROGRESSED ON OR AFTER LAST PRIOR BRAF INHIBITOR THERAPY (A) (D)	26 (96.3)	13 (100.0)	39 (97.5)	55 (93.2)	26 (96.3)	81 (94.2)
DOCUMENTATION OF PROGRESSION (E)						
RADIOGRAPHIC	23 (85.2)	12 (92.3)	35 (87.5)	51 (86.4)	24 (88.9)	75 (87.2)
CLINICAL	4 (14.8)	2 (15.4)	6 (15.0)	6 (10.2)	4 (14.8)	10 (11.6)

ORR population consists of randomized subjects with at least 6 months between randomization date and clinical cutoff date.
BRAF Mutant positive: CRF source.
(A) Any setting.
(B) Metastatic setting only.
(C) Duration of therapy is not derived for subject with missing start or stop date.
(D) Corresponds to progression information collected for the last prior BRAF inhibitor therapy.
(E) Any progression can be documented by both sources.

Numbers analysed

In this open-label study, a total of 35 of 405 subjects (8.6%) were randomized but did not continue in the treatment period of the study. More randomized subjects in the investigator's choice group did not receive treatment compared to the subjects in the nivolumab group (4 [1.5%] in the nivolumab group and 31 [23.3%] in the investigator's choice group). The most common reasons for a subject in the investigator's choice group not receiving treatment after randomisation was "subject request to discontinue study treatment" and "subject withdrew consent".

Table 33: Analyses Populations – Study CA209037

Population	Nivolumab Group	Investigator's Choice Group	Total
Enrolled: all subjects who signed an ICF and were registered into the IVRS			631
All Randomized Population: All subjects randomized to any treatment group	272	133	405
All Treated Population: All subjects treated with at least one dose of study drug	268	102	370
ORR Population: All randomized subjects to either treatment group with at least 6 months of follow-up at the time of the ORR analysis, which occurred when the first 120 nivolumab-treated subjects had a minimum of 6 months of follow-up.	122	60	182
Treated Subjects Among ORR Population: All subjects who received at least one dose of treatment (nivolumab or investigator's choice) and had at least 6 months of follow-up at the time of the ORR analysis. The primary endpoint analysis was restricted to subjects treated with nivolumab.	120	47	167
Immunogenicity: All subjects who received at least one dose of nivolumab and had baseline and at least one post-baseline assessment for ADA	180	N.A.	N.A.

ADA = anti-drug antibody; ICF = informed consent form; IVRS = interactive voice response system; ORR = objective response rate

Outcomes and estimation

The primary efficacy endpoint were ORR (as determined per IRRC) in nivolumab treated patients and OS.

Primary efficacy endpoint: IRRC- assessed ORR for treated patients among the ORR population

The primary analysis of ORR in the nivolumab treatment groups was performed when the first 120 patients were treated with nivolumab had a minimum follow-up of 6 months. The ORR was assessed both by the investigator and by the independent radiology review committee (IRRC).

The confirmed ORR as assessed by the IRRC was 31.7% (95% CI:23.5, 40.8) for the nivolumab arm and 10.6% (95% CI 3.5, 23.1) for the investigator's choice arm. A confirmed BOR of CR was achieved in 4 (3.3.%) of the patients in the nivolumab arm. No CRs were reported for patients treated with investigator's choice therapy. The unweighted difference in ORR between the two treatment groups was 21.0% (95% CI, 6.8, 31.7) (Table 34).

Best response of SD was observed in 28 (23.3%) of the patients in the nivolumab of whom 12 had ongoing SD at the time of the data cut-off. In the control arm, SD was observed in 16 (34.0%) of the patients.

The BOR could not be determined in 10% and 23.4% of the patients included in the nivolumab and investigator's choice arm respectively.

Table 34: Best Overall Response per IRRC - Treated Subjects Among the ORR Population - Study CA209037

	Nivolumab 3 mg/kg N=120		Investigator's choice (Reference arm) N=47
Confirmed BOR[a] Number (%) of Responders			
CR	4 (3.3)		0
PR	34 (28.3)		5 (10.6)
SD	28 (23.3)		16 (34.0)
PD	42 (35.0)		15 (31.9)
Unable to Determine	12 (10.0)		11 (23.4)
ORR[a]			
Number (%) of Responders	38 (31.7)		5 (10.6)
Exact 95% CI	(23.5, 40.8)		(3.5, 23.1)
Unweighted ORR difference		21.0	
95% CI		(6.8, 31.7)	
DOR[b]			
Median (95% CI) (Months)	Not reached (NA, NA)		3.6 (NA, NA)
Time to Response[c]			
Median (Min - Max) (Months)	2.1 (1.6, 7.4)		3.5 (2.1, 6.1)
Subjects with Response (Confirmed or Unconfirmed)			
Number (%) of Responders	43 (35.8%)		5 (10.6%)
ORR[d] (95% CI)	35.8% (27.3, 45.1)		10.6% (3.5, 23.1)

[a] Confirmed CR + PR as per RECIST v1.1 criteria, after imaging plus clinical review by the IRRC. Derived from the Kaplan-Meier estimation. Two-sided CI computed using the Clopper and Pearson method.

[b] Determined for subjects with confirmed CR or PR using the Kaplan-Meier product-limit method. Two-sided CI constructed based on a log-log transformed CI for the survivor function.

[c] Determined for subjects with confirmed CR or PR.

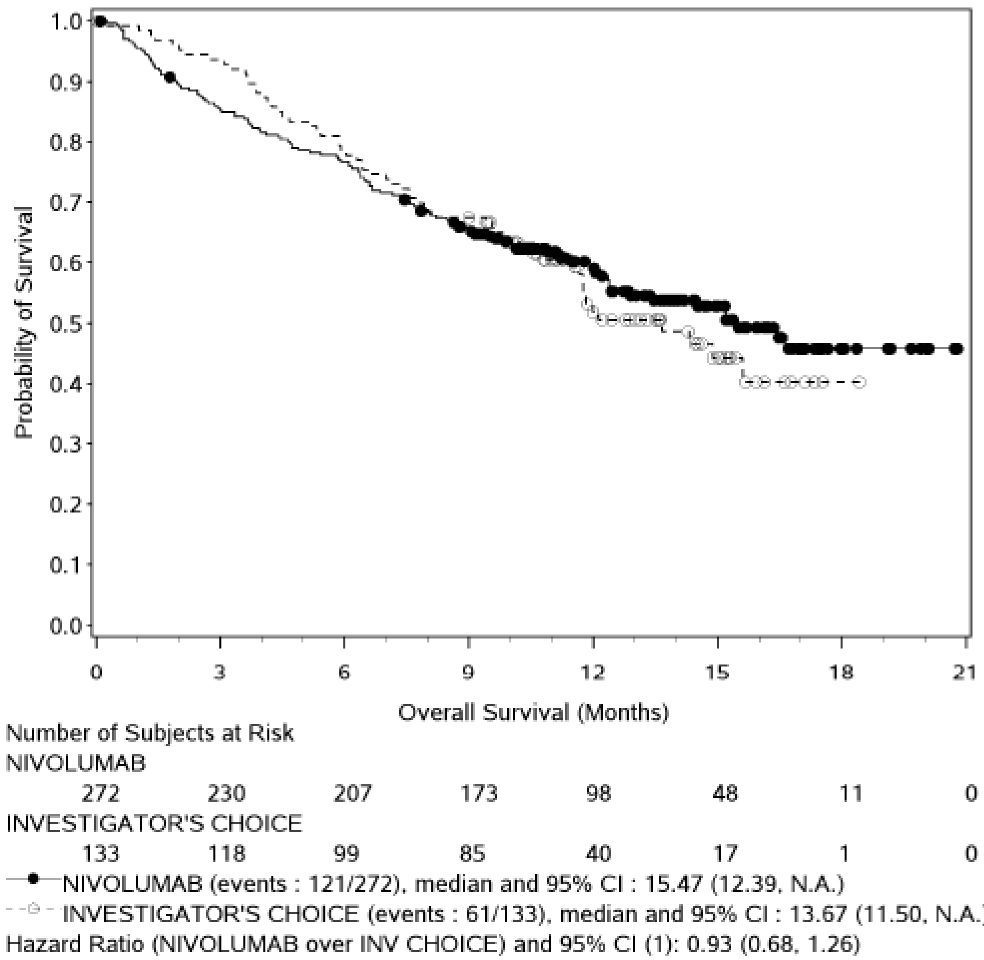
[d] Determined for subjects with confirmed CR or PR and subjects with unconfirmed CR or PR combined.

Abbreviations: BOR = best overall response; CI = confidence interval; CR = complete response; DOR = duration of response; NA = not available; ORR = objective response rate; PD = progressive disease; PR = partial response; SD = stable disease.

Primary efficacy endpoint: Overall survival

In December 2014, a pre-specified interim analysis for OS was performed, based on a 12-Nov-2014 database lock, and reviewed by an independent Data Monitoring Committee (DMC) (16-Dec-2014). The DMC recommended continuation of the study without modification. The final analysis of OS is expected to occur in late 2015. Focusing on the interim analysis for OS and with 70% [182/260] of the total death events required for final OS analysis, no benefit has been shown so far; HR of 0.93 for nivolumab over investigator's choice chemotherapy [95% CI: 0.68, 0.1.26]). Median OS in the nivolumab vs. investigator's choice chemotherapy group was 15.5 months (95% CI: 12.39, N.A.) vs. 13.7 months (95% CI: 11.5, N.A.). The 6-month OS rates were 76.7% (95% CI: 71,2, 81.3) and 78.6% (95% CI: 70.3, 84.8) for the nivolumab and investigator's choice groups, respectively.

Figure 22: Kaplan-Meier plot of OS – Study CA209037 (interim analyses)



(1) from stratified Cox proportional hazard model using randomized arm as a single covariate.
Symbols represent censored observations.

Secondary efficacy endpoint:

- **Investigator assessed ORR in all randomised and treated patients**

The investigator-assessed confirmed ORR (all randomised patients) for nivolumab monotherapy, was higher than that for investigator's choice chemotherapy (25.4% [95% CI: 20.3, 31.0] vs. 8.3% [95% CI: 4.2, 14.3]) with an ORR difference of 17.1% (95% CI: 9.5, 23.7; n=405). CR was achieved in 10 (3.7%) subjects, and confirmed PR was achieved in 59 (21.7%) subjects for nivolumab arm, whereas 0 subjects had confirmed CR, and 11 (8.3%) subjects had confirmed PR in the chemotherapy group. The investigator-assessed confirmed ORR for nivolumab (25.4% [95% CI: 20.3, 31.0] was numerically higher than that observed for dacarbazine- (4.4% [95% CI: 0.5, 15.1]) and carboplatin/paclitaxel-treated patients (15.8% [95% CI: 7.5, 27.9]).

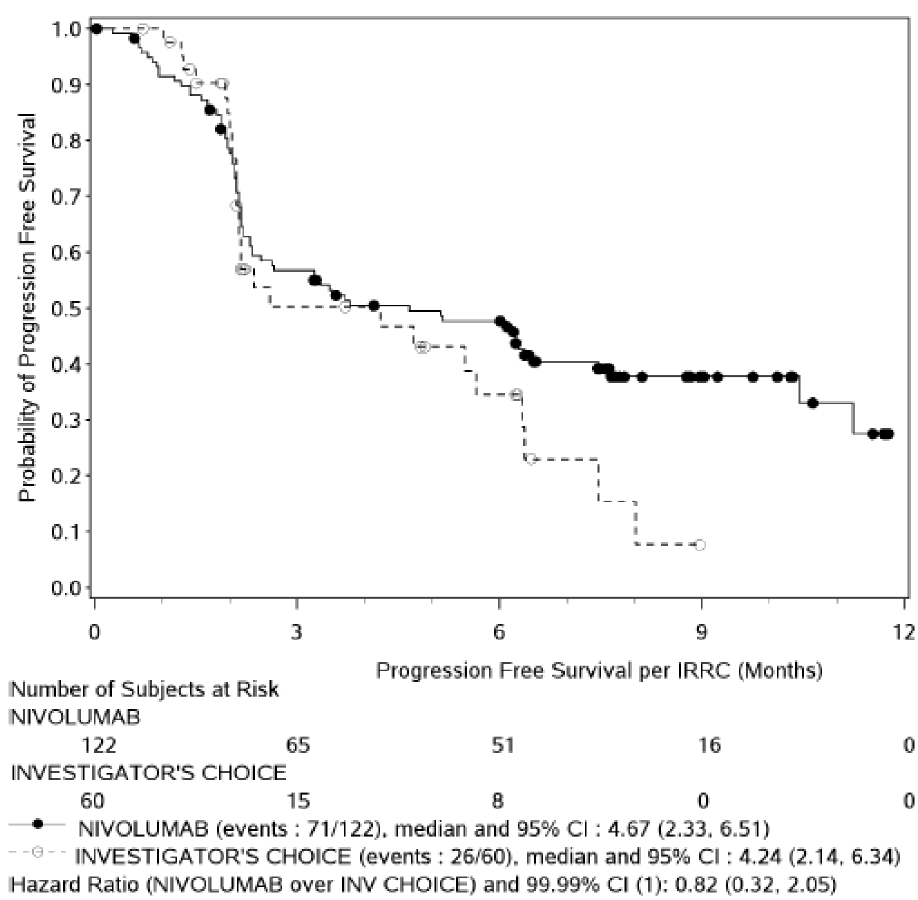
Investigator assessed, confirmed ORRs in all treated patients (n=370) were 25.7% [95% CI: 20.6, 31.4] in the nivolumab group (n=268) vs. 10.8% [95% CI: 5.5, 18.5]) in the chemotherapy group (n=102), with an ORR difference of 15.0% (95% CI: 6.0, 22.2).

- **Progression-Free Survival**

The descriptive analysis of PFS as assessed by investigator demonstrated a median PFS of 3.42 months (95% CI: 2.43, 4.90) and a 6-month PFS rate of 42% (95% CI: 0.36, 0.48) in the nivolumab group. In the investigators' choice group, the median PFS was 2.69 months (95% CI: 2.17, 3.71) and the 6-month PFS rate was 29% (95% CI: 0.20, 0.39). HR of 0.74 for PFS for nivolumab over investigator's choice chemotherapy [95% CI: 0.57, 0.97]. For the PFS analysis, a total of 31.3% of the nivolumab treated patient were censored. Most (21.3%) of these patients were censored while they were still on treatment without progression at the time of the data cut-off. Other reasons for censoring patients in the nivolumab arm were; receiving subsequent therapy (6.6%) and treatment is stopped, without signs of progression but patient is still in follow-up (1.8%).

In the investigator's choice arm, a total of 36.1% of the patients were censored, with 9.8% being never treated. Another common reason for censoring was receiving subsequent therapy (21.8%). Furthermore, patients were censored because they were still on treatment without progression at the time of the data cut-off (0.8%), treatment is stopped without signs of progression but still in follow-up (3.0%).

Figure 23: Kaplan-Meier curves of PFS per investigator, all randomised subjects – Study CA209037



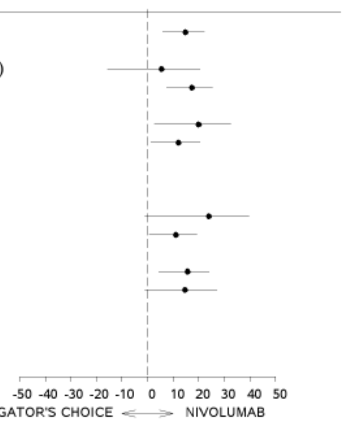
ORR population consists of randomized subjects with at least 6 months between randomization date and clinical cutoff date.
(1) from stratified Cox proportional hazard model using randomized arm as a single covariate.
Symbols represent censored observations.

Ancillary analyses

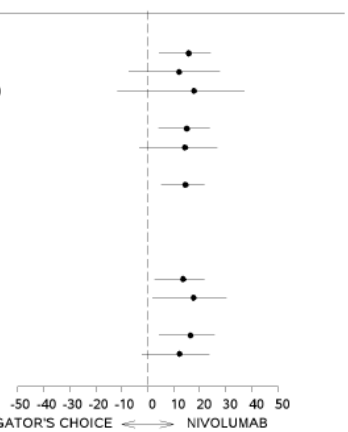
Subgroup analyses were conducted to assess the impact of age, gender, region, baseline ECOG performance status, PD-L1 status, prior anti-CTA-4 benefit, M stage at study entry, history of brain metastases, smoking status, baseline LDH, or AJCC stage (Figure 24).

Figure 24: Forest plot of treatment effect on objective response rate per investigator in pre-defined subsets (all Treated Subjects) – Study CA209037

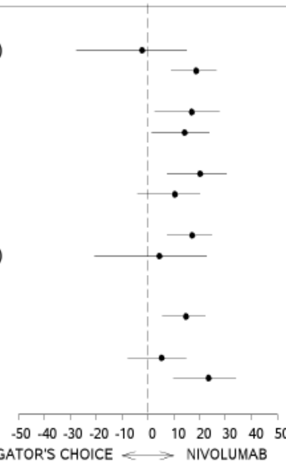
	N	NIVOLUMAB N of Response (N of subjects)	NIVOLUMAB ORR (95% Exact CI)	INVESTIGATOR'S CHOICE N of Response (N of subjects)	INVESTIGATOR'S CHOICE ORR (95% Exact CI)	Unweighted ORR difference (95% CI)
Overall	370	69(268)	25.7% (20.6, 31.4)	11(102)	10.8% (5.5, 18.5)	15.0% (6.0, 22.2)
CRF BRAF Status						
Mutant	79	11(57)	19.3% (10.0, 31.9)	3(22)	13.6% (2.9, 34.9)	5.7% (-15.7, 20.6)
Wild Type	291	58(211)	27.5% (21.6, 34.0)	8(80)	10.0% (4.4, 18.8)	17.5% (7.3, 25.5)
CRF Prior anti-CTLA-4 benefit						
Yes	126	30(93)	32.3% (22.9, 42.7)	4(33)	12.1% (3.4, 28.2)	20.1% (2.7, 32.6)
No	244	39(175)	22.3% (16.4, 29.2)	7(69)	10.1% (4.2, 19.8)	12.1% (1.3, 20.6)
M Stage at study entry						
M0	12	3(10)	30.0% (6.7, 65.2)	0(2)	0.0% (0.0, 84.2)	
M1A	23	7(15)	46.7% (21.3, 73.4)	1(8)	12.5% (0.3, 52.7)	
M1B	59	13(43)	30.2% (17.2, 46.1)	1(16)	6.3% (0.2, 30.2)	24.0% (-1.0, 39.7)
M1C	276	46(200)	23.0% (17.4, 29.5)	9(76)	11.8% (5.6, 21.3)	11.2% (0.6, 19.5)
Age category I						
<65	235	42(175)	24.0% (17.9, 31.0)	5(60)	8.3% (2.8, 18.4)	15.7% (4.4, 24.0)
>=65	135	27(93)	29.0% (20.1, 39.4)	6(42)	14.3% (5.4, 28.5)	14.7% (-1.1, 27.2)



	N	NIVOLUMAB N of Response (N of subjects)	NIVOLUMAB ORR (95% Exact CI)	INVESTIGATOR'S CHOICE N of Response (N of subjects)	INVESTIGATOR'S CHOICE ORR (95% Exact CI)	Unweighted ORR difference (95% CI)
Age category II						
<65	235	42(175)	24.0% (17.9, 31.0)	5(60)	8.3% (2.8, 18.4)	15.7% (4.4, 24.0)
>=65 AND <75	83	14(54)	25.9% (15.0, 39.7)	4(29)	13.8% (3.9, 31.7)	12.1% (-7.3, 27.6)
>=75	52	13(39)	33.3% (19.1, 50.2)	2(13)	15.4% (1.9, 45.4)	17.9% (-11.8, 37.1)
Gender						
Male	245	44(175)	25.1% (18.9, 32.2)	7(70)	10.0% (4.1, 19.5)	15.1% (4.2, 23.7)
Female	125	25(93)	26.9% (18.2, 37.1)	4(32)	12.5% (3.5, 29.0)	14.4% (-3.1, 26.7)
Race						
White	364	68(265)	25.7% (20.5, 31.4)	11(99)	11.1% (5.7, 19.0)	14.5% (5.4, 21.9)
Black	2	0(1)	0.0% (0.0, 97.5)	0(1)	0.0% (0.0, 97.5)	
Asian	2	1(2)	50.0% (1.3, 98.7)	0(0)	N.A.	
Other	2	0(0)	N.A.	0(2)	0.0% (0.0, 84.2)	
Region						
Rest of World	222	34(164)	20.7% (14.8, 27.7)	4(58)	6.9% (1.9, 16.7)	13.8% (2.8, 21.9)
US	148	35(104)	33.7% (24.7, 43.6)	7(44)	15.9% (6.6, 30.1)	17.7% (1.9, 30.2)
Baseline ECOG Performance Status						
0	218	43(160)	26.9% (20.2, 34.4)	6(58)	10.3% (3.9, 21.2)	16.5% (4.4, 25.7)
1	151	26(108)	24.1% (16.4, 33.3)	5(43)	11.6% (3.9, 25.1)	12.4% (-2.2, 23.5)



	N	NIVOLUMAB N of Response (N of subjects)	NIVOLUMAB ORR (95% Exact CI)	INVESTIGATOR'S CHOICE N of Response (N of subjects)	INVESTIGATOR'S CHOICE ORR (95% Exact CI)	Unweighted ORR difference (95% CI)
History of Brain Metastases						
Yes	70	9(54)	16.7% (7.9, 29.3)	3(16)	18.8% (4.0, 45.6)	-2.1% (-27.5, 15.0)
No	300	60(214)	28.0% (22.1, 34.6)	8(86)	9.3% (4.1, 17.5)	18.7% (9.0, 26.5)
Smoking Status						
Yes	156	30(109)	27.5% (19.4, 36.9)	5(47)	10.6% (3.5, 23.1)	16.9% (2.8, 27.7)
No	196	35(145)	24.1% (17.4, 31.9)	5(51)	9.8% (3.3, 21.4)	14.3% (1.5, 23.7)
Baseline LDH						
<=ULN	192	41(130)	31.5% (23.7, 40.3)	7(62)	11.3% (4.7, 21.9)	20.2% (7.6, 30.4)
>ULN	177	28(137)	20.4% (14.0, 28.2)	4(40)	10.0% (2.8, 23.7)	10.4% (-3.9, 20.1)
Baseline LDH						
<=2*ULN	307	59(222)	26.6% (20.9, 32.9)	8(85)	9.4% (4.2, 17.7)	17.2% (7.5, 24.8)
>2*ULN	62	10(45)	22.2% (11.2, 37.1)	3(17)	17.6% (3.8, 43.4)	4.6% (-20.7, 22.7)
AJCC Stage						
Stage III	13	3(11)	27.3% (6.0, 61.0)	0(2)	0.0% (0.0, 84.2)	
Stage IV	357	66(257)	25.7% (20.5, 31.5)	11(100)	11.0% (5.6, 18.8)	14.7% (5.6, 22.1)
PDL1 status per IVRS						
Negative	178	21(131)	16.0% (10.2, 23.5)	5(47)	10.6% (3.5, 23.1)	5.4% (-7.7, 14.8)
Positive	183	46(131)	35.1% (27.0, 43.9)	6(52)	11.5% (4.4, 23.4)	23.6% (9.8, 34.1)
Indeterminate	9	2(6)	33.3% (4.3, 77.7)	0(3)	0.0% (0.0, 70.8)	



PD-L1 Expression and Efficacy

All patients in the ORR population had a PD-L1 result in IVRS using the verified assay. Based on the IVRS reported results, where PD-L1 positive status is defined as ≥5% tumour cell membrane expression, the PD-L1 positivity rate in the ORR population was 45.1%. Objective responses were observed in patients from both the nivolumab and investigator's choice arm regardless of PD-L1 expression status. In the nivolumab arm, the ORR was numerically higher in PD-L1 positive vs negative patients (43.6% [95% CI 30.3, 57.7] vs 20.3% [95% CI: 11.3, 32.2]). Updated results in ORRs continued being higher for PD-L1 positive (35.1% [95% CI: 27.0, 43.9]) than for PD-L1 negative (16.0% [95% CI: 10.2, 23.5]).

BRAF status and Efficacy

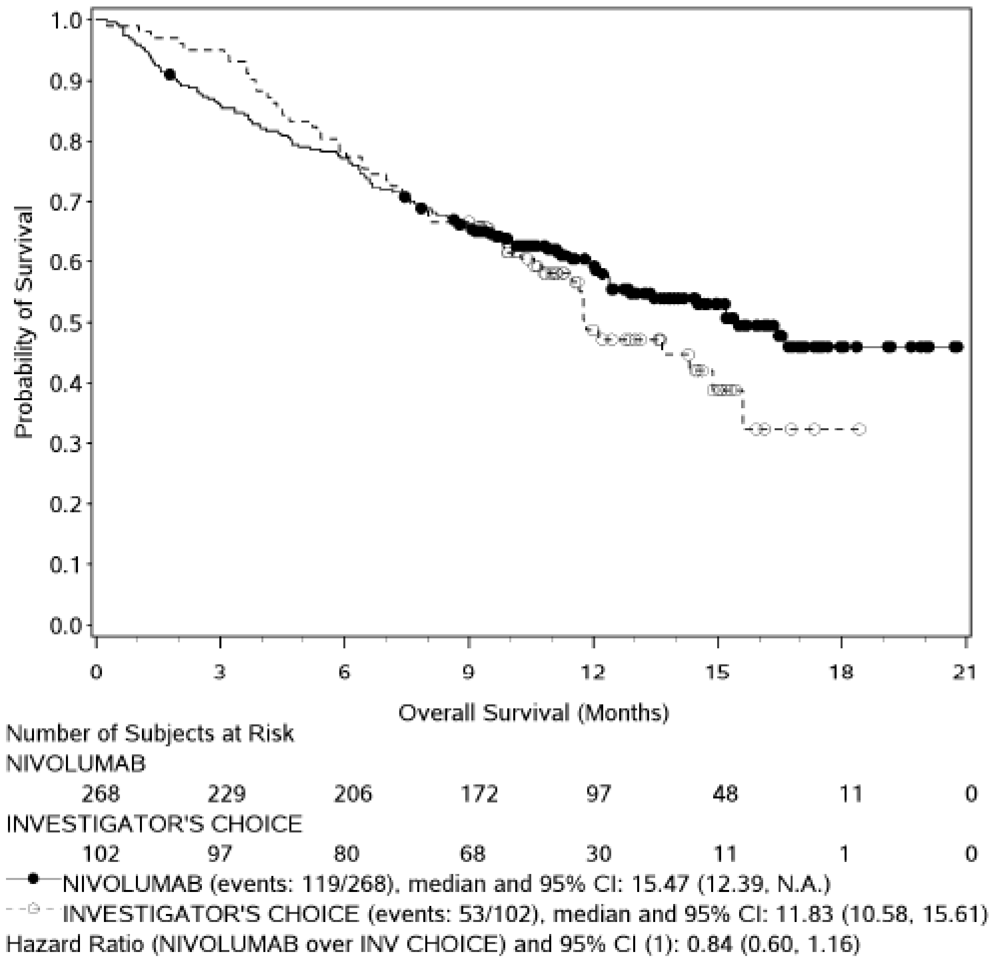
Objective responses were observed with nivolumab treatment regardless of BRAF status, with an ORR of 34.0% (95% CI: 24.6, 44%) in BRAF wild-type patients and 23.1% (95% CI: 9.0, 43.6) in BRAF mutant patient. Updated results highlighted the same trend (ORR of 27.5% [95% CI: 21.6, 34.0] in BRAF wild-type patients and 19.3% [95% CI: 10.0, 31.9] in BRAF mutant patient).

Sensitivity analyses of the co-primary endpoint

- ORR population (sensitivity analysis in the intention-to-treat population): the IRRC-assessed ORR in nivolumab-treated subjects was 31.1% (95% CI: 23.1, 40.2). The confirmed IRRC-assessed ORR in the investigator's choice group was 8.3% (95% CI: 2.8, 18.4).
- Investigator-assessed ORR in treated subjects among the ORR population (sensitivity analysis): the confirmed ORR was 25.8% [95% CI: 18.3, 34.6]) in the nivolumab group, which was consistent with the ORR based on the primary IRRC assessment (31.7% [95% CI: 23.5, 40.8]).
- A sensitivity analysis was performed on censoring for patients that were randomised to investigator's choice but were never treated. There were 31 patients that were randomised but did not receive the treatment. These patients likely received subsequent therapy relatively shortly after randomisation including other available anti-PD-1 agents (at least 9 subjects went on to receive an anti-PD-1 agent). An ad hoc sensitivity analysis of OS restricted to the All Treated population demonstrated a decrease in the HR from 0.93 (95%CI

0.68, 1.26) to 0.84 (95% CI: 0.60, 1.16) in favour of nivolumab treatment, and the median OS for the investigator's choice group decreased from 13.67 months to 11.83 months (Figure 25).

Figure 25: Kaplan-Meier Plot of OS (all treated patients) – Study CA209037



(1) from stratified Cox proportional hazard model using randomized arm as a single covariate.

- Symbols represent censored observations.

Time to Response and Duration of Response

Most responses reported after nivolumab treatment were rapid. The median time to response was 2.1 months (range 1.6 to 7.4 months). Thirteen of the 38 (34.2%) patients responded by the time of first scan (9 weeks), and the majority (26 patients, 68.4%) responded by 2.5 months.

The median time to response in the investigator's choice arm was 3.5 months (range 2.1 to 6.1 months).

Table 33: Time objective response and duration of response per IRR- patients with response among the ORR population – Study CA209037

	NIVOLUMAB N = 38	INVESTIGATOR'S CHOICE N = 5
TIME TO OBJECTIVE RESPONSE (MONTHS)		
NUMBER OF RESPONDERS	38	5
MEAN	2.71	4.05
MEDIAN	2.14	3.52
MIN, MAX	1.6, 7.4	2.1, 6.1
STANDARD DEVIATION	1.262	1.537
DURATION OF OBJECTIVE RESPONSE (MONTHS)		
MIN, MAX (A)	1.4+, 10.0+	1.3+, 3.5
MEDIAN (95% CI) (B)	N.A.	3.55 (N.A., N.A.)
N EVENT/N RESP (%)	2/38 (5.3)	1/5 (20.0)

ORR population consists of randomized subjects with at least 6 months between randomization date and clinical cutoff date.
RECIST v1.1 Response Criteria where confirmation of response is required.
(A) Symbol + indicates a censored value.
(B) Median computed using Kaplan-Meier method.

Of 120 treated subjects in the ORR population, 37 subjects that had progressed per RECIST v1.1 criteria as assessed by the investigators were subsequently considered eligible per protocol to receive continued nivolumab therapy. Of these 37 subjects, 10 (27.0%) subjects had a target lesion reduction of ≥ 30% compared to baseline after RECIST v1.1 progression.

Subgroup analysis in elderly population

Table 34: Summary of subjects in different age groups – Study CA209066 and CA209037

	Total number of subjects	Number (%) of Subjects Age 65-74	Number (%) of Subjects Age 75-84	Number (%) of Subjects Age 85+	Total number of subjects age ≥ 65 (%)
CA209063	115	43 (37.39)	15 (13.04)	1 (0.87)	59 (51.30)
CA209037	232	48 (20.69)	28 (12.07)	3 (1.29)	79 (34.05)

Analysis -Directory: /global/pkms/data/CA/209/C07/prd/ppk_eud120/final/
Program Source: Analysis-Directory/sp/scripts/sum-age.ssc
Source: Analysis-Directory/sp/export/ca209-mel-age-sum.xls

Summary of main study

The following tables summarise the efficacy results from the main studies supporting the present application. These summaries should be read in conjunction with the discussion on clinical efficacy as well as the benefit risk assessment (see later sections).

Table 35: Summary of efficacy for trials

Title: A Randomized, Open-label Phase 3 Trial of BMS-936558 (Nivolumab) versus Investigator's choice in Advanced (unresectable or metastatic) melanoma Patients progressing post anti-CTLA-4 therapy_	
Study identifier	CA209037

Design	Phase 3, randomized, open-label, multicenter study of nivolumab monotherapy (3 mg/kg Q2W) versus investigtor's choice in patients with advanced (unresectable or metastatic) melanoma who have progressed on or after anti-CTLA-4 therapy, and for those with BRAF V600 mutations, who have progressed on or after a BRAF inhibitor in addition to anti-CTLA-4 therapy		
	Duration of main phase:		FPFV: 12 Dec-2012 LPLV for the primary endpoint: 10-Mar-2014
	Duration of Run-in phase:		n/a
	Duration of Extension phase:		n/a
Hypothesis	Demonstrating a meaningful clinical activity as measured by ORR and/ or increased OS in comparison to investigator's choice treatment.		
Treatments groups	Nivolumab 3 mg/kg		Nivolumab was administered as an IV infusion of 60 minutes on day 1 of each 2-week cycle. 268 patients has received at least 1 infusion of nivolumab
	Investigator's choice		One of the following: DTIC at 1000 mg/m^2 as a 30 to 60 minute IV infusion on day 1 of a treatment cycle every 3 weeks Carboplatin (dose AUC 6) and paclitaxel (175 mg/m^2). Carboplatin was administered as a 30 minute IV infusion and paclitaxel was administered as a 180 minute IV infusion both drug were administered on day 1 of a treatment cycle every 3 weeks 102 patients received at least one dose of investigator's choice treatment
Endpoints and definitions	Primary endpoint	Non-comparative estimation of IRRC-assessed ORR	ORR: there percentage of patients in the analysis population with a confirmed best overall response (BOR) of complete response (CR) or partial response (PR) per IRRC using RECIST v 1.1
	Primary endpoint	OS	The time from the first dose of study drug until the date of death. Patients who did not die were censored at the last known alive date
	Secondary endpoint	DOR	The time between the date of first documented response (CR or PR) to the date of the first confirmed progression as determined by the IRRC (per RECIST v 1.1.), or death due to any cause, whichever occurred first. For patients who neither progressed nor died, duration of objective response was censored on the date of their last evaluable tumour assessment
	Secondary endpoint	TTR	Time from randomization to the date of the first confirmed documented response (CR or PR) as assessed by the IRRC

	Secondary endpoint	IRRC-assessed PFS	The time from randomization to the date of the first documented progression as assessed by the IRRC per RECIST v1.1., or death due to any cause whichever occurred first. Patients who did not progress or die were censored on the date of their last evaluable tumour assessment.
Database lock	30-Apr-2014 for clinical data, 30 May 2014 for IRRC data and 10 Mar 2014 for imaging cut off for both IRRC and investigator assessments		

Results and Analysis

Analysis description	**Primary Analysis**		
Analysis population and time point description	ORR population: All patients who received at least one dose of treatment and had at least 6 months of follow-up at the time of the ORR analysis		
Descriptive statistics and estimate variability	Treatment group	Nivolumab 3 mg/kg	Investigator's choice
	Number of subject	120	47
	ORR n (%) 95% CI	38 (31.7%) 23.5, 40.8	5 (10.6) 3.5, 23.1
	CR n (%) PR n (%) SD n (%)	4 (3.3) 34 (28.3) 28 (23.3)	0 5 (10.6) 16 (34.0)
	OS		
	Data not yet available		
	TTR Median (months)	2.1	3.5
	Min-max	27.3- 45.1	2.1-6.1
	DOR Median (months)	Not reached	3.6
	Min -max	1.4+ - 10.0+	1.3+ - 3.5
	Median PFS (months)	4.7	4.2
Effect estimate per comparison	Unweighted ORR difference	21.0%	
	95% CI	6.8, 31.7	

Analysis performed across trials (pooled analyses and meta-analysis)

The applicant did not provide analyses across studies.

Supportive studies

<u>Study CA209006</u>

In a Phase 1 study (CA209006: A Pilot Trial of a Vaccine Combining Multiple Class I Peptides and Montanide ISA 51 VG with Escalating Doses of Anti-PD-1 Antibody BMS-936558 for Patients with Unresectable Stages III/IV Melanoma), the clinical safety and efficacy of nivolumab was assessed in a cohort of ipilimumab-refractory subjects that were similar to the CA209037 population. In this cohort, subjects had unresectable Stage III or IV melanoma and had experienced progression without responding to prior ipilimumab treatment (41 subjects). These subjects received nivolumab (without the vaccine) at 3 mg/kg every 2 weeks for 24 weeks, then every 12 weeks for up to 2 years.

Key efficacy findings in response-evaluable subjects in the cohort (38 subjects) were consistent with results from CA209037, and were as follows:

- The confirmed investigator-assessed ORR was 26% (95% CI: 13.4, 43.1)
- 7 subjects (18%) had SD at ≥24 weeks
- The median DOR was not reached and ranged from 36+ to 48+ weeks
- The PFS rate at 24 weeks was 44%

Clinical efficacy by PD-L1 expression was not reported by cohort. Among the PD-L1 expression-evaluable subjects in all cohorts (44 subjects, including ipilimumab-refractory and -naive treated at 1, 3 or 10 mg/kg of nivolumab):

- 12 subjects (27%) were PD-L1 positive using a 5% cutoff.
- Objective responses were associated with PD-L1 positive expression (≥5% tumor cell membrane expression compared with PD-L1 negative expression (67% vs. 19%).

MDX1106-03

In melanoma, using the 5% threshold, 7/17 (41%) of PD-L1+ subjects and 3/21 (14%) of PD-L1(-) showed an objective response to nivolumab suggesting a better objective responses to nivolumab in PD-L1+ subjects (Table 36). Similar percentages were noted using the 1% threshold.

In NSCLC, using the 5% threshold, 5/31 (16%) of PD-L1+ subjects and 4/32 (13%) of PD-L1(-) subjects had objective responses to nivolumab. Similar percentages were noted using the 1% threshold. There was no apparent difference in objective response between PD-L1 positive and negative tumours.

Table 36: Objective Response Rates by PD-L1 Expression in Subjects with NSCLC and Melanoma – Study MDX1106-03

PD-L1 Cutoff	PD-L1 Status	ORR[b] n/N (%)
NSCLC Subjects, N = 63		
1% tumor membrane staining	Positive	5/35 (14)
	Negative	4/28 (14)
5% tumor membrane staining	Positive	5/31 (16)
	Negative	4/32 (13)
Melanoma Subjects, N = 38		
1% tumor membrane staining	Positive	8/25 (32)
	Negative	2/13 (15)
5% tumor membrane staining	Positive	7/17 (41)
	Negative	3/21 (14)

b ORR of PD-L1 evaluable patients; ORR includes complete or partial responders determined by RECIST

2.5.3. Discussion on clinical efficacy

Design and conduct of clinical studies

The applicant applied for the indication for the treatment of advanced melanoma (unresectable or metastatic) in adults. They submitted two pivotal studies to support their proposed indication. For first line in untreated patients, the study CA209066, a phase 3, randomized, double-blind study of nivolumab vs dacarbazine in subjects with BRAF wild type, previously untreated, unresectable or metastatic melanoma was submitted. For last line indication in patients who have progressed on or after anti-CTLA-4 therapy, and for those with BRAF V600 mutations, who have progressed on or after a BRAF inhibitor in addition to anti-CTLA-4-therapy, study CA209037, a randomized, open-label, phase 3 trial of nivolumab vs investigator's choice(dacarbazine or carboplatin and paclitaxel) in unresectable or metastatic melanoma patients progressing post anti-CTLA-4 therapy was submitted.

The study CA209066 was stopped early because of the results of the unplanned interim analysis, and as per DMC recommendation, the study was amended to allow subjects randomized to the dacarbazine group to receive nivolumab. The plan in the protocol with a single interim look at 218 deaths had a nominal significance level of 0.0455 for the final OS analysis at 312 deaths. The preliminary results showed that the analysis of OS had reached an HR 0.46 (95%CI 0.31-0.69) with a stratified log-rank test p-value 0.000085, based on 110 deaths.

The CHMP was concerned that from a statistical point of view, carrying out an unplanned interim analysis and altering the conduct of the study based on this analysis under the DMC which monitored every month the mortality, was questionable and would introduce uncertainties such as the potential for informative bias. The O'Brien-Fleming based-analysis in the original SAP did not take into account the 9 reviews that were conducted by the DMC. However, the CHMP concluded that the consistency of the data observed for overall survival, the subgroup analyses and the support of secondary variables, the effect observed was compelling and that major biases were unlikely.

Although the CHMP considered that the design and conduct of study CA209037 did not raise any major issues, the CHMP highlighted that use of ORR as a primary variable was not recommended and OS should have been favoured as primary endpoint in an open label trial in last-line setting in order to best assess the true magnitude of the clinical benefit of nivolumab treatment in this patient population.

Efficacy data and additional analyses

Dose rationale

The nivolumab monotherapy dose and schedule of 3 mg/kg Q2W was based most importantly on the observed clinical safety and efficacy from the 306 subjects treated in MDX1106-03 across different dose levels and tumour types. The data from study MDX1106-03 in support of the 3 mg/kg Q2W dosing were as follows:

- Nivolumab had dose-proportional exposure over the dose range studied; with no effect of tumour type on PK. Body weight based dosing produced approximately uniform exposure in subjects with body weight range 34-162 kg.
- Receptor occupancy was maintained at high levels across the dose range 0.1 to 10 mg/kg and over multiple cycles of drug administration.
- There was a greater percent of objective responses observed in NSCLC subjects treated with 3 mg/kg (24.3%) and 10 mg/kg (20.3%) nivolumab than with 1 mg/kg (3%) nivolumab. There was no apparent relationship between nivolumab dose and ORR in melanoma and RCC.

Efficacy

First line treatment – Study CA209066

Study CA209066 was a phase 3, randomised, double-blind study (CA209066) which enrolled adult patients with confirmed, treatment-naive, Stage III or IV BRAF wild-type melanoma and an Eastern Cooperative Oncology Group (ECOG) performance-status score of 0 or 1 which were treated with either nivolumab 3 mg/kg or dacarbazine. Patients with active autoimmune disease, ocular melanoma, or active brain or leptomeningeal metastases were excluded from the study.

A total of 418 patients were randomised to receive either nivolumab (n = 210) administered intravenously over 60 minutes at 3 mg/kg every 2 weeks or dacarbazine (n = 208) at 1000 mg/m^2 every 3 weeks. Randomisation was stratified by PD-L1 status and M stage (M0/M1a/M1b versus M1c). Treatment was continued as long as clinical benefit was observed or until treatment was no longer tolerated. Treatment after disease progression was permitted for patients who had a clinical benefit and did not have substantial adverse effects with the study drug, as determined by the investigator. Tumour assessments, according to the Response Evaluation Criteria in Solid Tumours (RECIST), version 1.1, were conducted 9 weeks after randomisation and continued every 6 weeks for the first year and then every 12 weeks thereafter. The primary efficacy outcome measure was overall survival (OS). Key secondary efficacy outcome measures were investigator-assessed progression-free survival (PFS) and objective response rate (ORR).

Baseline characteristics were balanced between the two groups. The median age was 65 years (range: 18-87), 59% were men, and 99.5% were white. Most patients had ECOG performance score of 0 (64%) or 1 (34%). Sixty-one percent of patients had M1c stage disease at study entry. Seventy-four percent of patients had cutaneous melanoma, and 11% had mucosal melanoma; 35% of patients had PD-L1 positive melanoma (≥5% tumour cell membrane expression). Sixteen percent of patients had received prior adjuvant therapy; the most common adjuvant treatment was interferon (9%). Four percent of patients had a history of brain metastasis, and 37% of patients had a baseline LDH level greater than ULN at study entry.

Nivolumab demonstrated a statistically significant improvement in OS over dacarbazine in treatment-naive patients with BRAF wild-type advanced (unresectable or metastatic) melanoma (HR = 0.42; 99.79%CI: 0.25, 0.73; p-value < 0.0001). Median OS was not reached for nivolumab and was 10.8 months (95% CI: 9.33, 12.09) for dacarbazine. The estimated OS rates at 12 months were 73% (95% CI: 65.5, 78.9) and 42% (95% CI: 33.0, 50.9), respectively. The observed OS benefit was consistently demonstrated across subgroups of patients including baseline ECOG performance status, M stage, history of brain metastases, and baseline LDH level. Survival benefit was observed regardless of whether PD-L1 expression was above or below a PD-L1 tumour membrane expression cut-off of 5% or 10%. The results for OS show that there is a clear survival advantage for patients in the nivolumab treatment arm compared to patients in the dacarbazine arm for which the magnitude of the effect is clinically relevant. As of 07-Jan-2015, approximately 24 patients (22.2%) on the control arm have switched to treatment with nivolumab after the database lock for the final CSR. No data on the possible impact of this crossover has been provided yet. Therefore, in order to ascertain the magnitude of the OS survival as the proportion of number of events was small, the CHMP has imposed a condition to the marketing authorisation in Annex II, for the submission of updated results for OS.

Nivolumab also demonstrated statistically significant improvement in PFS compared with dacarbazine (HR = 0.43; 95%CI: 0.34, 0.56; $p < 0.0001$). The median PFS was 5.1 months (95%CI: 3.48, 10.81) for nivolumab and 2.2 months (95%CI: 2.10, 2.40) for dacarbazine. The estimated PFS rates at 6 months were 48% (95%CI: 40.8, 54.9) and 18% (95%CI: 13.1, 24.6), respectively. The estimated PFS rate at 12 months was 42% (95% CI: 34.0, 49.3) for nivolumab.

Response rates were low but the duration of response were pronounced reaching up to 12.5 months in the nivolumab arm. The investigator-assessed ORR using RECIST v1.1 criteria was significantly higher in the nivolumab group than in the dacarbazine group (OR: 4.06; $p<0.0001$). The ORR was 40.0% (95% CI: 33.3, 47.0) in the nivolumab group vs 13.9% (95% CI: 9.5, 19.4) in the dacarbazine group. A higher proportion of subjects had CR in the nivolumab group than the dacarbazine group (7.6% vs 1.0%, respectively). The median time to response was similar for both groups, 2.10 months (range: 1.2 to 7.6) in the nivolumab group and 2.10 months (range 1.8 to 3.6) in the dacarbazine group.

It is important to note that the benefit of nivolumab was seen regardless of the presence of distant metastases and the positivity of PD-L1 tumours. The differences observed for OS and PFS in patients with PD-L1 positive and negative tumours are not considered important at this point since patients which have negative PD-L1 tumours also appear to derive some clinically relevant benefit from nivolumab. Therefore, a restriction of the population to patients expressing PD-L1 is not justified. The expression of PD-L1 and PD-L2 in the tumour microenvironment and the relationship with tumour responses needs to be further investigated. The CHMP has imposed a condition to the marketing authorization in Annex II to perform further analyses to ascertain the potential role of the PD-L2 biomarker, to further explore the relationship between PD-L1 and PD-L2 expression on the efficacy of nivolumab, to continue the exploration of the optimal cut-off for PD-L1 positivity and to further investigate the possible change in PD-L1 status of the tumour during treatment and/or tumour progression. Study CA209066 only enrolled patients which tumours tested BRAF WT but no data is available for first line treatment in patients bearing tumours which are BRAFV600 mutated. However, results from the study CA209037 where patients with BRAFV600 tumours that have been previously treated with a BRAF inhibitor show treatment benefit, it is reasonable to assume that patients with BRAFV600 tumours which have not yet been treated will also derive benefit from nivolumab treatment.

Last line treatment – Study CA209037

Study CA209037 was a phase 3, randomised, open-label study which enrolled adult patients with unresectable or metastatic melanoma who had progressed on or after ipilimumab and if BRAF V600 mutation positive had also progressed on or after BRAF kinase inhibitor therapy. Patients were treated with nivolumab 3 mg/kg or with the investigator’s choice of therapy. Patients with active autoimmune disease, ocular melanoma or a known history of prior ipilimumab-related high-grade (Grade 4 per CTCAE v4.0) adverse reactions, except for resolved nausea, fatigue, infusion reactions, or endocrinopathies, were excluded from the study.

A total of 405 patients were randomised to receive either nivolumab (n = 272) administered intravenously over 60 minutes at 3 mg/kg every 2 weeks or chemotherapy (n = 133) which consisted of the investigator's choice of either dacarbazine (1000 mg/m^2 every 3 weeks) or carboplatin (AUC 6 every 3 weeks) and paclitaxel (175 mg/m^2 every 3 weeks). Randomisation was stratified by BRAF and PD-L1 status and best response to prior ipilimumab. The co-primary efficacy outcome measures were confirmed ORR, as measured by independent radiology review committee (IRRC) using RECIST 1.1, and comparison of OS of nivolumab to chemotherapy. Additional outcome measures included duration and timing of response. The median age was 60 years (range: 23-88). Sixty-four percent of patients were men and 98% were white. ECOG performance scores were 0 for 61% of patients and 1 for 39% of patients. The majority (75%) of patients had M1c stage disease at study entry. Seventy-three percent of patients had cutaneous melanoma and 10% had mucosal melanoma. The number of prior systemic regimen received was 1 for 27% of patients, 2 for 51% of patients, and > 2 for 21% of patients. Twenty-two percent of patients were BRAF mutation positive and 50% of patients were PD-L1 positive. Sixty-four percent of patients had no prior clinical benefit (CR/PR or SD) on ipilimumab. Baseline characteristics were balanced between groups except for the proportions of patients who had a history of brain metastasis (19% and 13% in the nivolumab group and chemotherapy group, respectively) and patients with LDH greater than ULN at baseline (51% and 35%, respectively). The baseline characteristics reflect an advanced metastatic population, in general evenly distributed, with a higher rate of elevated LDH and brain metastases for nivolumab arm vs investigator's choice group (51% vs 35% and 19.5% vs 13.5% respectively) and a majority of subjects with M1c status (75%).

For ORR, Investigator-assessed confirmed ORR for nivolumab 25.4% [95% CI: 20.3, 31.0] vs. 8.3% [95% CI: 4.2, 14.3]) for investigator's choice, with an ORR difference of 17.1% (95% CI: 9.5, 23.7). CR was achieved in 10 (3.7%) subjects, and confirmed PR was achieved in 59 (21.7%) subjects for nivolumab arm, whereas 0 subjects had confirmed CR, and 11 (8.3%) subjects had confirmed PR in the chemotherapy group.

Objective responses (in all treated subjects) to nivolumab were observed in patients with or without BRAF mutation-positive melanoma. The number of patients included with a mutated BRAF was low (57 vs. 22 in the nivolumab and investigator's choice arm, respectively). Of the patients who received nivolumab, the ORR in the BRAF mutant subgroup was 19.3% (95% CI: 10.0, 31.9), and 27.5% (95% CI: 21.6, 34.0) in patients whose tumours were BRAF wild-type. The CHMP asked the applicant to commit to a post-authorisation measure to collect data and perform further analyses for PFS and tumour response in patients with BRAFV600 mutated tumours. The measure is included as part of the RMP.

Objective responses to nivolumab were observed regardless of whether PD-L1 expression was above or below a PD-L1 tumour membrane expression cut-off of 5% or 10%.

Regarding the results in PFS, descriptive results for PFS appear to show a benefit for nivolumab with HR=0.74 over investigator's choice chemotherapy (95% CI: 0.57, 0.97). A median PFS of 3.42 months (95% CI: 2.43, 4.90) and 2.69 months (95% CI: 2.17, 3.71) was observed in the nivolumab and comparator's arm. The difference in median PFS between the nivolumab and comparator arm was only 0.73 months, however after 3 months of treatments, the two curves in the Kaplan-Meier Plot separate in favour of nivolumab.

The OS data were not mature at the time of the PFS analysis. The OS HR was 0.93 (95%CI: 0.68, 0.1.26). The median OS in the nivolumab vs. investigator's choice chemotherapy group was 15.5 months (95% CI: 12.39, N.A.) vs. 13.7 months (95% CI: 11.5, N.A.). The 6-month OS rates were 76.7% (95% CI: 71,2, 81.3) and 78.6% (95% CI: 70.3, 84.8) for the nivolumab and investigator's choice groups, respectively. There was no statistically significant difference between nivolumab and chemotherapy in the preliminary OS analysis that was not adjusted for the potentially confounding effects of subsequent therapy. Of the 133 patients in the chemotherapy arm, 42 (31.6%) subsequently received an anti-PD1 treatment. The CHMP had concerns over the shape of the OS curves, where nivolumab and the investigator's choice of therapy crossed at around 9 months and it appeared that the OS was unfavourable for nivolumab early during treatment where there was an increase in deaths. The applicant provided a discussion on the possible confounding factors, mainly that there were baseline imbalances between the nivolumab arm and chemotherapy arm. There was a higher proportion of patients with brain metastases and elevated LDH, two risk factors that are known to negatively affect the outcome in melanoma patients. Another confounding factor was the use of post-progression therapies in the comparator arm, where 31.6% (42/133) of patients who had received investigator's choice of therapy had subsequent systemic therapy with an anti-PD-1 therapy compared to 5.5% (15/272) in the nivolumab arm. The incidence of subsequent systemic therapy in the nivolumab vs. investigator's choice group was 31.3% (85/272) vs. 54.9% (73/133), respectively. Therefore, the CHMP is of the opinion that a detrimental effect on OS does not appear likely. The same antibody has shown a clear survival benefit in the randomised phase 3 trial CA209066 and all the sensitivity analyses carried out by the applicant support the main endpoints. Even though the final analysis of OS remains to be submitted, a different profile of the curves is not expected.

The final analysis of OS is expected to occur in late 2015. Therefore, the CHMP has imposed a condition to the marketing authorisation in Annex II for the submission of updated results for OS.

2.5.4. Conclusions on the clinical efficacy

The pivotal study CA209066 provided satisfactory evidence that nivolumab prolonged OS and PFS in melanoma patients in first line treatment. Therefore, the CHMP considers that efficacy of nivolumab in patients previously untreated with unresectable and metastatic melanoma has been demonstrated. The pivotal study CA209037 provided satisfactory evidence that nivolumab treatment provided a clinically relevant benefit in terms of objective response and response duration to patients that had progressed on ipilimumab and for patients with BRAFV600 mutation, a BRAF inhibitor. This is a patient population that has few treatment options available. Hence, the CHMP considers that efficacy of nivolumab in patients previously treated with BRAFV600 inhibitors and/or ipilimumab with unresectable and metastatic melanoma has been demonstrated. Although OS was not statistically significant, patients who responded to treatment had a prolonged response which was considered clinically meaningful. In addition, PFS data, although not completely mature (66% of events) suggests a delay in the progression of the tumour as compared to chemotherapy. Nonetheless, the benefit is limited to a proportion of patients. The challenge remains to identify this patient population that derives the greatest chance to benefit from nivolumab,

Therefore, as there are concerns relating to some aspects of efficacy of the medicinal product that have been identified and can be resolved only after the medicinal product has been marketed, the CHMP considers the following measures should be provided as a condition for the marketing authorisation:

- Post-authorisation efficacy study (PAES): Final Study report for study CA209037: a Randomized, Open-Label, Phase 3 Trial of nivolumab vs Investigator's Choice in Advanced (Unresectable or Metastatic) Melanoma Patients Progressing Post Anti-CTLA-4 Therapy. This post-authorisation measure is included in the Annex II.
- Post-authorisation efficacy study (PAES): Updated OS data for Study CA209066: a Phase 3, randomized, double-blind study of nivolumab vs dacarbazine in subjects with BRAF wild type, previously untreated, unresectable or metastatic melanoma. This postauthorisation measure is included in the Annex II.
- The value of biomarkers to predict the efficacy of nivolumab should be further explored, specifically:

- To continue the exploration of the optimal cut-off for PD-L1 positivity based on current assay method used to further elucidate its value as predictive of nivolumab efficacy. These analysis will be conducted in Studies CA 209037 and CA209066 in patients with advanced melanoma.
- To further investigate the value biomarkers other than PD-L1 expression status at tumour cell membrane level by IHC (e.g., other methods / assays, and associated cut-offs, that might prove more sensitive and specific in predicting response to treatment based on PD-L1, PD-L2, tumour infiltrating lymphocytes with measurement of CD8+T density, RNA signature, etc.) as predictive of nivolumab efficacy. These additional biomarker analyses are occurring in the context of Study CA209-038 and Study CA209-066.
- To further investigate at post-approval the relation between PDL-1 and PDL-2 expression in Phase 1 studies (CA209009, CA209038 and CA209064).
- To further investigate the associative analyses between PDL-1 and PDL-2 expression conducted in Study CA209-066.
- To further investigate at post-approval the possible change in PD-L1 status of the tumour during treatment and/or tumour progression in Studies CA209-009, CA209-038 and CA209-064.

2.6. Clinical safety

Patient exposure

The estimated total number of subjects treated with nivolumab 3 mg/kg monotherapy Q2W across multiple studies and indications is approximately 1800 as of the cut-off dates for the submission (Table 37). The cumulative dose and dose intensities are shown in Table 38.

Table 37: Estimated Number of Subjects Treated with Nivolumab 3 mg/kg Monotherapy Every 2 Weeks in BMS-Sponsored Studies

Study Number	Phase	Study Design	No. of Nivolumab-treated Subjects at 3 mg/kg Q2W (No. of Total Treated Subjects)	Status at Time of SCS	SCS Database Lock Date
Melanoma					
CA209037	3	Randomized, open-label vs. investigator's choice	268 (370)	Ongoing[a]	30-Apr-2014
CA209038	1	Exploratory study of nivolumab	85 (85)	Ongoing	01-Apr-2014
CA209066	3	Randomized, double-blind vs. dacarbazine	206 (411)	Ongoing[c]	23-May-2014
CA209067	3	Randomized, double-blind, nivolumab monotherapy or combined with ipilimumab vs. ipilimumab monotherapy	302 (906)	Ongoing	27-Mar-2014
Refractory and advanced malignancies, including melanoma					
MDX1106-03 (NSCLC, melanoma, RCC, CRC, mCRPC)	1	Open-label, multicenter, multidose, dose escalation (0.1, 0.3, 1, 3, 10 mg/kg nivolumab)	54 (306)	Completed	04-Feb-2013
NSCLC					
CA209063	2	Single arm with nivolumab	117 (117)	Completed	06-Mar-2014
CA209017	3	Randomized, open-label vs. docetaxel	130 (259)	Ongoing	03-Feb-2014
CA209057	3	Randomized, open-label vs. docetaxel	278 (555)	Ongoing	30-Jan-2014
RCC[b]					
CA209025	3	Randomized, open-label vs. everolimus	395 (790)	Ongoing	11-Mar-2014
TOTAL			**1835 (3799)**		

[a] An interim CSR is available for this study. CA209037 is not considered completed because analyses of both primary endpoints are not yet available. OS data will be reported in a subsequent final CSR, estimated to be available in 4Q 2014/1Q 2015.
[b] Completed Study CA209010 in RCC is not included in this table because the 3 mg/kg dose of nivolumab was not assessed in CA209010.
[c] A Data Monitoring Committee (DMC) report is available for CA209066. A CSR is estimated to be available in 4Q 2014.
Abbreviations: CRC:= colorectal cancer; mCRPC = metastatic castrate-resistant prostate cancer; No. = number; NSCLC = non-small cell lung cancer; RCC = renal cell carcinoma; SCS = summary of clinical safety.

Table 38: Cumulative Dose and Relative Dose Intensity Summary - All Treated Subjects - CA209037

	NIVOLUMAB N = 268	DACARBAZINE N = 45	CARBOPLATIN N = 57	PACLITAXEL N = 57
RELATIVE DOSE INTENSITY				
>= 110%	0	0	0	0
90% TO < 110%	225 (84.0)	32 (71.1)	19 (33.3)	31 (54.4)
70% TO < 90%	38 (14.2)	10 (22.2)	26 (45.6)	21 (36.8)
50% TO < 70%	5 (1.9)	2 (4.4)	9 (15.8)	5 (8.8)
< 50%	0	0	3 (5.3)	0
NOT REPORTED	0	1 (2.2)	0	0
NUMBER OF DOSES RECEIVED				
MEAN (SD)	9.4 (6.70)	3.7 (2.21)	4.6 (2.18)	4.6 (2.18)
MEDIAN (MIN - MAX)	8.0 (1 - 31)	3.0 (1 - 11)	5.0 (1 - 11)	5.0 (1 - 11)
CUMULATIVE DOSE (1)				
MEAN (SD)	28.10 (20.093)	3655.72 (2279.556)	23.70 (12.572)	752.18 (358.987)
MEDIAN (MIN - MAX)	23.77 (3.0 - 93.1)	3002.51 (993.2 - 11045.6)	22.73 (4.9 - 54.0)	705.57 (171.4 - 1611.2)

(1) Dose units are mg/m2 for Dacarbazine and Paclitaxel, AUC for Carboplatin and mg/kg for Nivolumab.

Adverse events

The following table described the adverse reactions in the pooled safety population.

Table 39: Adverse reaction by worse CTC grade - Pooled CA209066 and CA209037

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	352 (74.3)	57 (12.0)	0
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS	193 (40.7)	4 (0.8)	0
FATIGUE	155 (32.7)	3 (0.6)	0
PYREXIA	28 (5.9)	0	0
OEDEMA	21 (4.4)	1 (0.2)	0
CHILLS	12 (2.5)	0	0
INFLUENZA LIKE ILLNESS	5 (1.1)	0	0
PAIN	3 (0.6)	0	0
CHEST PAIN	2 (0.4)	0	0
XEROSIS	2 (0.4)	0	0
FEELING COLD	1 (0.2)	0	0
GAIT DISTURBANCE	1 (0.2)	0	0
INFUSION SITE EXTRAVASATION	1 (0.2)	0	0
INFUSION SITE PRURITUS	1 (0.2)	0	0
INFUSION SITE RASH	1 (0.2)	0	0
MALAISE	1 (0.2)	0	0
NODULE	1 (0.2)	0	0
TEMPERATURE INTOLERANCE	1 (0.2)	0	0
SKIN AND SUBCUTANEOUS TISSUE DISORDERS	187 (39.5)	6 (1.3)	0
RASH	97 (20.5)	4 (0.8)	0
PRURITUS	86 (18.1)	1 (0.2)	0
VITILIGO	46 (9.7)	0	0
DRY SKIN	23 (4.9)	0	0
ERYTHEMA	13 (2.7)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
ALOPECIA	10 (2.1)	0	0
ECZEMA	7 (1.5)	0	0
NIGHT SWEATS	7 (1.5)	0	0
PHOTOSENSITIVITY REACTION	6 (1.3)	0	0
HAIR COLOUR CHANGES	5 (1.1)	0	0
HYPERHIDROSIS	4 (0.8)	0	0
PAPULE	3 (0.6)	0	0
SKIN HYPOPIGMENTATION	3 (0.6)	0	0
ACTINIC KERATOSIS	2 (0.4)	0	0
MACULE	2 (0.4)	0	0
NAIL DISORDER	2 (0.4)	0	0
PEMPHIGOID	1 (0.2)	1 (0.2)	0
SKIN LESION	2 (0.4)	0	0
BLISTER	1 (0.2)	0	0
COLD SWEAT	1 (0.2)	0	0
ERYTHEMA MULTIFORME	1 (0.2)	0	0
GRANULOMA SKIN	1 (0.2)	0	0
HAIR TEXTURE ABNORMAL	1 (0.2)	0	0
HYPERKERATOSIS	1 (0.2)	0	0
KERATOSIS PILARIS	1 (0.2)	0	0
NAIL DYSTROPHY	1 (0.2)	0	0
PAIN OF SKIN	1 (0.2)	0	0
PRURIGO	1 (0.2)	0	0
PSORIASIS	1 (0.2)	0	0
ROSACEA	1 (0.2)	0	0
SCAR PAIN	1 (0.2)	0	0
SEBORRHOEIC DERMATITIS	1 (0.2)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
SKIN DISORDER	1 (0.2)	0	0
SKIN IRRITATION	1 (0.2)	0	0
SKIN WARM	1 (0.2)	0	0
SOLAR DERMATITIS	1 (0.2)	0	0
GASTROINTESTINAL DISORDERS	165 (34.8)	9 (1.9)	0
DIARRHOEA	75 (15.8)	3 (0.6)	0
NAUSEA	67 (14.1)	0	0
CONSTIPATION	33 (7.0)	0	0
VOMITING	25 (5.3)	2 (0.4)	0
ABDOMINAL PAIN	19 (4.0)	1 (0.2)	0
DRY MOUTH	12 (2.5)	0	0
COLITIS	5 (1.1)	3 (0.6)	0
DYSPEPSIA	8 (1.7)	0	0
STOMATITIS	6 (1.3)	0	0
FLATULENCE	3 (0.6)	0	0
PANCREATITIS	2 (0.4)	1 (0.2)	0
FAECES DISCOLOURED	2 (0.4)	0	0
FREQUENT BOWEL MOVEMENTS	2 (0.4)	0	0
ABDOMINAL DISTENSION	1 (0.2)	0	0
DYSPHAGIA	1 (0.2)	0	0
FAECES HARD	1 (0.2)	0	0
GASTROINTESTINAL PAIN	1 (0.2)	0	0
GINGIVAL PAIN	1 (0.2)	0	0
HAEMORRHOIDS	1 (0.2)	0	0
LIP SWELLING	1 (0.2)	0	0
ORAL PAIN	1 (0.2)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
PAINFUL DEFAECATION	1 (0.2)	0	0
SENSITIVITY OF TEETH	1 (0.2)	0	0
TEETH BRITTLE	1 (0.2)	0	0
INVESTIGATIONS	56 (11.8)	15 (3.2)	0
TRANSAMINASES INCREASED	23 (4.9)	6 (1.3)	0
LIPASE INCREASED	6 (1.3)	5 (1.1)	0
WEIGHT DECREASED	10 (2.1)	0	0
BLOOD ALKALINE PHOSPHATASE INCREASED	7 (1.5)	1 (0.2)	0
AMYLASE INCREASED	5 (1.1)	2 (0.4)	0
GAMMA-GLUTAMYLTRANSFERASE INCREASED	3 (0.6)	1 (0.2)	0
LIVER FUNCTION TEST ABNORMAL	2 (0.4)	2 (0.4)	0
BLOOD CREATININE INCREASED	3 (0.6)	0	0
BLOOD THYROID STIMULATING HORMONE INCREASED	3 (0.6)	0	0
HEPATIC ENZYME INCREASED	2 (0.4)	1 (0.2)	0
BLOOD LACTATE DEHYDROGENASE INCREASED	2 (0.4)	0	0
BLOOD PHOSPHORUS DECREASED	2 (0.4)	0	0
PANCREATIC ENZYMES ABNORMAL	1 (0.2)	1 (0.2)	0
BLOOD GLUCOSE FLUCTUATION	1 (0.2)	0	0
BLOOD INSULIN INCREASED	1 (0.2)	0	0
BLOOD THYROID STIMULATING HORMONE DECREASED	1 (0.2)	0	0
BLOOD UREA INCREASED	1 (0.2)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	 Grade 3-4	 Grade 5
LYMPHOCYTE COUNT INCREASED	1 (0.2)	0	0
NEUTROPHIL COUNT INCREASED	1 (0.2)	0	0
THYROXINE FREE INCREASED	1 (0.2)	0	0
TRI-IODOTHYRONINE FREE DECREASED	1 (0.2)	0	0
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS	66 (13.9)	4 (0.8)	0
MUSCULOSKELETAL PAIN	36 (7.6)	1 (0.2)	0
ARTHRALGIA	28 (5.9)	1 (0.2)	0
MUSCLE SPASMS	6 (1.3)	1 (0.2)	0
GROIN PAIN	3 (0.6)	0	0
JOINT SWELLING	1 (0.2)	1 (0.2)	0
MUSCULOSKELETAL STIFFNESS	2 (0.4)	0	0
JOINT WARMTH	1 (0.2)	0	0
MUSCULAR WEAKNESS	1 (0.2)	0	0
OSTEOARTHRITIS	1 (0.2)	0	0
PERIARTHRITIS	1 (0.2)	0	0
SYNOVITIS	1 (0.2)	0	0
TENOSYNOVITIS	1 (0.2)	0	0
NERVOUS SYSTEM DISORDERS	61 (12.9)	4 (0.8)	0
HEADACHE	19 (4.0)	0	0
DYSGEUSIA	16 (3.4)	0	0
NEUROPATHY PERIPHERAL	13 (2.7)	1 (0.2)	0
DIZZINESS	8 (1.7)	1 (0.2)	0
MEMORY IMPAIRMENT	3 (0.6)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	 Grade 3-4	 Grade 5
DEMYELINATION	1 (0.2)	1 (0.2)	0
GUILLAIN-BARRE SYNDROME	1 (0.2)	1 (0.2)	0
LETHARGY	2 (0.4)	0	0
SCIATICA	2 (0.4)	0	0
APHONIA	1 (0.2)	0	0
PAROSMIA	1 (0.2)	0	0
PRESYNCOPE	1 (0.2)	0	0
SENSORY LOSS	1 (0.2)	0	0
SYNCOPE	1 (0.2)	0	0
TREMOR	1 (0.2)	0	0
VIITH NERVE PARALYSIS	1 (0.2)	0	0
BLOOD AND LYMPHATIC SYSTEM DISORDERS	46 (9.7)	6 (1.3)	0
ANAEMIA	29 (6.1)	3 (0.6)	0
LYMPHOPENIA	14 (3.0)	2 (0.4)	0
LEUKOPENIA	6 (1.3)	0	0
THROMBOCYTOPENIA	2 (0.4)	1 (0.2)	0
EOSINOPHILIA	2 (0.4)	0	0
LEUKOCYTOSIS	2 (0.4)	0	0
HAEMOGLOBINAEMIA	1 (0.2)	0	0
METABOLISM AND NUTRITION DISORDERS	45 (9.5)	5 (1.1)	0
DECREASED APPETITE	29 (6.1)	0	0
HYPERGLYCAEMIA	6 (1.3)	4 (0.8)	0
HYPONATRAEMIA	5 (1.1)	0	0
HYPERKALAEMIA	3 (0.6)	0	0
DIABETIC KETOACIDOSIS	1 (0.2)	1 (0.2)	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
DEHYDRATION	1 (0.2)	0	0
DIABETES MELLITUS	1 (0.2)	0	0
HYPERNATRAEMIA	1 (0.2)	0	0
HYPOALBUMINAEMIA	1 (0.2)	0	0
HYPOCALCAEMIA	1 (0.2)	0	0
HYPOGLYCAEMIA	1 (0.2)	0	0
HYPOKALAEMIA	1 (0.2)	0	0
RESPIRATORY, THORACIC AND MEDIASTINAL DISORDERS	43 (9.1)	1 (0.2)	0
DYSPNOEA	19 (4.0)	0	0
COUGH	16 (3.4)	0	0
PNEUMONITIS	11 (2.3)	0	0
HYPOXIA	2 (0.4)	0	0
PULMONARY MASS	1 (0.2)	1 (0.2)	0
DYSPHONIA	1 (0.2)	0	0
HAEMOPTYSIS	1 (0.2)	0	0
HYPERVENTILATION	1 (0.2)	0	0
LOWER RESPIRATORY TRACT INFLAMMATION	1 (0.2)	0	0
NASAL CONGESTION	1 (0.2)	0	0
OROPHARYNGEAL PAIN	1 (0.2)	0	0
PLEURAL EFFUSION	1 (0.2)	0	0
RESPIRATORY TRACT CONGESTION	1 (0.2)	0	0
THROAT IRRITATION	1 (0.2)	0	0
ENDOCRINE DISORDERS	38 (8.0)	2 (0.4)	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
HYPOTHYROIDISM	27 (5.7)	0	0
HYPERTHYROIDISM	12 (2.5)	1 (0.2)	0
HYPOPITUITARISM	3 (0.6)	0	0
HYPOPHYSITIS	1 (0.2)	1 (0.2)	0
THYROIDITIS	2 (0.4)	0	0
ADRENAL INSUFFICIENCY	1 (0.2)	0	0
HYPOGONADISM	1 (0.2)	0	0
INFECTIONS AND INFESTATIONS	22 (4.6)	3 (0.6)	0
UPPER RESPIRATORY TRACT INFECTION	7 (1.5)	0	0
FOLLICULITIS	4 (0.8)	0	0
HERPES ZOSTER	3 (0.6)	1 (0.2)	0
PNEUMONIA	2 (0.4)	1 (0.2)	0
SKIN INFECTION	1 (0.2)	1 (0.2)	0
CYSTITIS	1 (0.2)	0	0
EAR INFECTION	1 (0.2)	0	0
FUNGAL SKIN INFECTION	1 (0.2)	0	0
ORAL CANDIDIASIS	1 (0.2)	0	0
RESPIRATORY TRACT INFECTION VIRAL	1 (0.2)	0	0
URINARY TRACT INFECTION	1 (0.2)	0	0
VASCULAR DISORDERS	18 (3.8)	3 (0.6)	0
HYPERTENSION	7 (1.5)	2 (0.4)	0
FLUSHING	6 (1.3)	0	0
HYPOTENSION	4 (0.8)	1 (0.2)	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
VASCULAR PAIN	1 (0.2)	0	0
EYE DISORDERS	20 (4.2)	0	0
UVEITIS	4 (0.8)	0	0
VISION BLURRED	4 (0.8)	0	0
DRY EYE	3 (0.6)	0	0
LACRIMATION INCREASED	2 (0.4)	0	0
PHOTOPSIA	2 (0.4)	0	0
VITREOUS FLOATERS	2 (0.4)	0	0
CATARACT	1 (0.2)	0	0
EYE IRRITATION	1 (0.2)	0	0
EYE PRURITUS	1 (0.2)	0	0
EYELASH DISCOLOURATION	1 (0.2)	0	0
MACULAR HOLE	1 (0.2)	0	0
OCULAR ICTERUS	1 (0.2)	0	0
PHOTOPHOBIA	1 (0.2)	0	0
PRESBYOPIA	1 (0.2)	0	0
RETINAL OEDEMA	1 (0.2)	0	0
VISUAL IMPAIRMENT	1 (0.2)	0	0
INJURY, POISONING AND PROCEDURAL COMPLICATIONS	17 (3.6)	1 (0.2)	0
INFUSION RELATED REACTION	13 (2.7)	1 (0.2)	0
CONTUSION	1 (0.2)	0	0
POST PROCEDURAL DIARRHOEA	1 (0.2)	0	0
POSTOPERATIVE WOUND COMPLICATION	1 (0.2)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
SKIN ABRASION	1 (0.2)	0	0
RENAL AND URINARY DISORDERS	10 (2.1)	3 (0.6)	0
RENAL FAILURE	4 (0.8)	2 (0.4)	0
POLLAKIURIA	2 (0.4)	0	0
TUBULOINTERSTITIAL NEPHRITIS	1 (0.2)	1 (0.2)	0
HAEMATURIA	1 (0.2)	0	0
INCONTINENCE	1 (0.2)	0	0
URINARY INCONTINENCE	1 (0.2)	0	0
IMMUNE SYSTEM DISORDERS	12 (2.5)	0	0
HYPERSENSITIVITY	12 (2.5)	0	0
CARDIAC DISORDERS	10 (2.1)	1 (0.2)	0
TACHYCARDIA	4 (0.8)	0	0
PALPITATIONS	3 (0.6)	0	0
ARRHYTHMIA	1 (0.2)	1 (0.2)	0
ATRIAL FLUTTER	1 (0.2)	0	0
BRADYCARDIA	1 (0.2)	0	0
PERICARDIAL EFFUSION	1 (0.2)	0	0
PSYCHIATRIC DISORDERS	9 (1.9)	0	0
INSOMNIA	3 (0.6)	0	0
MOOD ALTERED	2 (0.4)	0	0
ANXIETY	1 (0.2)	0	0
ATTENTION DEFICIT/HYPERACTIVITY DISORDER	1 (0.2)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474		
	Any Grade	Grade 3-4	Grade 5
CONFUSIONAL STATE	1 (0.2)	0	0
FOOD AVERSION	1 (0.2)	0	0
NIGHTMARE	1 (0.2)	0	0
REPRODUCTIVE SYSTEM AND BREAST DISORDERS	5 (1.1)	0	0
GENITAL RASH	1 (0.2)	0	0
PENILE SWELLING	1 (0.2)	0	0
PERINEAL PAIN	1 (0.2)	0	0
PRURITUS GENITAL	1 (0.2)	0	0
SCROTAL OEDEMA	1 (0.2)	0	0
EAR AND LABYRINTH DISORDERS	4 (0.8)	0	0
EAR PRURITUS	2 (0.4)	0	0
MOTION SICKNESS	1 (0.2)	0	0
TINNITUS	1 (0.2)	0	0
HEPATOBILIARY DISORDERS	3 (0.6)	0	0
HYPERBILIRUBINAEMIA	2 (0.4)	0	0
HEPATOTOXICITY	1 (0.2)	0	0
NEOPLASMS BENIGN, MALIGNANT AND UNSPECIFIED (INCL CYSTS AND POLYPS)	3 (0.6)	0	0
MALIGNANT NEOPLASM PROGRESSION	1 (0.2)	0	0
SEBORRHOEIC KERATOSIS	1 (0.2)	0	0
TUMOUR HAEMORRHAGE	1 (0.2)	0	0

System Organ Class (%) Preferred Term (%)	NIVOLUMAB N = 474		
	Any Grade	Grade 3-4	Grade 5
CONGENITAL, FAMILIAL AND GENETIC DISORDERS	1 (0.2)	0	0
CONGENITAL NAEVUS	1 (0.2)	0	0

Serious adverse event/deaths/other significant events

<u>Serious adverse events for Study CA209037</u>

The frequency of SAEs, regardless of causality, was higher in the nivolumab group (53.0% [33.2% Grade 3-4]) vs. the investigator's choice group (23.5% [16.7% Grade 3-4]). Grade 5 SAEs were reported in 29 (10.8%) subjects in the nivolumab group and 3 (2.9%) subjects in the investigator's choice group. The most frequently-reported SAEs (all causality, > 2% of subjects) were malignant neoplasm progression (13.8%), pneumonia (2.6%), and back pain (2.2%) in the nivolumab group and malignant neoplasm progression (4.9%), vomiting (2.9%), and dyspnea (2.9%) in the investigator's choice group. The frequency of serious cardiac events (all grades) was 4.9% and 0 in the nivolumab and investigator's choice group, respectively. In the nivolumab group, only one (0.4%) subject had a serious cardiac event that was considered by the investigator to be related to study drug.

The only SAE of any grade, regardless of causality, reported by ⩾ 2% of subjects treated with nivolumab was malignant neoplasm progression (10.1%).

<u>Serious adverse events for CA209066</u>

The overall frequency of SAEs was lower in the nivolumab group than in the dacarbazine group (31.1% vs 38.0%). No individual SAE, except malignant neoplasm progression, was reported with a frequency ≥ 2% in the nivolumab group (4.9% vs 8.3% nivolumab and DTIC).

In order to characterize AEs of clinical importance that are potentially associated with the use of nivolumab, select AE categories were defined and consistent criteria were applied across the nivolumab clinical development program. Select AEs included endocrinopathies, diarrhea/colitis, hepatitis, pneumonitis, nephritis, and rash, with the multiple event terms that may describe each of these grouped into endocrine, GI, liver, pulmonary, renal, and skin select AE categories, respectively. According to the data submitted, the immunological AEs related to nivolumab (pooled studies CA209003, CA209037, CA209063, CA209066) include skin (31.3%), gastrointestinal (15.0%) endocrine (7.9%; 7.1% within thyroid disorder subcategory), hepatic (5.1%), pulmonary (2.8%), and renal (2.2%) events, with the following terms reported: pruritis (14.9%), diarrhea (14.4%), hypothyroidism (4.8%), ALT increased (2.9%), pneumonitis (2.5%), and blood creatinine increased (0.9%).

Table 40: Summary of drug related select adverse events (treated subjects) – study CA209037 and CA209066

	Study CA209037				Study CA209066			
	% of Subjects				% of Subjects			
	Nivolumab (N = 268)		Investigator's Choice (N = 102)		Nivolumab (N = 206)		Dacarbazine (N = 205)	
Select AE Category- All Drug-related AEs Most frequent drug-related AEs in select AE category[a]	All Grades	Grade 3-4	All Grades	Grade 3-4	All Grades	Grade 3-4	All Grades	Grade 3-4
Skin	**94 (35.1)**	**1 (0.4)**	**13 (12.7)**	**0**	**77 (37.4)**	**3 (1.5)**	**29 (14.1)**	**0**
Pruritus	51 (19.0)	0	2 (2.0)	0	35 (17.0)	1 (0.5)	11 (5.4)	0
Rash	34 (12.7)	1 (0.4)	5 (4.9)	0	31 (15.0)	1 (0.5)	6 (2.9)	0
Vitiligo	24 (9.0)	0	0	0	22 (10.7)	0	1 (0.5)	0
Rash maculo-papular	16 (6.0)	0	2 (2.0)	0	6 (2.9)	1 (0.5)	0	0
Gastrointestinal	**43 (16.0)**	**3 (1.1)**	**16 (15.7)**	**2 (2.0)**	**35 (17.0)**	**3 (1.5)**	**32 (15.6)**	**1 (0.5)**
Diarrhea	42 (15.7)	1 (0.4)	16 (15.7)	2 (2.0)	33 (16.0)	2 (1.0)	32 (15.6)	1 (0.5)
Endocrine	**24 (9.0)**	**0**	**1 (1.0)**	**0**	**15 (7.3)**	**2 (1.0)**	**1 (0.5)**	**0**
Hypothyroidism	18 (6.7)	0	0	0	9 (4.4)	0	1 (0.5)	0
Hyperthyroidism	5 (1.9)	0	1 (1.0)	0	7 (3.4)	1 (0.5)	0	0
Hepatic	**22 (8.2)**	**6 (2.2)**	**6 (5.9)**	**0**	**7 (3.4)**	**3 (1.5)**	**8 (3.9)**	**2 (1.0)**
AST increased	15 (5.6)	2 (0.7)	2 (2.0)	0	2 (1.0)	1 (0.5)	4 (2.0)	1 (0.5)
ALT increased	14 (5.2)	3 (1.1)	1 (1.0)	0	3 (1.5)	2 (1.0)	3 (1.5)	1 (0.5)
Pulmonary	**8 (3.0)**	**0**	**0**	**0**	**3 (1.5)**	**0**	**0**	**0**
Pneumonitis	7 (2.6)	0	0	0	3 (1.5)	0	0	0
Hypersensitivity/infusion reactions	**10 (3.7)**	**1 (0.4)**	**10 (9.8)**	**0**	**15 (7.3)**	**0**	**13 (6.3)**	**0**
Hypersensitivity	6 (2.2)	0	1 (1.0)	0	6 (2.9)	0	5 (2.4)	0
Infusion-related reaction	4 (1.5)	1 (0.4)	9 (8.8)	0	9 (4.4)	0	8 (3.9)	0
Select AE Category- All Drug-related AEs Most frequent drug-related AEs in select AE category[a]	All Grades	Grade 3-4	All Grades	Grade 3-4	All Grades	Grade 3-4	All Grades	Grade 3-4
Renal	**5 (1.9)**	**2 (0.7)**	**1 (1.0)**	**0**	**4 (1.9)**	**1 (0.5)**	**1 (0.5)**	**0**
Blood creatinine increased	2 (0.7)	0	0	0	1 (0.5)	0	1 (0.5)	0

a The most frequently reported AE in each category is presented along with any other AEs reported in more than 5 subjects in either treatment group.

Abbreviations: AE: adverse event; ALT: alanine aminotransferase; AST: aspartate aminotransferase.

Endocrine Select Adverse Events

The endocrine select AE category included the following sub-categories: adrenal disorders, diabetes, pituitary disorders, and thyroid disorders. In order to monitor for endocrine select AEs in the thyroid disorder and diabetes sub-categories, routine testing of thyroid stimulating hormone and blood glucose every 6 weeks while receiving nivolumab, was performed.

Adverse event terms belonging to the endocrine select AE category were reported in 32 (11.9%) of the patients in the nivolumab group and in 2 (2.0%) of the patients in the investigator's choice group. The majority of the patients with AEs belonging to the endocrine select AE category were grade 1-2, the only grade 3 AE was a SAE of adrenal insufficiency. This AE was considered not related to nivolumab by the investigator. The most frequently reported terms (≥2% of the patients), regardless of causality, were hypothyroidism and hyperthyroidism in in the nivolumab group. No terms were reported in more than 1 patient in the investigator's choice group.

The median time to onset of any grade AE belonging to the endocrine select AE category was 8.1 weeks (range: 0.1 weeks to 46.9+ weeks), while the time to onset of the single grade 3 event of adrenal insufficiency was 2.1 weeks.

Of the 32 patients in the nivolumab group who experienced AEs belonging to the endocrine select AE category, 14 (43.8%) had events that resolved with a median time to resolution of 24.1 weeks. The single grade 3 event resolved after 0.6 weeks. Four patients received immunosuppressive medication (systemic corticosteroids) for treatment of the AE.

Table 41: Summary of adverse events belonging to the endocrine select adverse event category

Sub Category (%) Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	40 (8.4)	3 (0.6)	0
THYROID DISORDER	36 (7.6)	1 (0.2)	0
HYPOTHYROIDISM	27 (5.7)	0	0
HYPERTHYROIDISM	12 (2.5)	1 (0.2)	0
BLOOD THYROID STIMULATING HORMONE INCREASED	3 (0.6)	0	0
THYROIDITIS	2 (0.4)	0	0
BLOOD THYROID STIMULATING HORMONE DECREASED	1 (0.2)	0	0
THYROXINE FREE INCREASED	1 (0.2)	0	0
DIABETES	2 (0.4)	1 (0.2)	0
DIABETES MELLITUS	1 (0.2)	0	0
DIABETIC KETOACIDOSIS	1 (0.2)	1 (0.2)	0
ADRENAL DISORDER	1 (0.2)	0	0
ADRENAL INSUFFICIENCY	1 (0.2)	0	0
PITUITARY DISORDER	1 (0.2)	1 (0.2)	0
HYPOPHYSITIS	1 (0.2)	1 (0.2)	0

Gastrointestinal Select Adverse Events

The GI select AE category included the following terms: colitis, colitis ulcerative, diarrhoea, enteritis, enterocolitis, frequent bowel movements, and GI perforation.

Adverse event terms belonging to the GI select AE category were reported in 20.5% of the patients treated with nivolumab and in 16.7% of the patients in the investigator's choice arm. The most frequently reported AE (≥2% of the patients), regardless of causality, was diarrhoea in both the nivolumab and the investigator's choice groups. Two of patients in both the nivolumab group and the investigator's choice group, experienced grade 3 diarrhoea. In the nivolumab group, two other patients experienced Grade 3 drug-related colitis, which was a SAE and led to treatment discontinuation.

In the nivolumab group, the median time to onset of any grade AE belonging to the GI select AE category was 6.3 weeks (range: 0.1 to 37.6 weeks), while the median time to onset of the 4 Grade 3 events was 10.8 weeks (range: 4.1 to 26.7 weeks).

The majority (90.9%) of GI AEs resolved. Four of the patients in the nivolumab group were treated with immunosuppressive medications (corticosteroid treatment). The median total duration of immunosuppressive medication given was 6.1 weeks. For all AEs belonging to the GI select AE category, the median time to resolution was 1.1 weeks.

Table 42: Summary of adverse events belonging to the gastrointestinal select adverse event category

	NIVOLUMAB N = 474		
Preferred Term (%)	Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	78 (16.5)	6 (1.3)	0
DIARRHOEA	75 (15.8)	3 (0.6)	0
COLITIS	5 (1.1)	3 (0.6)	0
FREQUENT BOWEL MOVEMENTS	2 (0.4)	0	0

Hepatic select adverse events

The hepatic select AE category included the following terms: acute hepatic failure, ALT increased, AST increased, bilirubin conjugate increased, blood bilirubin increased, gamma-glutamyl-transferase (GGT) increased, hepatic enzyme increased, hepatic failure, hepatitis, hepatitits acute, hyperbilirubinemia, liver disorder, liver function test abnormal, liver injury, and transaminases increased.

Adverse event terms belonging to the hepatic select AE category were reported in 10.4% of the patients treated with nivolumab and in 6.9% of the patients in the investigator's choice arm. The most frequently reported AE (≥2% of the patients), regardless of causality, were AST increased and ALT increased in the nivolumab group and AST increased , ALT increased and GGT increased in the investigator's choice group. Nine of patients in the nivolumab group experienced grade 3-4 AE.

In the nivolumab group, the median time to onset of any grade AE belonging to the hepatic select AE category was 4.1 weeks (range: 0.1 to 23.0 weeks), while the median time to onset of the 4 Grade 3 events was 2.3 weeks (range: 0.9 to 20.3 weeks).

In the nivolumab group, 46.4% of the patients experienced a hepatic AE, resolved. Four of the patients in the nivolumab group were treated with immunosuppressive medications (corticosteroid treatment). The median total duration of immunosuppressive medication given was 7.9 weeks. For all AEs belonging to the hepatic select AE category, the median time to resolution was 6.1 weeks for any-grade AEs and 2.9 weeks for Grade 3-4 AEs.

Table 43: Summary of adverse events belonging to the hepatic select adverse event category

Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	29 (6.1)	9 (1.9)	0
ALANINE AMINOTRANSFERASE INCREASED	17 (3.6)	5 (1.1)	0
ASPARTATE AMINOTRANSFERASE INCREASED	17 (3.6)	3 (0.6)	0
GAMMA-GLUTAMYLTRANSFERASE INCREASED	3 (0.6)	1 (0.2)	0
BLOOD BILIRUBIN INCREASED	2 (0.4)	0	0
HEPATIC ENZYME INCREASED	2 (0.4)	1 (0.2)	0
LIVER FUNCTION TEST ABNORMAL	2 (0.4)	2 (0.4)	0
TRANSAMINASES INCREASED	2 (0.4)	0	0
HEPATOTOXICITY	1 (0.2)	0	0

Pulmonary select Adverse Events

The pulmonary select AE category included the following terms: acute respiratory distress syndrome, acute respiratory failure, interstitial lung disease, lung infiltration, and pneumonitis. Pneumonia was not included as a term in the pulmonary select AE category because of the high frequency with which it was expected to be reported in the study population, especially to describe infectious rather than non-infectious pneumonitis. According to the applicant, inclusion of pneumonia as a select AE term would hinder characterization of the true frequency of pneumonitis.

AEs belonging to the pulmonary select AE category were reported in 8 patients (3.0%) in the nivolumab group and in no patients in the investigator's choice group. Pneumonitis was the most frequently reported term in the nivolumab group (7 patients, 2.6%). No grade 3 or higher AEs belonging to the pulmonary select AE category were reported in either treatment group.

In the nivolumab group, the median time to onset of any grade AE, regardless of causality, was 8.7 weeks (range: 3.6 to 26.0 weeks).

Pulmonary AEs resolved in fifty percent of the patients treated with nivolumab. Five of the patients with a pulmonary AE were treated with immunosuppressive medications (corticosteroid treatment). Four of the patients who received immunosuppressive medication the pulmonary AE resolved. The median time to resolution among these patients was 3.3 weeks (range: 2.3 to 8.1+ weeks).

Table 44: Summary of adverse events belonging to the pulmonary select adverse event category

Preferred Term (%)	NIVOLUMAB N = 474 Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	11 (2.3)	0	0
PNEUMONITIS	10 (2.1)	0	0
INTERSTITIAL LUNG DISEASE	1 (0.2)	0	0

Renal select adverse events

The renal select AE category included the following terms: blood creatinine increased, blood urea increased, blood urea increased, creatinine renal clearance decreased, hypercreatinemia, nephritis, nephritis allergic, nephritis autoimmunune, renal failure, acute renal failure, renal tubular necrosis, tubulointerstitial nephritis, and urine output decreased.

Adverse events terms belonging to the renal select AE category were reported in 18 (6.7%) patients in the nivolumab group and 4 (3.9%) patients in the investigator's choice group. The most frequently reported terms (≥2% of the patients), regardless of causality, were blood creatinine increased in the nivolumab group and renal failure acute in the investigator's choice group.

The majority of the AEs belonging to the renal select AE category were Grade 1-2. Grade 3-4 AEs were reported in 4 (1.5%) and 1 (1.0%) of the patients in the nivolumab and investigator's choice groups, respectively. No grade 5 renal AEs or AEs leading to discontinuation were reported.

In the nivolumab group, the median time to onset of any grade AE was 6.1 weeks (range:0.9 to 45.3 weeks), while the median time to onset of the 4 Grade 3-4 events was 6.9 weeks (range: 2.1 to 15.4 weeks).

In the nivolumab group, 61.1% of the patients experienced a renal AE, resolved. Two of the patients in the nivolumab group were treated with immunosuppressive medications (corticosteroid treatment). The median total duration of immunosuppressive medication given was 2.3 weeks. For all AEs belonging to the renal select AE category, the median time to resolution was 5.1 weeks for any-grade AEs and 2.4 weeks for Grade 3-4 AEs.

Table 45: Summary of adverse events belonging to the renal select adverse event category

	NIVOLUMAB N = 474		
Preferred Term (%)	Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	9 (1.9)	3 (0.6)	0
BLOOD CREATININE INCREASED	3 (0.6)	0	0
RENAL FAILURE	2 (0.4)	0	0
RENAL FAILURE ACUTE	2 (0.4)	2 (0.4)	0
BLOOD UREA INCREASED	1 (0.2)	0	0
TUBULOINTERSTITIAL NEPHRITIS	1 (0.2)	1 (0.2)	0

Skin select adverse events

The skin select AE category included the following terms, blister, dermatitis, dermatitis exfoliative, drug eruption, eczema, erythema, erythema multiform, exfoliative rash, palmarplantar erythordysesthesia, syndrome, photosensititvity reaction, pruritius, pruritus allergic, pruritus generalized, psoriasis, rash, rash erythematous, rash generalized, rash macular, rash maculo-papular, rash popular, rash pruritic, skin exfoliation, skin hypopimantation, skin irritation, Steve-Johnson Syndrome, toxic epidermal necrolysis, urticarial, and vitiligo.

Adverse events terms belonging to the skin select AE category were reported in 96 (35.8%) patients in the nivolumab group and 15 (14.7%) patients in the investigator's choice group. The most frequently reported terms (≥2% of the patients), regardless of causality, were pruritis, rash, rash maculo-papular, vitiligo, and dermatitis in the nivolumab group and pruritus, rash, rash maculo-papular, and photosensitivity reaction in the investigator's choice group.

The majority of the AEs belonging to the skin select AE category were Grade 1-2 (95 of 96 patients in the nivolumab group and all 15 patients in the investigator's choice group). For one patient treated with nivolumab a grade 3 AE of rash was reported.

In the nivolumab group, the median time to onset of any grade AE was 4.3 weeks (range: 0.1 to 42.7 weeks).

In the nivolumab group, 45.7% of the patients experienced a skin AE, resolved. Thirty-two of the patients in the nivolumab group were treated with immunosuppressive medications. The majority were dermatological corticosteroid preparations. Three patients received systemic corticosteroid treatment. The median total duration of immunosuppressive medication given was 11.1 weeks. For all AEs belonging to the skin select AE category, the median time to resolution was 16.1.

Table 46: Summary of adverse events belonging to the skin select adverse event category

	NIVOLUMAB N = 474		
Preferred Term (%)	Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	171 (36.1)	4 (0.8)	0
PRURITUS	86 (18.1)	1 (0.2)	0
RASH	65 (13.7)	2 (0.4)	0
VITILIGO	46 (9.7)	0	0
RASH MACULO-PAPULAR	22 (4.6)	1 (0.2)	0
ERYTHEMA	13 (2.7)	0	0
ECZEMA	7 (1.5)	0	0
RASH PAPULAR	7 (1.5)	0	0
PHOTOSENSITIVITY REACTION	6 (1.3)	0	0
DERMATITIS	5 (1.1)	0	0
RASH ERYTHEMATOUS	5 (1.1)	0	0
RASH MACULAR	5 (1.1)	0	0
RASH PRURITIC	3 (0.6)	0	0
SKIN HYPOPIGMENTATION	3 (0.6)	0	0
DERMATITIS EXFOLIATIVE	2 (0.4)	0	0
PRURITUS GENERALISED	2 (0.4)	0	0
BLISTER	1 (0.2)	0	0
ERYTHEMA MULTIFORME	1 (0.2)	0	0
PSORIASIS	1 (0.2)	0	0
SKIN IRRITATION	1 (0.2)	0	0

Hypersensitivity/infusion Reactions

The hypersensitivity and infusion reactions are shown in Table 47.

Table 47: Summary of Hypersensitivity/infusion Reactions

	NIVOLUMAB N = 474		
Preferred Term (%)	Any Grade	Grade 3-4	Grade 5
TOTAL SUBJECTS WITH AN EVENT	25 (5.3)	1 (0.2)	0
INFUSION RELATED REACTION	13 (2.7)	1 (0.2)	0
HYPERSENSITIVITY	12 (2.5)	0	0

Deaths

In CA209037, the total number of deaths in the ORR analysis was 24% vs 25% for investigator's choice versus nivolumab groups; however, with further follow-up, the total number of deaths at the time of the interim OS analysis was higher in the chemotherapy group compared to nivolumab (52% vs 44%), which is consistent with CA209066 where a greater number of deaths occurred in the dacarbazine group than the nivolumab group (47% vs 23%), respectively.

Of the 119 deaths in the nivolumab group, 113 were due to disease progression, 2 were of unknown causes, and 4 were due to "other" reasons. In the four patients, in the nivolumab group within the study CA209037, with the reason for death reported as 'other', the relationship between the use of nivolumab and the cause of death as a consequence of cardio-pulmonary AEs, is not clear. Either the use of other treatments before the death (trametinib, vemurafenib, chemotherapy, steroids) or the time after the last dose of nivolumab, make difficult to reach any conclusion. In the study CA209066, disease progression seems to be the cause of death in 2 out of 3 patients with the reason for death reported as 'other', whereas the history of cardiac disorders seemed to be the cause of death in the last one. For the time being, the relationship between death due to cardio-pulmonary AEs and the use of nivolumab has not been demonstrated.

Laboratory findings

Study CA209037

Among all treated patients, any grade shifts from baseline value were reported for selected laboratory tests including ALP, total bilirubin, AST, ALT, absolute neutrophils, creatine, haemoglobin, leukocytes, lymphocytes, and platelet count. Laboratory measurements were recorded regardless of causality and some were correlated with reported laboratory-based AEs.

Hematology

Abnormalities in haematology tests performed within 30 days of last nivolumab or investigator's choice dose were primarily Grade 1-2 in severity. Grade 3-4 hematologic abnormalities (haemoglobin, platelet count, leukocytes, lymphocytes, and absolute neutrophil count) were less frequently reported in the nivolumab group compared with the investigator's choice group.

In the nivolumab group, Grade 3-4 hematologic abnormalities (all decreases) occurred in <15% of the patients, with the most frequent being absolute lymphocyte count decreased (23/258 patients, 8.9%) and haemoglobin decreased (17/261 patients, 6.5%).

In the investigator's choice group, the most frequently occurring Grade 3-4 hematologic abnormalities (>15% of the patients) were absolute lymphocyte count decreased (19/99 patients, 19.2%) and absolute neutrophil count decreased (21/99 patients, 21.2%).

Liver Function tests

The majority of patients in both treatment groups had normal ALP, ALT, AST, and total bilirubin levels during the treatment reporting period. In the nivolumab and investigator's choice groups, abnormalities in select liver function tests were primarily Grade 1-2 in severity.

Grade 3-4 increases in liver function tests (ALP, ALT, AST and total bilirubin) were reported more frequently in the nivolumab group compared to the investigator's choice group.

Table 48: Summary on-treatment liver function Tests by Worst CTC grade (SI Units) (all treated patients) – Study CA209037

Lab Test Description / Toxicity Grade	NIVOLUMAB N = 268	INVESTIGATOR'S CHOICE N = 102
ALKALINE PHOSPHATASE	N = 256	N = 96
GRADE 0	157 (61.3)	65 (67.7)
GRADE 1	75 (29.3)	27 (28.1)
GRADE 2	17 (6.6)	3 (3.1)
GRADE 3	7 (2.7)	1 (1.0)
GRADE 4	0	0
ASPARTATE AMINOTRANSFERASE	N = 255	N = 97
GRADE 0	172 (67.5)	83 (85.6)
GRADE 1	71 (27.8)	12 (12.4)
GRADE 2	6 (2.4)	1 (1.0)
GRADE 3	5 (2.0)	1 (1.0)
GRADE 4	1 (0.4)	0
ALANINE AMINOTRANSFERASE	N = 256	N = 97
GRADE 0	203 (79.3)	89 (91.8)
GRADE 1	46 (18.0)	7 (7.2)
GRADE 2	3 (1.2)	1 (1.0)
GRADE 3	4 (1.6)	0
GRADE 4	0	0
BILIRUBIN, TOTAL	N = 256	N = 96
GRADE 0	230 (89.8)	95 (99.0)
GRADE 1	15 (5.9)	1 (1.0)
GRADE 2	10 (3.9)	0
GRADE 3	1 (0.4)	0
GRADE 4	0	0

Toxicity Scale: CTC Version 4.0
Includes laboratory results reported after the first dose and within 30 days of last dose of study therapy.
Percentages are based on subjects with on-treatment laboratory results and CTC grade criteria available.

Kidney Function Tests

In the nivolumab and investigator's choice groups, the majority (83.6% vs 86.5% respectively) of patients with at least 1 on-treatment measurement had normal (Grade 0) creatine values during the treatment reporting period. The incidence of a 2-grade or greater shift in creatinine elevation was higher in the nivolumab group than in the investigator's choice group (4.3% vs. 2.1%, respectively).

Thyroid function tests

In the nivolumab group, a higher proportion of patients with TSH levels ≤ULN at baseline experienced TSH elevation (>TSN) compared with the investigator's choice group (21.2% vs. 5.5.%, respectively). A higher proportion of patients in the nivolumab group had at least one on-study elevated TSH and one T3 or T4<lower limit of normal (LLN) compared with the investigator's choice group (11.0% vs. 1.1%).

The frequency of patients with TSH≥LLN at baseline that experienced low TSH levels (<LLN) was similar between the nivolumab and investigator's choice groups (13.1% vs 13.2% respectively), as was the frequency of patients with at least 1 on-study low TSH and 1 free T3 or free T4>ULN.

Immunogenicity

An integrated summary of immunogenicity of nivolumab in treatment refractory solid tumour population from studies MDX1106-03, CA209063, and CA209037 is presented in Table 49.

Table 49: Integrated Summary of anti-nivolumab anti-drug antibody (ADA) assessments – Studies MDX1106-03, CA209063, and CA209037

	Number of Subjects (%)
	NIVOLUMAB all doses (N = 524)
BASELINE ADA POSITIVE	27 (5.2)
ADA POSITIVE	45 (8.6)
PERSISTENT POSITIVE	4 (0.8)
ONLY AT LAST SAMPLE	24 (4.6)
NEUTRALIZING POSITIVE	2 (0.4)
ADA NEGATIVE	479 (91.4)

Overall, out of 524 nivolumab treated subjects who had available ADA assessment at baseline and post-baseline from MDX1106-03, CA209063, and CA209037 studies, there was detectable neutralising antibodies in 2 (0.4%) subjects. Neutralising antibodies were not detectable in subsequent ADA assessments in these patients; one of the subjects with neutralising positive ADA was also persistent positive, but subsequent assessment was ADA positive with lower titer and undetectable neutralising antibodies. The safety profiles of these 2 neutralising antibody positive subjects were no different than those seen in other subjects. There were no acute infusion reactions, hypersensitivity events, new or additional AEs observed in subjects with neutralising antibodies. In addition, patients with neutralising antibodies continued treatment with benefit from therapy and thus no clear association can be established between presence of neutralising antibodies and loss of efficacy.

In summary, the immunogenic potential of nivolumab was minimal with relatively low titers, low persistent positive rates, low incidences of neutralising antibodies and no impact of immunogenicity on safety.

Safety in special populations

The safety profile of nivolumab was similar between the subgroups based on intrinsic (race, age, gender) and extrinsic (geographical location) factors.

A summary of adverse events in the elderly population is presented below.

Table 50: Summary of Adverse Events by Age Groups – Pooled studies CA209037 and CA209066

MedDRA Terms	Age < 65 years (N=279) n (%)	Age 65 - 74 years (N=129) n (%)	Age 75-84 years (N=61) n (%)	Age 85+ years (N=5) n (%)
Total AEs	269 (96.4)	124 (96.1)	59 (96.7)	5 (100.0)
Serious AEs – Total	126 (45.2)	53 (41.1)	25 (41.0)	2 (40.0)
- Fatal	28 (10.0)	13 (10.1)	3 (4.9)	0
- Hospitalization/prolong existing hospitalization	111 (39.8)	46 (35.7)	21 (34.4)	2 (40.0)
- Life-threatening	5 (1.8)	3 (2.3)	0	0
- Disability/incapacity	0	0	0	0
- Other (medically significant)	10 (3.6)	3 (2.3)	2 (3.3)	0
AE leading to drop-out	25 (9.0)	10 (7.8)	13 (21.3)	0
Psychiatric disorders	57 (20.4)	17 (13.2)	8 (13.1)	1 (20.0)
Nervous system disorders	107 (38.4)	43 (33.3)	21 (34.4)	2 (40.0)
Accidents and injuries	15 (5.4)	13 (10.1)	3 (4.9)	1 (20.0)
Cardiac disorders	24 (8.6)	15 (11.6)	4 (6.6)	2 (40.0)
Vascular disorders	47 (16.8)	27 (20.9)	11 (18.0)	2 (40.0)
Cerebrovascular disorders	6 (2.2)	3 (2.3)	0	1 (20.0)
Infections and infestations	95 (34.1)	64 (49.6)	26 (42.6)	3 (60.0)
Anticholinergic syndrome	84 (30.1)	44 (34.1)	18 (29.5)	1 (20.0)
Quality of life decreased	0	0	0	0
Sum of postural hypotension, falls, black outs, syncope, dizziness, ataxia, fractures	23 (8.2)	18 (14.0)	6 (9.8)	1 (20.0)

Abbreviations: AEs = adverse events; MedDRA = Medical Dictionary for Regulatory Activities

Safety related to drug-drug interactions and other interactions

The applicant did not submit studies on drug-drug interaction (see safety discussion).

Discontinuation due to adverse events

Study CA209037

The frequency of AEs leading to discontinuation of study drug (all grades), regardless of causality, was lower in the nivolumab group (12.7%) vs. the investigator's choice group (15.7%), whereas the frequency of Grade 3-4 AEs leading to discontinuation, regardless of causality, was higher in the nivolumab group (9.7%) than the investigator's choice group (2.9%).

The most frequently reported (> 2% of subjects) AEs leading to discontinuation, regardless of causality, were malignant neoplasm progression in the nivolumab group (4.5%) and peripheral neuropathy and fatigue in the investigator's choice group (2.9% each). The frequency of cardiac events leading to discontinuation, regardless of causality, was 12.7% and 0% in the nivolumab and investigator's choice groups, respectively.

The frequency of drug-related AEs leading to discontinuation (all grades) was lower in the nivolumab group (4.5%) vs. the investigator's choice group (10.8%), whereas the frequency of Grade 3-4 AEs leading to discontinuation was higher in the nivolumab group (4.1%) vs. the investigator's choice group (2.0%). No drug related AEs leading to discontinuation were reported by > 2% of subjects in the nivolumab group.

Study CA209066

The overall frequency of AEs leading to discontinuation was lower in the nivolumab group compared with the dacarbazine group (6.8% vs 11.7%). Most AEs leading to discontinuation were Grade 3-4. Malignant neoplasm progression was the most frequently reported AE leading to discontinuation in both groups and no other individual event was reported with a frequency ≥ 2% in either group.

Post marketing experience

The applicant did not submit post-marketing experience.

2.6.1. Discussion on clinical safety

The safety profile of nivolumab was characterized in 3 studies where nivolumab was given as monotherapy: 2 Phase 3 studies (CA209037 and CA209066) in advanced melanoma and 1 Phase 1b study (MDX1106-03) in multiple tumour types, including melanoma.

The administration of nivolumab is associated with an increase in the rate of fatigue, nausea, diarrhoea, pruritus, anaemia, cough, rash and dyspnoea. The most frequently reported ADRs of any grade were fatigue, pruritus, nausea, diarrhea, and rash. Select AEs on the basis of its mechanism of action and rate of frequency, include endocrinopathies, diarrhoea/colitis, hepatitis, pneumonitis, nephritis, and rash, with the multiple event terms that may describe each of these grouped into endocrine, GI, hepatic, pulmonary, renal, and skin select AE categories, respectively.

According to the data submitted, the immunological ADRs related to nivolumab (pooled studies CA209003, CA209037, CA209063, CA209066) and identified as important identified risks include skin (31.3%), gastrointestinal (15.0%) endocrine (7.9%; 7.1% within thyroid disorder subcategory), hepatic (5.1%), pulmonary (2.8%), and renal (2.2%) events, with the following terms reported: pruritis (14.9%), diarrhea (14.4%), hypothyroidism (4.8%), ALT increased (2.9%), pneumonitis (2.5%), and blood creatinine increased (0.9%). The SmPC section 4.2 and 4.4 and 4.8 contain the recommendations on how to manage the immunologic ADRs. There is a lack of information on patients with autoimmune disease and patients already receiving systemic immunosuppressants before starting nivolumab. A warning in section 4.4 on special population and in 4.5 on the use of systemic immunosuppressants has been included in the SmPC. In addition, these populations have been included in the RMP as missing information.

In the first line setting, the overall frequency of SAEs was lower in the nivolumab group than in the dacarbazine group (31.1% vs 38.0%). No individual SAE, except malignant neoplasm progression, was reported with a frequency ≥ 2% in the nivolumab group (4.9% vs 8.3% nivolumab and DTIC). The incidence of SAE, adjusted by incidence rate per 100 person-years of exposure, balances the groups of the study, with a pretty similar rate (156 vs 145 nivilumab and chemo respectively). This fact seems to highlight that the longer exposure could be responsible of the higher incidence of SAE in the nivolumab group (44% vs 21.6%). In study CA209037, the frequency of cardiac events, regardless of causality, was 12.7% (34 subjects) and 0% in the nivolumab and investigator's choice groups, respectively but was lower in the nivolumab group than dacarbazine group in the metastatic melanoma without prior treatment population. All were considered not related to nivolumab by investigators except arrhythmia (atrial fibrillation, tachycardia and ventricular arrhythmia). Overall, nivolumab does not have QT prologation potential in the studied dose range. Therefore, the risk of cardiac arrhythmias has been included in the RMP as an important potential risk. The Applicant plans to conduct a PASS which will provide further insight in the safety of nivolumab under clinical practice conditions for use to investigate postmarketing use safety profile, management and outcome of immune-related pneumonitis, colitis, hepatitis, nephritis or renal dysfunction, endocrino-pathies, and other immune-related adverse reactions. Details of the design of this PASS will be further discussed within 3 months of the EC Decision with PRAC. It is expected data from all these studies will be provided as part of the PSURs, which is considered adequate. This is reflected in the RMP.

Regarding deaths, in study CA209037, a total of 119 (44.4%) subjects in the nivolumab group and 53 (52%) subjects in the investigator's choice group died at the time of the database lock for the Interim Ad hoc Report. Regarding the profile of deaths during the first 9 weeks, this seems to extend over the first 3months, even though disease progression appears to be the cause of the vast majority of deaths in the first 9 weeks (28 deaths out of 29 were due to disease progression). Importantly, during the first 3months of nivolumab treatment, a higher frequency of death due to malignant neoplasm progression than for the investigator's choice treatment was reported. These results are probably a reflection of worse prognostic factors in the nivolumab arm (brain metastases and LDH levels) than in the chemotherapy group in the CA209066 study, a total of 47 (22.8%) subjects in the nivolumab group and 96 (46.8%) subjects in the dacarbazine group died prior to database lock.

The laboratory findings seem to highlight the higher haematological toxicity associated to chemotherapy and the higher number of hepatic events grade 3-4 and thyroid test abnormalities related to nivolumab.

As with any other intravenous administered drugs, infusion-related reactions can occur with nivolumab. Severe infusion reactions have been reported in clinical trials (see section 4.8 of the SmPC). This risk has been included in the RMP as an important identified risk.

Nivolumab showed a low immunogenicity potential, overall from the 524 evaluable subjects only 4 (0.8%) patients were persistently positive of whom 2 (0.4%) were positive for neutralizing antibodies. There was no evidence of acute infusion reactions, hypersensitivity reactions or altered safety profile in patients with persistent or neutralizing antibodies. The number of patients tested for ADA was low, therefore, the risk of developing ADA was considered not yet been fully investigated. Results will be presented in the final clinical study report for each respective study. The risk of immunogenicity has been included in the RMP as an important potential risk and is included in the SmPC in section 4.8.

Like most therapeutic proteins, nivolumab is not metabolised by liver cytochrome (CYP) P450 metabolising enzymes or other drug-metabolising enzymes, and is not expected to have an effect on cytochrome P450 or other drug-metabolising enzymes in terms of inhibition or induction. In addition, nivolumab treatment did not result in any meaningful change in cytokines known to have indirect effect on CYP enzymes across the dose range 0.3 to 10 mg/kg. The lack of cytokine modulation suggests nivolumab has no or low potential for modulating CYP enzymes and therefore, there is a low risk of a therapeutic protein-drug interaction. Therefore, the lack of studies investigating the safety related to drug-drug interaction is acceptable. There is missing information for patients below 18 years of age, patients with severe hepatic and/or renal impairment. The missing information has been included as part of the RMP and is also described in section 4.2 and 5.2 of the SmPC.

From the safety database all the adverse reactions reported in clinical trials have been included in the Summary of Product Characteristics.

2.6.2. Conclusions on the clinical safety

The ADRs reported for patients being treated with nivolumab appear to be mostly of low grade and manageable. It was noted that immunological ADRs include skin, gastrointestinal, endocrine, hepatic, pulmonary and renal events. These are managed appropriately with the recommendations as stated in the SmPC and are also addressed in the RMP. Therefore, the CHMP considers that the safety and tolerability of nivolumab has been described appropriately and is acceptable.

The CHMP considers the following measures necessary to address issues related to safety:

- To generate additional information on AEs of special interest (eg. immune-related pneumonitis, colitis, hepatitis, nephritis or renal dysfunction, endocrinopathies, rash, and other immune-related adverse reactions and infusion reactions) in routine oncology practice during post-marketing use. The study protocol will be discussed at PRAC within 3 months after the EC decision. The applicant should submit study CA209234, a non-interventional PASS. This post-authorisation measure is included in the RMP.

The CHMP recommends the following the following measure to address issues related to safety:

- To further evaluate the immunogenicity and the impact of ADA on efficacy and safety.

2.7. Pharmacovigilance

Detailed description of the pharmacovigilance system

The CHMP considered that the Pharmacovigilance system as described by the applicant fulfils the legislative requirements.

2.8. Risk Management Plan

The CHMP received the following PRAC Advice on the submitted Risk Management Plan:

The PRAC considered that the risk management plan version 1.2 is acceptable. The PRAC endorsed PRAC Rapporteur assessment report is attached.

The CHMP endorsed this advice without changes.

The applicant submitted an updated RMP to reflect some requested changes to the Product Information in the RMP.

The CHMP endorsed the Risk Management Plan version 1.3 with the following content:

Safety concerns

The Applicant identified the following safety concerns in the RMP:

Table 51: Summary of the Safety Concerns

Summary of safety concerns	
Important identified risks	Immune-related pneumonitis Immune-related colitis Immune-related hepatitis Immune-related nephritis or renal dysfunction Immune-related endocrinopathies Immune-related rash Other immune-related ARs Severe infusion reactions
Important potential risks	Embryofetal toxicity

	Immunogenicity Cardiac arrhythmias
Missing information	Pediatric patients <18 years of age Patients with severe hepatic and/or renal impairment Patients with autoimmune disease Patients already receiving systemic immunosuppressants before starting nivolumab

The PRAC agreed.

Pharmacovigilance plan

Table 52: Ongoing and planned studies in the PhV development plan

Activity/Study title (category 1-3)*	**Objectives**	**Safety concerns addressed**	**Status**	**Date for submission of interim or final reports**
CA209234: Pattern of Use, Safety, and Effectiveness of Nivolumab in Routine Oncology Practice. Category 3	To assess the effectiveness and safety of nivolumab, and management of important identified risks of nivolumab in patients with lung cancer or melanoma in routine oncology practice	Post-marketing use safety profile, management and outcome of immune-related pneumonitis, colitis, hepatitis, nephritis or renal dysfunction, endocrinopathies, rash, and other immune-related adverse reactions, and infusion reactions	Planned	Final CSR submission: 4Q2024

*Category 1 are imposed activities considered key to the benefit risk of the product.
Category 2 are specific obligations
Category 3 are required additional PhV activity (to address specific safety concerns or to measure effectiveness of risk minimisation measures)

The PRAC, having considered the data submitted, was of the opinion that the proposed post-authorisation PhV development plan is sufficient to identify and characterise the risks of the product.

The PRAC also considered that routine PhV is sufficient to monitor the effectiveness of the risk minimisation measures.

Risk minimisation measures

Table 53: Ongoing and planned studies in the PhV development plan

Safety concern	Routine risk minimisation measures	Additional risk minimisation measures
Important identified risks		
Immune-related pneumonitis Immune-related colitis Immune-related hepatitis Immune-related nephritis or renal dysfunction Immune-related endocrinopathies Immune-related rash Other immune-related ARs	The SmPC warns the risks of immune-related pneumonitis, immune-related colitis, immune-related hepatitis, immune-related nephritis and renal dysfunction, immune-related endocrinopathies, immune-related rash, and other immune-related adverse reactions in Section 4.4 (Special warnings and precautions for use), and provides specific guidance on their monitoring and management, including treatment delay or discontinuation and intervention with corticosteroids in Sections 4.2, 4.4 and 4.8, as appropriate. Further ADRs are included in Section 4.8. In addition, the package leaflet also includes specific warnings and descriptions of the most important safety information in the language suitable for patients.	To further raise awareness of HCPs on important risks and their appropriate management, additional risk minimization activity includes a Communication Plan. The Plan comprising 2 tools to be distributed to potential prescribers at launch by BMS: • Adverse Reaction Management Guide • Patient Alert Card
Severe infusion reactions	The SmPC warns the risk of severe infusion reactions in Section 4.4 and ADR in Section 4.8.	None
Important potential risks		
Embryofetal toxicity	SmPC includes Embryofetal Toxicity in Section 4.6 Fertility, pregnancy and lactation, Section 5.3 Preclinical safety data The package leaflet also includes specific description on the safety information in the language suitable for patients.	None
Immunogenicity	SmPC Section 4.8 Immunogenicity	None
Cardiac arrhythmias	SmPC Section 4.8 Undesirable effects	None
Missing information		
Pediatric patients	SmPC Section 4.2 Posology and method of administration, subsection on pediatric population	None
Severe hepatic and/or renal	SmPC Section 4.2 Posology and	None

impairment	method of administration: Patients with hepatic or renal impairment; Section 5.2 Pharmacokinetic properties: Hepatic or renal impairment	
Patients with autoimmune disease	SmPC Section 4.4 provides warning and cautionary information for patients with a history of autoimmune disease	None
Patients already receiving systemic immunosuppressants before starting nivolumab	SmPC Sections 4.4 Special populations and 4.5 Systemic Immunosuppressants	None

The PRAC, having considered the data submitted, was of the opinion that the proposed post-authorisation PhV development plan is sufficient to identify and characterise the risks of the product.

The PRAC also considered that routine PhV is sufficient to monitor the effectiveness of the risk minimisation measures.

2.9. Product information

2.9.1. User consultation

The results of the user consultation with target patient groups on the package leaflet submitted by the applicant show that the package leaflet meets the criteria for readability as set out in the *Guideline on the readability of the label and package leaflet of medicinal products for human use.*

3. Benefit-Risk Balance

Benefits

Nivolumab, an antibody that binds to PD-1 and blocks binding to the natural ligand PD-L1 and PD-L2, has shown superior efficacy compared to dacarbazine in untreated metastatic or unsectable melanoma patients in the pivotal phase 3 study CA209066. The trial showed a statistical significant benefit in terms of OS with a HR=0.42 (99.79%CI: 0.25 – 0.73; p-value <0.0001). The estimated OS rate at 12 months was 73% (95%CI: 9.33 – 12.09) and 42% (95%CI: 65.5 – 78.9) for nivolumab and dacarbazine, respectively. The secondary efficacy endpoint was supportive of the primary analyses: the PFS HR=0.43 (95%CI: 0.34-0.56; p<0.0001) with a median PFS of 5.1 months for nivolumab compared to 2.2 month for dacarbazine. The confirmed ORR was 40% compared to 13.9% for nivolumab and dacarbazine, respectively.

The second pivotal study, CA209037, showed a confirmed ORR of 31.7% and 10.6% for nivolumab and investigator's choice, respectively. A descriptive result for PFS highlights a benefit for nivolumab with HR= 0.74 (95% CI: 0.57, 0.97) for nivolumab over investigator's choice chemotherapy. The CHMP considered that the ORR and PFS data results were clinically relevant for a patient population with advanced disease and which has few treatment options. Subgroup analyses for both studies supported the main findings and were consistent with regard to the whole study population.

It is important to note that the study excluded certain patients which had active brain metastases or leptomeningeal metastases, ocular melanoma, patients which had known or suspected autoimmune disease and patients with conditions requiring systemic treatment with immunosuppressive agents. Therefore, caution must be used when treating patients with these risk factors. This is stated in the smPC section 4.4. Moreover, there were no patients with BRAF mutated tumours enrolled in this study. However, considering the data from study CA209037 and the fact that the mechanism of action of BRAFV600 and PD-1 appear to be independent of each other, it is believed that this patient population would still respond and benefit from nivolumab treatment.

It is of note that for both studies, the benefit of nivolumab was observed regardless of the presence of distant metastases and PD-L1 expression.

Beneficial effects

Uncertainty in the knowledge about the beneficial effects

There is uncertainty in the knowledge of nivolumab in the long term efficacy in previously untreated patients. Study CA209066 was stopped early upon the recommendation of the DMC following assessment of the results of an unplanned interim analysis. The study was unblinded and patients receiving dacarbazine were allowed to cross over to nivolumab treatment, creating a confounding factor for OS. Although there is no concern over the validity of the data, follow up data are required as the OS analysis was based on a small proportion of events at the time of this application. This is reflected as a condition in the Annex II.

Concerning the treatment of previously treated melanoma patients (study CA209037), more mature data are required to further describe the clinical benefit of nivolumab treatment. This is reflected as a condition in the Annex II.

The role of the biomarkers PD-L1 or PD-L2 expression as potential predictive or prognostic biomarkers remains undetermined. The CHMP has imposed a condition to the marketing authorization in Annex II to perform further analyses to ascertain the potential role of the PD-L2 biomarker and other markers, to further explore the relationship between PD-L1 and PD-L2 expression on the efficacy of nivolumab, to continue the exploration of the optimal cut-off for PD-L1 positivity and to further investigate the possible change in PD-L1 status of the tumour during treatment and/or tumour progression.

Risks

Unfavourable effects

The safety and tolerability of nivolumab was characterized in 3 studies where nivolumab was given as monotherapy: 2 Phase 3 studies (CA209037 and CA209066) in advanced melanoma and 1 Phase 1b study (MDX1106-03) in multiple tumour types, including melanoma. The most frequently reported drug-related AEs of any grade were fatigue, pruritus, nausea, diarrhea, and rash. Nivolumab is associated with immune-related adverse reactions. AEs on the basis of its mechanism of action and rate of frequency included endocrinopathies, diarrhoea/colitis, hepatitis, pneumonitis, nephritis, and rash, with the multiple event terms that may describe each of these.

The rate of ADA is very low and does not lead to any concern at this point. However, given the low number of ADA positive patients, the risk of developing ADA was not considered as fully investigated. Thus, the CHMP recommended to further evaluate the immunogenicity and the impact of ADA on efficacy and safety in patients treated with nivolumab (see clinical safety discussion).

Uncertainty in the knowledge about the unfavourable effects

AEs of special interest will be systematically assessed within ongoing and planned studies (see RMP).

Benefit-risk balance

Importance of favourable and unfavourable effects

The two pivotal phase 3 studies comparing nivolumab with dacarbazine or investigator's choice of treatment in a first line and last line patient population, respectively, has shown a convincing clinical benefit of major relevance for patients with advanced melanoma. In the first line treatment, nivolumab has shown an important clinical benefit with a statistically significant increase in survival of untreated melanoma patients. The data was considered robust and clinically compelling. In the last line indication, nivolumba showed a clinically significant improvement in overall response rate in patients who had received previous therapy. The toxicity of nivolumab was considered manageable. Appropriate wording can be found in the SmPC on the recommendations and management of treatment-related immune ADRs and adequate RMP measures have been implemented.

Benefit-risk balance

Based on the results of the pivotal trials CA209066 and CA209037, the benefits of nivolumab treatment in of advanced (metastatic or unresectable) melanoma patients outweighed the adverse events. Therefore, the CHMP considers that the benefit-risk balance for nivolumab in the proposed indication is positive.

4. Recommendations

Outcome

Based on the CHMP review of data on quality, safety and efficacy, the CHMP considers by consensus decision that the risk-benefit balance of Opdivo as monotherapy for the treatment of advanced (unresectable or metastatic) melanoma in adults is favourable and therefore recommends the granting of the marketing authorisation subject to the following conditions:

Conditions or restrictions regarding supply and use

Medicinal product subject to restricted medical prescription (see Annex I: Summary of Product Characteristics, section 4.2).

Conditions and requirements of the Marketing Authorisation

- **Periodic Safety Update Reports**

The marketing authorisation holder shall submit the first periodic safety update report for this product within 6 months following authorisation. Subsequently, the marketing authorisation holder shall submit periodic safety update reports for this product in accordance with the requirements set out in the list of Union reference dates (EURD list) provided for under Article 107c(7) of Directive 2001/83/EC and published on the European medicines web-portal.

Conditions or restrictions with regard to the safe and effective use of the medicinal product

- **Risk Management Plan (RMP)**

The MAH shall perform the required pharmacovigilance activities and interventions detailed in the agreed RMP presented in Module 1.8.2 of the Marketing Authorisation and any agreed subsequent updates of the RMP.

An updated RMP should be submitted:

- At the request of the European Medicines Agency;
- Whenever the risk management system is modified, especially as the result of new information being

received that may lead to a significant change to the benefit/risk profile or as the result of an important (pharmacovigilance or risk minimisation) milestone being reached.

If the dates for submission of a PSUR and the update of a RMP coincide, they can be submitted at the same time.

- **Additional risk minimisation measures**

Prior to launch of OPDIVO in each Member State the Marketing Authorisation Holder (MAH) must agree about the content and format of the educational programme, including communication media, distribution modalities, and any other aspects of the programme, with the National Competent Authority.

The educational programme is aimed at increasing the awareness about the potential immune mediated adverse events associated with OPDIVO use, how to manage them and to enhance the awareness of patients or their caregivers on the signs and symptoms relevant to the early those adverse events.

The MAH shall ensure that in each Member State where OPDIVO is marketed, all healthcare professionals and patients/carers who are expected to prescribe and use OPDIVO have access to/are provided with the following educational package:

- Physician educational material
- Patient alert card

The physician educational material should contain:

- The Summary of Product Characteristics
- Adverse Reaction Management Guide

The Adverse Reaction Management Guide shall contain the following key elements:

- Relevant information (e.g. seriousness, severity, frequency, time to onset, reversibility of the AE as applicable) for the following safety concerns:
    - Immune-related pneumonitis
    - Immune-related colitis
    - Immune-related hepatitis
    - Immune-related nephritis or renal dysfunction
    - Immune-related endocrinopathies
    - Immune related rash
    - Other immune-related ARs
- Details on how to minimise the safety concern through appropriate monitoring and management
- **The patient alert card** shall contain the following key messages:
- That OPDIVO treatment may increase the risk of:
    - Immune-related pneumonitis
    - Immune-related colitis
    - Immune-related hepatitis

- Immune-related nephritis or renal dysfunction
- Immune-related endocrinopathies
- Immune related rash
- Other immune-related ARs

- Signs or symptoms of the safety concern and when to seek attention from a HCP
- Contact details of the OPDIVO prescriber

- **Obligation to complete post-authorisation measures**

The MAH shall complete, within the stated timeframe, the below measures:

Description	**Due date**
1. Post-authorisation efficacy study (PAES): The MAH should submit the final study report for study CA209037: a Randomized, Open-Label, Phase 3 Trial of nivolumab vs Investigator's Choice in Advanced (Unresectable or Metastatic) Melanoma Patients Progressing Post Anti-CTLA-4 Therapy.	The final clinical study report should be submitted by 30th June 2016
2. Post-authorisation efficacy study (PAES): The MAH should submit an updated OS data for study CA209066: a Phase 3, randomized, double-blind study of nivolumab vs dacarbazine in subjects with BRAF wild type, previously untreated, unresectable or metastatic melanoma.	The updated data/study report should be submitted by 31st December 2015
3. The value of biomarkers to predict the efficacy of nivolumab should be further explored, specifically: 1) To continue the exploration of the optimal cut-off for PD-L1 positivity based on current assay method used to further elucidate its value as predictive of nivolumab efficacy. These analyses will be conducted in studies CA 209037 and CA209066 in patients with advanced melanoma. 2) To further investigate the value biomarkers other than PD-L1 expression status at tumour cell membrane level by IHC (e.g., other methods / assays, and associated cut-offs, that might prove more sensitive and specific in predicting response to treatment based on PD-L1, PD-L2, tumour infiltrating lymphocytes with measurement of CD8+T density, RNA signature, etc.) as predictive of nivolumab efficacy. These additional biomarker analyses are occurring in the context of study CA209-038 and study CA209-066. 3) To further investigate at post-approval the relation between PDL-1 and PDL-2 expression in Phase 1 (CA209009, CA209038 and CA209064). 4) To further investigate the associative analyses between PDL-1 and PDL-2 expression conducted in study CA209-066. 5) To further investigate at post-approval the possible change in PD-L1 status of the tumour during treatment and/or tumour progression in	30th September 2015 30th September 2017 31st March 2017 31st December 2017 30th September 2017

studies CA209-009, CA209-038 and CA209-064.	

Conditions or restrictions with regard to the safe and effective use of the medicinal product to be implemented by the Member States

Not applicable.

New Active Substance Status

Based on the CHMP review of data on the quality properties of the active substance, the CHMP considers that nivolumab is qualified as a new active substance.